



MINISTRY OF HEALTH MALAYSIA

MANAGEMENT GUIDELINES OF MALARIA IN MALAYSIA



**Vector Borne Disease Sector,
Disease Control Division**



MINISTRY OF HEALTH MALAYSIA

Management Guidelines of Malaria in Malaysia

**VECTOR BORNE DISEASE SECTOR
DISEASES CONTROL DIVISION
MINISTRY OF HEALTH MALAYSIA**

2025

Management Guidelines of Malaria in Malaysia (2025)



National Strategic Framework for
Prevention of Malaria Re-establishment
(2021-2026)

MINISTRY OF HEALTH MALAYSIA

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MANAGEMENT GUIDELINES OF MALARIA IN MALAYSIA

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Any enquiries should be addressed to:

Disease Control Division,
Ministry of Health, Malaysia
Level 3, Block E10, Complex E, Precinct 1,
Federal Government Administration Centre,
62590 Putrajaya, MALAYSIA.
Tel: +03-88834418
Email: unitmalaria@moh.gov.my
Website: www.moh.gov.my

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FOREWORD BY THE DEPUTY DIRECTOR-GENERAL OF HEALTH



Malaysia's efforts in malaria prevention and control programme started during the Malaria Eradication Programme in 1960. In 2011, Malaysia has participated in the Malaria Elimination programme and started to report zero indigenous human malaria in 2018. After reporting zero indigenous human malaria for three (3) consecutive years in 2021, Malaysia has moved into prevention of malaria re-establishment phase.

In reference to the previous guidelines, numerous clinical development has been updated particularly the expansion of the artemisinin combination therapy (ACT) usage among children as recommended by the World Health Organization (WHO). Over the past decade, the knowlesi malaria has garnered international attention with Malaysia as the key international player for international scientists and experts to collaborate for better malaria management and treatment, including diagnostics and molecular surveillance.

The Management Guidelines of Malaria 2025 strengthens the diagnostics protocols, laboratorial capacities and concepts, and drug response surveillance for all anti-malarials. It is our sincere hope that all health care personnel, private and public fraternities will utilize it optimally to prevent malaria from re-establishing in our country. Lastly, I would like to acknowledge the contribution of the technical working group in developing a consensus policy on malaria case management for Malaysia.

DR ISMUNI BIN BOHARI

Deputy Director-General of Health Malaysia (Public Health)

PREFACE

Malaria remains as a major public-health concern, with estimated 249 million malaria cases with 608 000 malaria deaths in 85 countries in 2022.¹ Malaria has been reported in Malaysia even before 1900's. In 1990, there were 50,500 cases in Malaysia. In 2022, there were 2,869 cases which is a 94.3% reduction of malaria cases compared to the figures in 1990's. There has also been a reduction in the number of malaria deaths from 43 in 1990 to 11 deaths in 2022. The mortality rate due to malaria has been around 0.001 per 1,000 population since 2006. All figures and rates are inclusive of *P. knowlesi*.

Over the decade, early diagnosis and prompt treatment remain the fundamental of malaria control and need to be readily available. Intravenous anti-malarial treatment has been noted to effective and fast treatment for all-forms of malaria. In many occasions, the emergency treatment has saved many lives and preventable large number of malaria deaths.

Since 2018, Malaysia has reported zero indigenous human malaria cases. However, this guideline caters for both human and knowlesi malaria that have been detected in Malaysia. This guideline has been updated with the latest World Health Organization (WHO) and revised from previous national guidelines (published in year 1993 and 2015).

DRAFTING COMMITTEE

ADVISORS

Datuk Dr Norhayati binti Rusli

Deputy Director-General of Health
(Public Health)

Dr Thilaka A/P Chinayah @ Nadarajah

Director
Disease Control Division

AUTHORS

Dr Zailiza binti Suli

Sector Head
Vector Borne Diseases
Disease Control Division

Dr Tam Jenn Zhueng

Senior Principal Assistant Director
Disease Control Division

Dr Mohd Hafizi bin Abdul Hamid

Senior Principal Assistant Director
Disease Control Division

Dr. Shamilah Hisam,

Parasitology Unit Head,
Institute for Medical Research

Dr Lee Heng Gee,

Infectious Disease Consultant, Ministry of
Health, Malaysia

Dr. Chua Hock Hin,

Infectious Disease Consultant, Ministry of
Health, Malaysia

Kumuthamalar Sangaran,

Science Officer, National Public Health
Laboratory, Ministry of Health, Malaysia

Nur Ziana Binti Abdullah Sani

Senior Entomologist
Disease Control Division

REVIEWERS

**Prof. Datuk Dr. Lokman Hakim
Sulaiman**

Professor of Public Health,
International Medical University

Datuk Dr Sethuraman Nagappan

Clinical Associate Professor of
Internal Medicine,
International Medical University

CONTRIBUTORS

Prof. Dr Jenarun Jelip

Professor of Public Health
University Malaysia of Sabah (UMS)

ABBREVIATIONS

ACT	Artemisinin-based Combination Therapy
ASMQ	Artesunate / Mefloquine
BD	Bis Die (Latin for twice a day)
BFMP	Blood Film for Malaria Parasite
CVP	Central Venous Pressure
EBT	Exchange Blood Transfusion
EDTA	Ethylenediaminetetraacetic Acid
FDC	Fixed-dose Combination
GCS	Glasgow Coma Scale
G6PD	Glucose-6-Phosphate Dehydrogenase
GI	Gastrointestinal
HDU	High Dependency Unit
ICU	Intensive Care Unit
LDH	Lactate dehydrogenase
NSPEM	National Strategic Plan for Elimination of Malaria
OD	Omne in Die (Latin for once daily)
RDT	Rapid Diagnostic Test
TNF- α	Tumor Necrosis Factor Alpha

TABLE OF CONTENT

Foreword	i
Preface	ii
Drafting Committee	iii
Abbreviations	iv
Table of Content	v
Introduction	1
Chapter 1 National Policy on Management of Malaria In Malaysia:	
1.1 Agencies in Malaria Control Program and the Linkages	5
1.2 Policy statement regarding malaria control	6
1.3 List of related Act and Regulatory Bodies	6
Chapter 2 Malaria Situation in Malaysia	
2.1 Fact Checks On Malaria In Malaysia	7
2.2 Geography in Malaysia	7
2.3 Malaysia Weather and Climate	8
2.4 Malaria Cases Terminology	9
2.5 Acquired Malaria Cases	10
2.6 Distribution according to malaria species	11
2.7 Zoonotic Malaria in Malaysia	12
Chapter 3 Malaria Parasites:	
3.1 Malaria Parasite	13
3.2 Transmission Cycle of Malaria	15
3.3 Malaria Species in Man and Primates	15
3.4 Relapse and Recrudescence	17
3.5 Role of Gametocytes in Transmission of Malaria	20
3.6 Fever in Malaria	21
Chapter 4 Mosquito Vectors In Malaysia:	
4.1 The Anopheline mosquitoes	23
4.2 Anophelines as Vectors of Malaria in Malaysia	23
4.3 Intergrated Vector Control Strategies in Malaysia	24
Chapter 5 Human and Simian Hosts of Malaria In Malaysia:	
5.1 Risk Groups and Risk Activities to acquire malaria	25
5.2 Innate Vs Adaptive Immunity in Malaria	26
5.3 Primates as Natural Host of Zoonotic Malaria	28

Chapter 6	Management of Malaria for Public Health	
6.1	Individual Case management	29
6.2	Suspected Malaria Definition	30
6.3	Case Detection	31
6.4	Malaria Diagnosis Method	32
6.5	Mandatory Hospital Admission	32
6.6	Notification	32
6.7	Malaria Treatment	33
6.8	Case Investigation	33
6.9	Case Classification	34
6.10	Case Registration	34
6.11	Post Discharge Follow-up	34
6.12	Focus Investigation and Response	35
6.13	Human Malaria : Case classification & management	36
6.14	Intervention matrix for Human Malaria foci management	36
6.15	Zoonotic Malaria : Case classification & management	37
6.16	Intervention matrix for Zoonotic Malaria management	37
6.17	Malaria Case Management Data Flow	38
Chapter 7	Clinical Features of Malaria:	
7.1	Symptoms of Malaria	39
7.2	Definition of Uncomplicated Malaria	40
7.3	Definition of Severe / Complicated Malaria	40
7.4	Definition of Severe P. knowlesi Malaria	42
7.5	Chronic Malaria	43
Chapter 8	Diagnosis of Malaria:	
8.1	Malaria diagnosis system	45
8.2	Malaria suspected definition	47
8.3	Clinical diagnosis	47
8.4	Laboratory diagnosis	48

Chapter 9	Malaria Treatment:	
	9.1 Treatment of uncomplicated malaria	53
	9.2 Treatment of severe and complicated malaria	54
	9.3 Treatment of special risk groups	59
	9.4 Field treatment	71
	9.5 Presumptive treatment	71
	9.6 Discharge and follow-up	71
Chapter 10	Prevention of Malaria (Chemoprophylaxis)	75
Chapter 11	Vector Control Strategies In Malaysia	81
Chapter 12	Antimalaria Drug Resistance Surveillance & Monitoring	83
Chapter 13	Quality Assurance Programme for Malaria Microscopy	85
Chapter 14	Enhanced Case Surveillance To Prevent Re-establishment of Malaria in Malaysia	89
Chapter 15	Malaria Training Program	93
Chapter 16	Research (Potential Area)	95
	16.1 Anti-malarial Drugs	
	16.2 Vaccines	

INTRODUCTION

Concise Notes on Malaria In Malaysia 2024

- *There are two types of malaria in Malaysia: 1) human Malaria, due to infection by Plasmodium falciparum, P. vivax, P. malariae and P. ovale; and 2) zoonotic malaria, attributed predominantly to Plasmodium knowlesi*
- *Plasmodium knowlesi is a malaria species known to exist in its natural host, the macaques, locally named as 'kera', 'beruk', and 'lotong'.*
- *Transmission of malaria is via mosquito bites from various species of the Anopheles spp. which breeds in the near fringes and deep forested areas in our tropical climate*
- *From 2018 to 2024, there are no longer indigenous human malaria cases in Malaysia*
- *Zoonotic malaria due to P. knowlesi, however remains in the 2,000 – 4,000 cases per year, thus becoming a high priority area in the national malaria elimination agenda*
- *Since there are no longer indigenous human malaria in Malaysia, we intend to sustain this record by planning and enforcing several strategies for the prevention of re-establishment of malaria*
- *Publication of this Guideline on The Management of Malaria is primarily intended to guide and assist the medical and public health personnel of the MOH Malaysia involved in the routine management of malaria*
- *The Guideline also serves as a necessary MUST READ to the private healthcare institutions and related government and non-government agencies directly and/or indirectly involved in the control and prevention of malaria in Malaysia.*

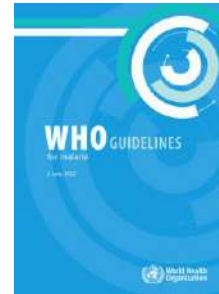
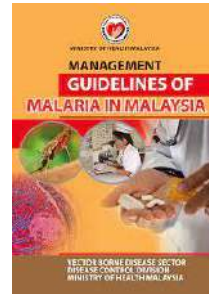
Background

Although malaria control activities over the last few decades have greatly reduced the incidence of malaria in Malaysia, it is still a major public health problem in the hinterland and less developed parts of the country and accounts for a considerable number of cases treated at health clinics and admissions to hospitals. In addition, frequent and affordable travel especially to malaria endemic countries and influx of workers from endemic countries in search of work have kept the number of imported malaria cases on the constant high and pose a real threat of re-introducing malaria into areas which have eliminated the disease¹.

Specific elimination's strategies have been outlined in accordance with the WHO's guidelines. Consistent political, financial and human resource commitments have contributed to Malaysia's success in maintaining the pre-elimination phase of malaria with positive progression towards elimination of malaria in the country.

This guideline on the management of malaria has been revised and an updated based on previous national guidelines (published in year 2013) and the WHO guidelines for malaria (1st edition, 2022).

It is intended primarily for healthcare personnel in Malaysia for the management of malaria patients. The Management Guidelines of Malaria in Malaysia 2022 made available on Ministry of Health web-page as referral and the hard-copy of the guideline had been produced and distributed to all health facilities in Malaysia.



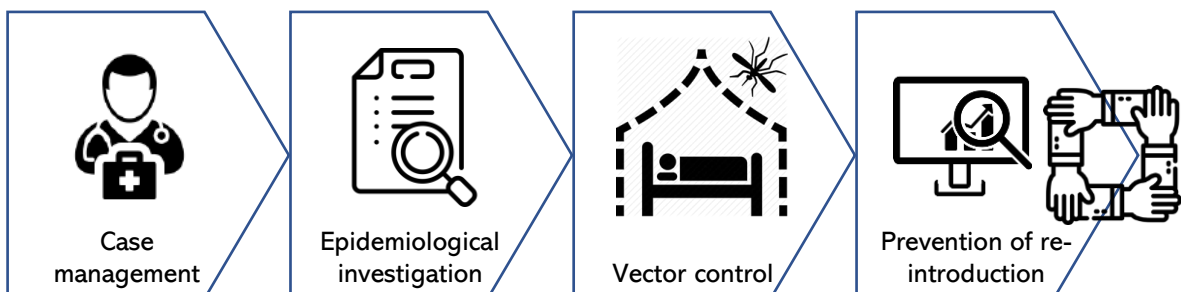
Objectives

This guideline aims to assist medical and health personnel in the management of malaria and its complications. It also aims to promote the use of safe, effective and updated methods to manage malaria and sustain currently used treatment and preventive actions that proves beneficial to the prevention of drug and/or insecticide resistance.

Scope

This guideline provides a structured guidance to manage malaria cases at different level of health services starting from detection and diagnosis of malaria cases until the management of individual (i.e the infected patient) and the community interlinked with such individual. The Guideline, in short, covers:

- Immediate actions to care for an infected person,
- Immediate and reactive actions taken to link/connect illness to source of infection,
- Responsive actions to prevent spread of infection in the forms of public health interventions including the use of intergrated vector control measures
- Planned preventive actions to mitigate future outbreak situations and re-emergence of indigenous malaria



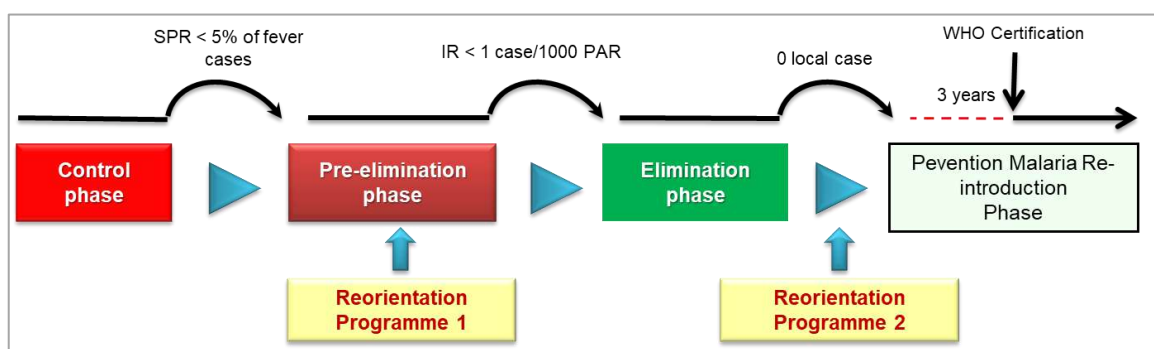
Intended User

This Guideline is targeted to serve the needs of several key players in the management of malaria such as:

Health personnel in the Medical / Clinical Departments in Government and Private Healthcare Institutions	Clinical Specialists, Medical Officers, Nurses in Infectious Diseases, Pathologists; Science Officers, Medical Lab Technologists
Health Personnel in the Health Clinics at primary healthcare facilities in Govt and Private agencies	Medical / Science Officers, Medical Lab technologists; Medical Lab Assistants,
Public Health personnel in the Malaria Control Program of NMOH	Malaria Program Managers; State Malaria Units; State Vector Units; Public Health Laboratories consortium; Inst. Medical Research; Health Promotion departments
Advocates of National Malaria Control Program	Malaria Ambassadors / Volunteers in communities; Non-Govt organisations; the public

Publication of the Guideline is also to support strategies and activities undertaken to achieve malaria elimination

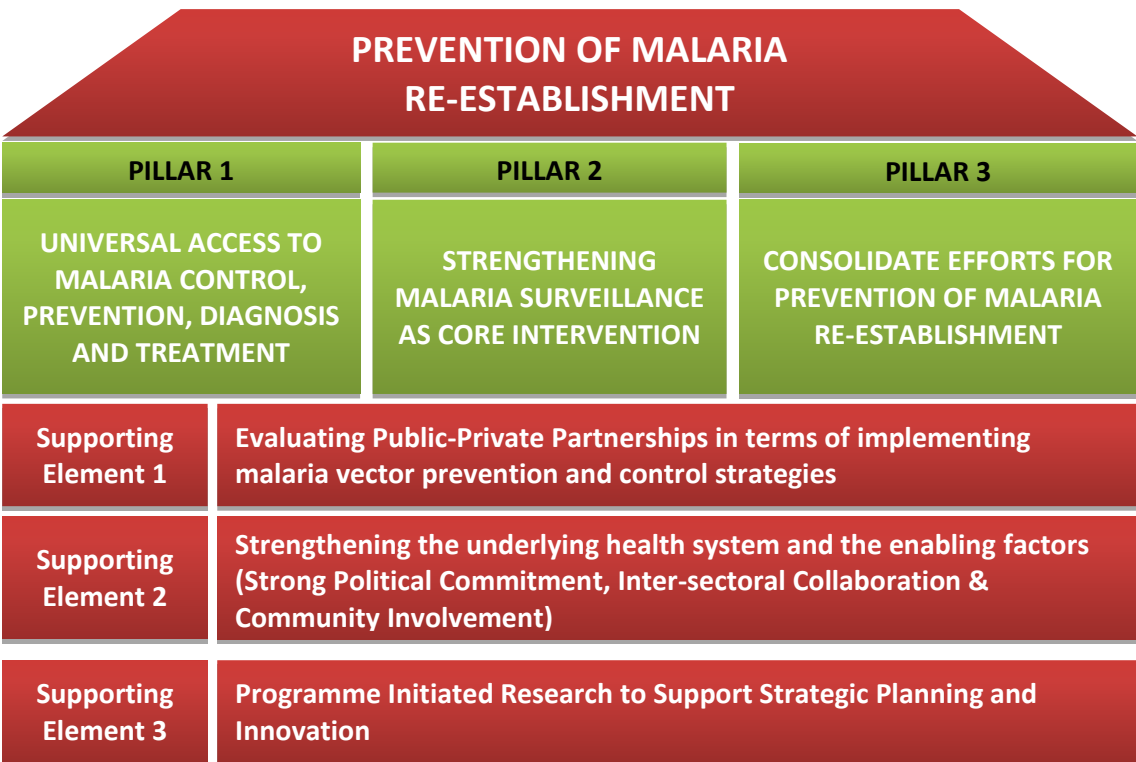
In 2011, the national malaria program strategy was reoriented from control to elimination, and Malaysia has achieved zero indigenous human malaria for five consecutive years since 2018. Malaysia has embarked into Malaria Elimination Program since 2011 and built on its experience and past achievements to strengthen current efforts and secure continuous political commitments towards malaria elimination certification from WHO². MOH Malaysia has conducted another reorientation programme and develop the National Strategic Framework for Prevention of Malaria Re-establishment (NSFPOR) 2021-2025, which is one of the key requirements for malaria elimination certification by WHO³.



Birth of National Strategic Framework for Prevention of Malaria Re-establishment (NSFPOR) 2021-2026

MOH Malaysia has conducted a programme review and develop the National Strategic Framework for Prevention of Malaria Re-establishment (NSFPOR) 2021-2026, which is one of the key requirements for malaria elimination certification by WHO.

The NSFPOR consists of 3 pillars and 3 supporting elements³.



References:

1. Eliminating Malaria, Progress Towards elimination in Malaysia. APMEN case study 8, 2012.
2. National Strategic Plan For Elimination Of Malaria 2011 – 2020. MOH Malaysia, 2010.
3. National Strategic Framework for Prevention of Re-establishment for Malaria (NSFPOR) 2021-2026. MOH Malaysia, 2024.



CHAPTER 1 :

National Policy On Management Of Malaria In Malaysia

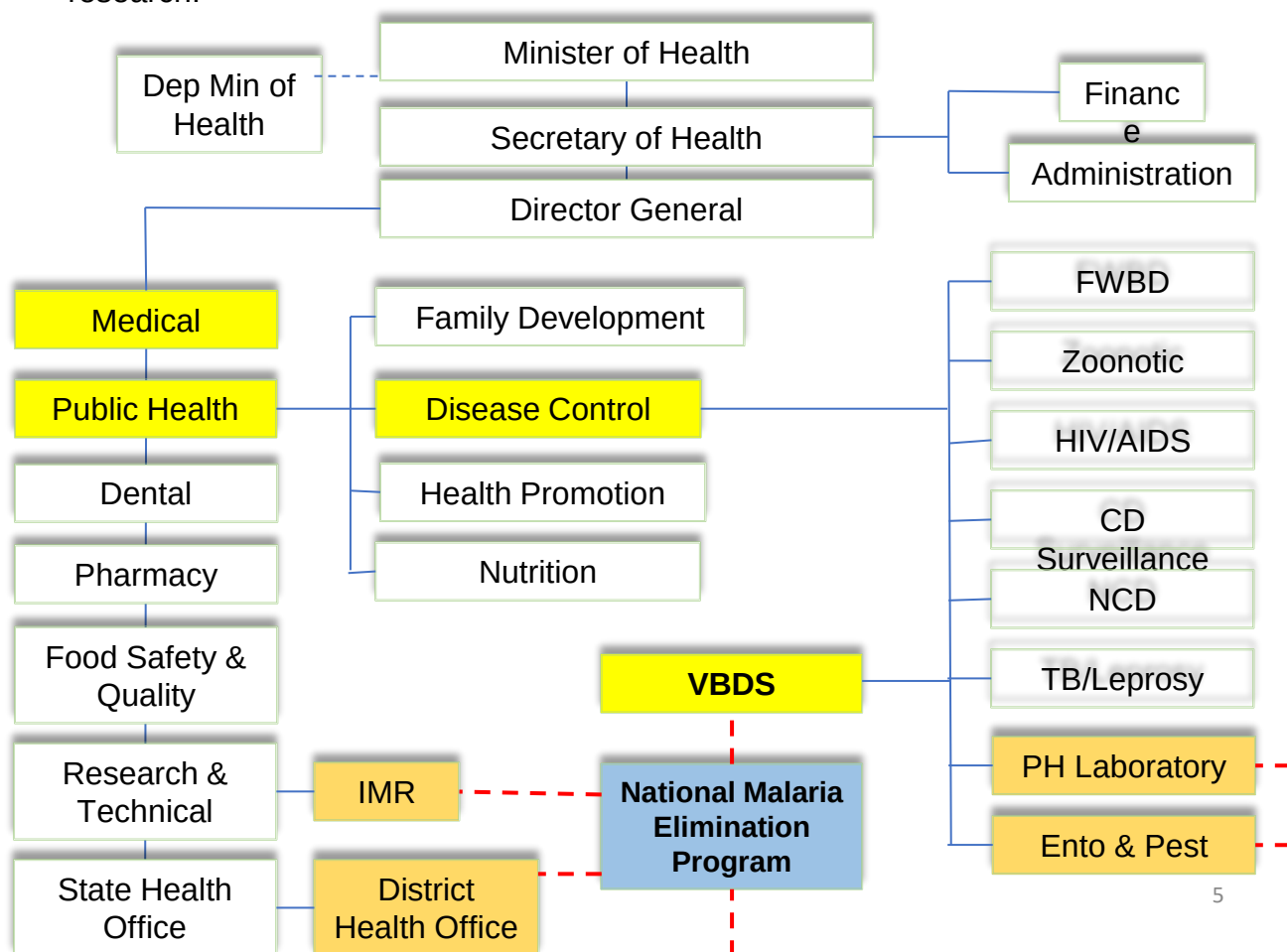
1.1 Agencies involved in Malaria Elimination Program and the linkages

The National Malaria Elimination Program is managed within the Vector-Borne Disease Sector (VBDS) under Disease Control Division and is responsible for developing policies, strategies, guidelines, and providing technical support for state and district level malaria programs. VBDS coordinates with the Entomology and Pest Sector to develop policy and provide supervision for vector control activities and entomological surveillance.

Each state and district has its own Malaria Unit, where states provide additional oversight and assistance for their districts.

District level malaria personnel are responsible for carrying out activities in their localities and providing feedback to state and national level programs to inform strategic planning; emphasis is on increased strengthening for program management and supervision and transitioning from elimination to control.

Other partner agency include National Public Health Laboratory which involve in the malaria diagnosis quality control, quality assessment and training for the microscopists in the country. While Institute for Medical Research support the program in the molecular diagnosis expertise, molecular surveillance and malaria research.

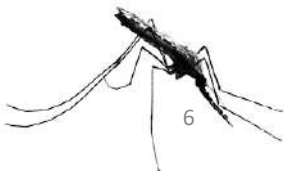


1.2 Policy statement regarding malaria control

Malaysia has the Prevention and Control of Infectious Diseases Act 1988 with regards to legislative provisions for sustainable governmental activities in communicable diseases. Malaria is one of the 30 diseases that need to be notified by medical practitioner. Various platforms have been developed under this regulation such as via paper based fax machine, email, online system-eNotification, emails, and etc. Punitive actions will be taken on practitioner that fails to comply. Besides the Act, the Destruction of Disease-Bearing Insect Act 1975 is another legislation that has relevance with malaria. Any large land clearing that would likely propagate or harbour disease-bearing insects without prior approval is an offence under this act.

1.3 Related Act and Regulatory Bodies

1.	CDC Act 1998	Malaria is an infectious disease that must be notified within seven (7) days under this Act. This allows all reported malaria cases be treated in any health facilities including private clinics.
2.	DDBIA Act 1975	This Act is to enforce the destruction of vectors.
3.	EHIA	Environmental Quality Act 1974 provides the legal framework for environmental protection, including EIA requirements. The Environmental Quality (Prescribed Activities) (Environmental Impact Assessment) Order 2015 specifies which activities require an EIA, which includes health impact assessment
4.	Pesticides Board	The registration authority for pesticides under the Pesticides Act 1974. It's responsible for ensuring that pesticides sold and used in the country are effective, safe, and meet quality standards. The Board also regulates the importation, manufacture, sale, and use of pesticides, including highly toxic ones.
5.	National Pharmaceutical Regulatory Agency (NPRA)	A regulatory body under the Ministry of Health Malaysia, plays a crucial role in ensuring the quality, safety, and efficacy of pharmaceutical products, including medicines, natural health supplements, and cosmetic products available in the Malaysian market.
6.	Medical Device Authority	A statutory body under the Ministry of Health Malaysia which was established under the Medical Device Authority Act 2012 (Act 738) to control, regulate medical device, its industry and activities as well as to enforce medical device law under Medical Device Act 2012 (Act 737).



CHAPTER 2 : Malaria Situation In Malaysia

2.1 Fact Checks On Malaria In Malaysia

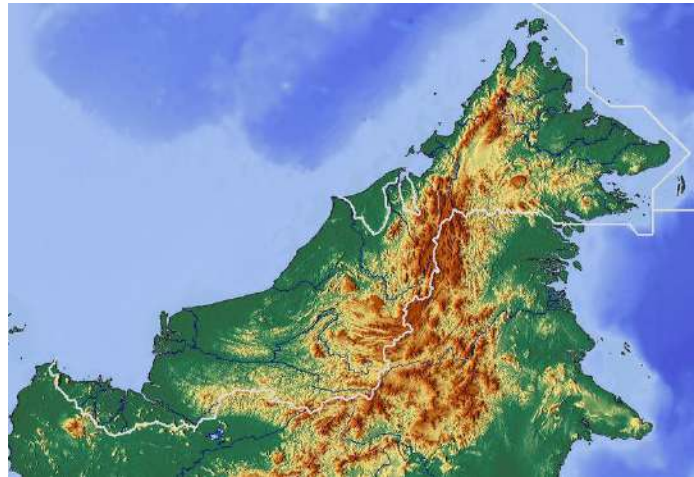
- Human malaria parasites in Malaysia : *P.falciparum*, *P.vivax*, *P.ovale* & *P.malariae*
- Zoonotic malaria parasite reported in Malaysia : *P.knowlesi*
- There are no longer incidence of local (indigenous) malaria reported in Malaysia from 2018 – 2022, leading us to commence application of certification of malaria-free status from WHO
- In 2022, there were a total of 286 cases of imported human malaria and 2,514 cases of malaria attributed to *P. knowlesi*.
- Incidence of *P. knowlesi* malaria has been recognised in Malaysia since 2013, with an average of 1,500 to 2,500 cases each year.
- Distribution of imported malaria cases is represented predominantly in Sarawak (53% of total imported malaria cases in 2022)
- Distribution of simian malaria cases is also largely represented in Sabah and Sarawak (74.1% of total malaria cases in Malaysia)
- Malaria cases in Peninsular Malaysia are mainly in the states of Kelantan, Perak, Pahang and Selangor, representing 13.5% of total malaria cases in Malaysia
- The highest incidence of imported malaria is attributed to *P. vivax* (70.7%) followed by *P. falciparum* (25.5%), *P. Ovale* (2.4%) and *P. malariae* (1.4%)
- Regrettably, malaria case fatality recorded in 2022 was excessively due to *P. knowlesi*. Factors contributing to this event are further discussed.

2.2 Geography of Malaysia

Malaysia is a tropical country situated in South East Asia, consisting of Peninsular Malaysia and north Borneo states of East Malaysia, divided by the South China Sea. Peninsular Malaysia extends from south of Thailand's Isthmus of Kra until the borders of Singapore and the Indonesian Archipelago. States of Sabah and Sarawak which represent the East Malaysia lie from the borders of Kalimantan Indonesia (Borneo) until the southern islands of the Philippines¹.



A central mountain range with high peaks and dense jungle named the Banjaran Titiwangsa divides Peninsular Malaysia into the East Coast, facing the South China Sea; and the West Coast facing the Straits of Malacca and Sumatra Islands of Indonesia. A steep chain of mountains also exist separating the east and west coast of Sabah named the Crocker Range, covered in dense green rainforest. Mount Kinabalu, one of the highest mountains in Southeast Asia, is part of this range. Most of Malaysian populations live in the plains and foothills of the West Coast of Peninsular Malaysia compared to East Coast and Sabah, Sarawak. Primary forest covers at least 60% of the land with tropical vegetations and forested areas thriving even in the cities. An abundance of not only the flora, but also wildlife such as the monkeys, gibbons, tigers, elephants and limitless species of birds and insects inhabit the jungle with frequent encounters with wild monkeys in fringes of urban settlements¹.



2.3 Malaysia Weather & Climate

Malaysia enjoys tropical weather all year round. Due to her proximity to water resources, the climate is often quite humid. Despite this, the weather is never too hot and temperatures range from a mild 20°C to 30°C average throughout the year. However, the highlands experience cooler temperatures¹.

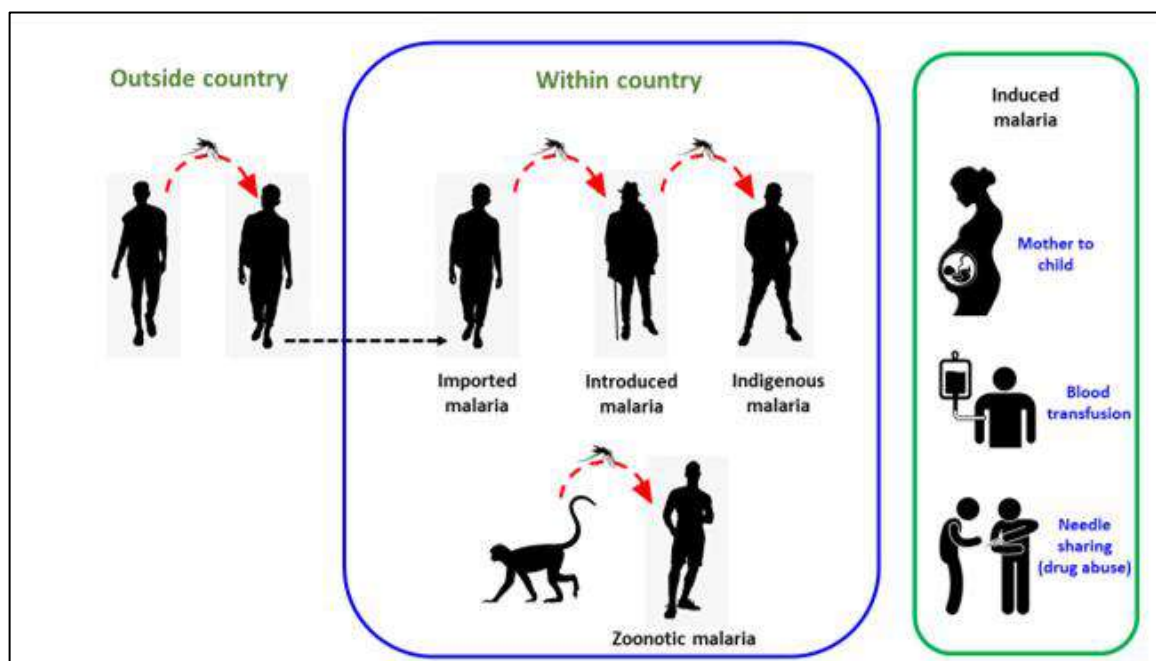
The nature of Malaysia's weather and climate means that it experiences a monsoon season. However, this varies depending on the destination. The south-west experiences its monsoon season from May to September while November to March is the wettest in the north-east¹.

The monsoon also varies on the Peninsular Malaysia coastline where Kuala Lumpur experiences rainfall from March to April and September to November. Peninsular Malaysia generally receives a rainfall average of 2500 mm annually while East Malaysia experiences 5080 mm of annual rainfall¹.

2.4 Malaria Cases Terminology

There are two sources of malaria infection in Malaysia: 1) from man to man, and 2) from monkeys to man, attributed to *P. knowlesi*. In the well-known man to man transmission of malaria, the parasite could originate from the local population or acquired via people movement inter-country². Since 2018 until 2022, there were no local (indigenous) cases of malaria recorded (Fig. 1) while 301 acquired cases were recorded in 2022². Acquired malaria is further categorized as imported or introduced according to published WHO Malaria Terminology (2021)³.

Malaria Terminology (WHO, 2021)	
Indigenous Case	A case contracted locally with no evidence of importation and no direct link of transmission from an imported case
Imported case	Where infection was acquired outside the area in which it was diagnosed
Introduced case	A case contracted locally, with strong epidemiological evidence linking it directly to a known imported case (1 st generation local transmission)
Induced case	A case where the origin can be traced to a blood transfusion or other forms of parenteral inoculation of the parasite BUT NOT due to a natural mosquito-borne inoculation
Zoonotic malaria (Definition by CDC)	malaria that is transmitted from animal to human
Simian or primate or monkey malaria	Malaria that is transmitted from monkeys / macaques to human
Malaria Death	Number of death due to malaria



Distribution according to classification of malaria and parasite species

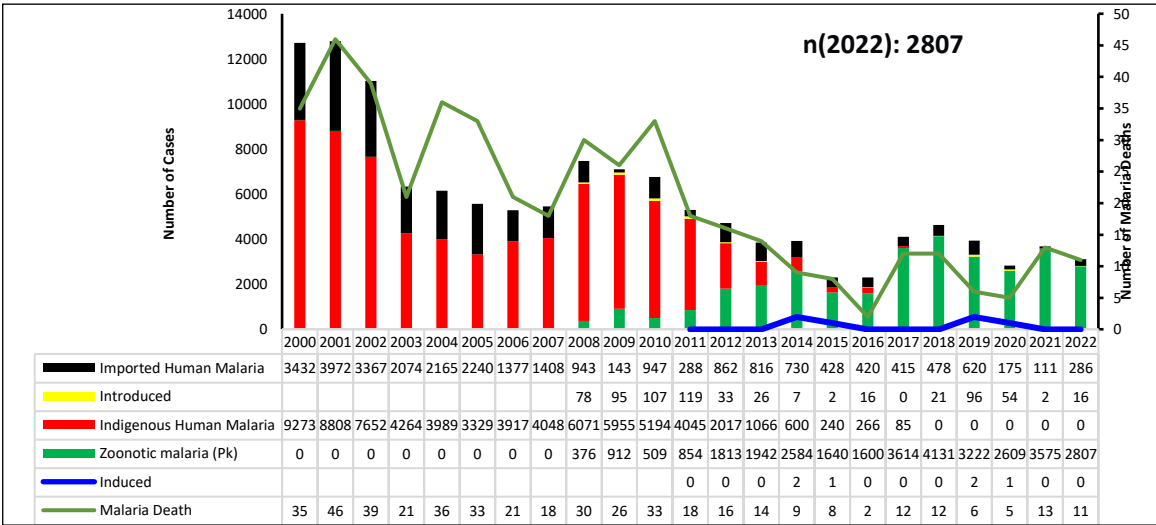


Figure 1: Number of Malaria Cases (by case classification) and Death in Malaysia (2000–2022)

The reduction of indigenous malaria to zero cases and its zero sustainability for the three consecutive years were not accidental. This achievement was made possible through a well-planned disease control program that encompass continuous surveillance and monitoring of active and passive detections of malaria cases, clear guidelines in treatment and follow-ups of cases and the execution of an integrated vector control strategies².

Referring to Figure 1, Malaysia gradually recorded an 87.3% reduction in human malaria incidence from 0.55 per 1000 populations in year 2000 to 0.03 per 1000 populations in year 2021. Malaria mortality recorded an 87% reduction from 0.15 per 100,000 populations in year 2000 to 0.04 per 100,000 populations in 2022².

2.5 Acquired Malaria Cases

As the Covid-19 pandemic regressed and enters the endemic phase, our borders are now opened to influx of visitors and foreign workers that are badly needed to support economic re-building. According to the MITI Weekly Bulletin (Vol. 715, 15 Nov 2022), Malaysian economic performance in production and expenditure showed sectors such as in the service, construction and manufacturing dominate the economy much more than in mining and quarrying and agriculture. The Statistics Department has recorded 2.1 million foreigners working in various sectors in Malaysia up to June 2022 as compared to 1.98 million regular migrant workers hired in 2019. This constitutes about 20% of the country's labor force in 2022. According to the International Labor Organization (ILO), in 2020 there were 2,090,900 migrant workers in Malaysia that constitutes an approximate 12% of labor force in Malaysia. The World Bank report in 2019 indicate that approximately one-third of workers in the service sector and 25 percent in agriculture were migrant workers. From the health perspective, this large number of people pose some challenges in malaria control and elimination due to:

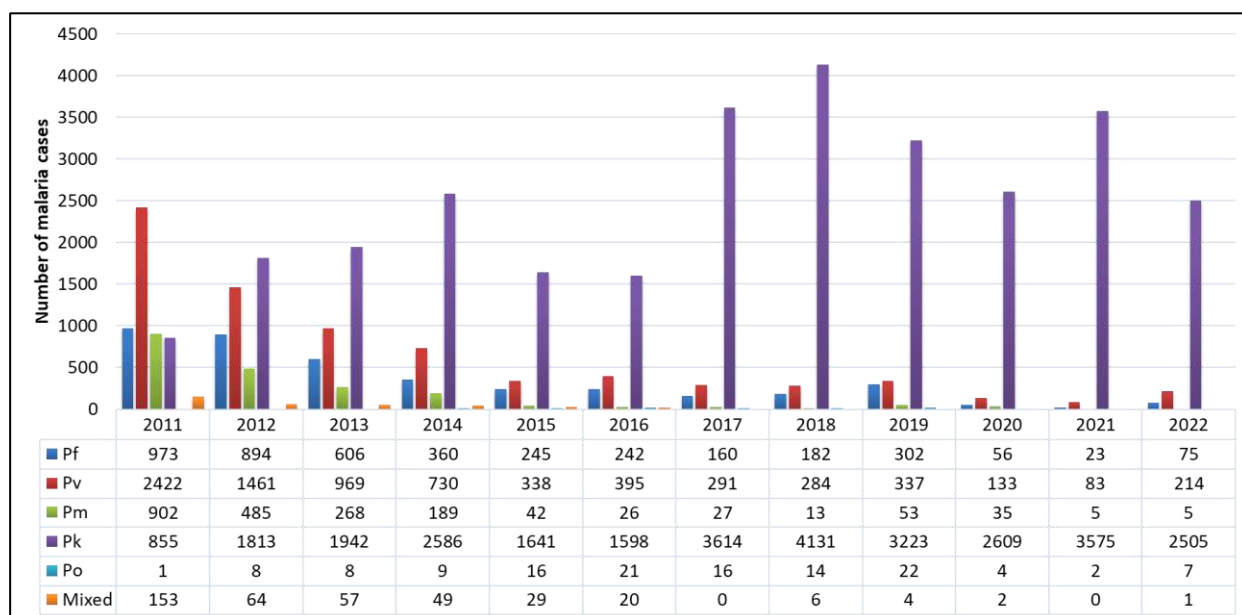
Migrant workers and visitors from known malaria endemic countries unknowingly carry parasitemia at recrudescence level and in some instances, harbor reservoirs of dormant hypnozoites of *P. vivax*, predisposing to relapse malaria, and

- A dynamic people movement in economic sectors that predispose to land clearing and increased exposure to malaria vectors
- Variable immune response to illness in people with continuous exposure to malaria versus people with low exposure to malaria is one factor that contributes to good or poor treatment response to antimalarial agents and directly impact cost of hospitalization
- Though foreign workers entering Malaysia via the registered foreign worker recruitment agencies are routinely screen for malaria before permitted to work in Malaysia, a sizeable proportion of these workers enter Malaysia via the porous border of the country and recruited without undergoing malaria screening.

In 2022, 286 (10.2%) of the total malaria in Malaysia could be traced to source of infection outside Malaysia. These cases resulted in introduction of infection to 16 local people. In short, above challenges in dealing with a dynamic people movement create pockets of imported and introduced malaria into the country².

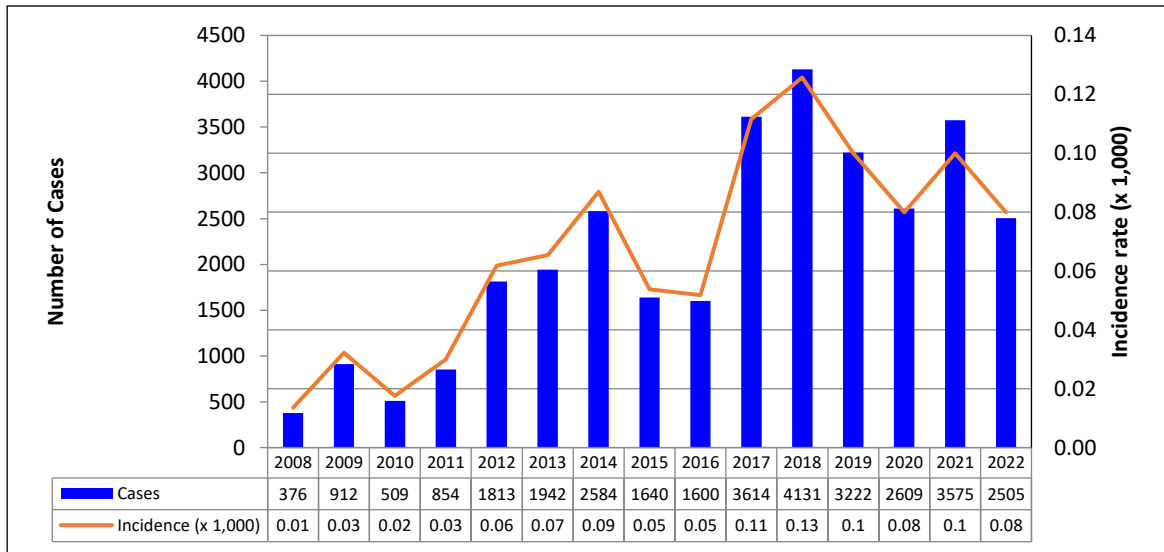
2.6 Distribution according to malaria species

In terms of malaria species classification, *P. knowlesi* has been the dominant species since 2012. In 2022, *P. knowlesi* infection remained the highest with 2,505 (89.2%) cases, followed by *P. vivax* 214 (7.6%) cases, *P. falciparum* 75 (2.7%) cases, *P. ovale* 7 (0.25%) cases, *P. malariae* 5 (0.18%) cases, and mixed infections 1 (0.04%) cases².



2.7 Zoonotic Malaria In Malaysia

In 2022, zoonotic malaria (*Plasmodium knowlesi*) contributed to 89.2% of total malaria cases in Malaysia and the biggest challenge with this new infection is the inadequate effective and mosquito species-specific vector control strategies and control of the macaque population².



References:

1. SPBV (2020). E-2020 Malaria Elimination Report (Malaysia)
2. National Strategic Planning for Prevention of Malaria in Malaysia (NSFPOR) 2021-2026.
3. World Health Organization (WHO), Malaria Terminology. Geneva.

Author: Dr. Mohd Hafizi Abdul Hamid, MD, MPH. Malaria Unit, Vector Borne Disease Sector, Disease Control Division, Ministry of Health, Malaysia



CHAPTER 3

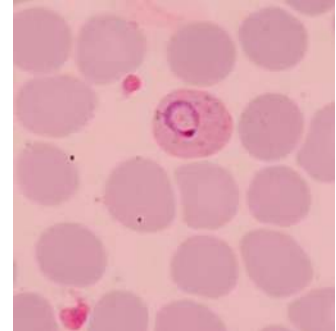
MALARIA PARASITES

3.1 Malaria Parasite

The malaria parasite, **Plasmodium spp.**

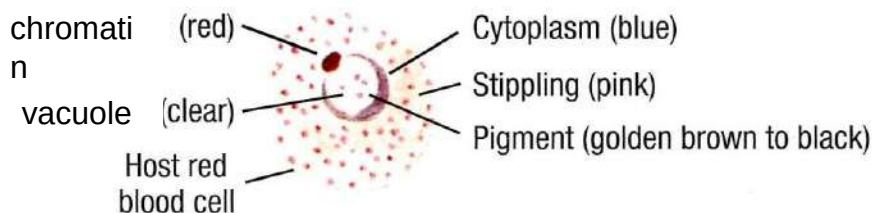
genus

- A blood protozoa, resides in red blood cells (RBCs) and in liver cells
- Phylum Apicomplexa
- Hosts: humans, primates, birds, reptiles, mammals (vertebrate hosts)



When seen under the compound microscope at x100 objective lens:

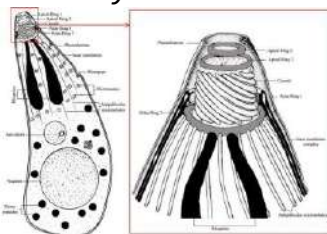
- ✓ Malaria parasite within an RBC
- ✓ Ring form (i.e. trophozoite) consist of
 - Nucleus = chromatin dot
 - Cytoplasm = ring
 - Pigment = remnants of hemoglobin digested by the parasite
 - Stippling = host cell reaction to parasite



Within the same Phylum Apicomplexa, there are other protozoa, consist of:

☐ Coccidia in Intestine and Tissues

Cryptosporidium, *Isospora*, *Cyclospora*
Toxoplasma gondii
Sarcocystis



☐ Blood Protozoa

Plasmodium

Babesia

**Eimeria* spp (in cattle, chicken)

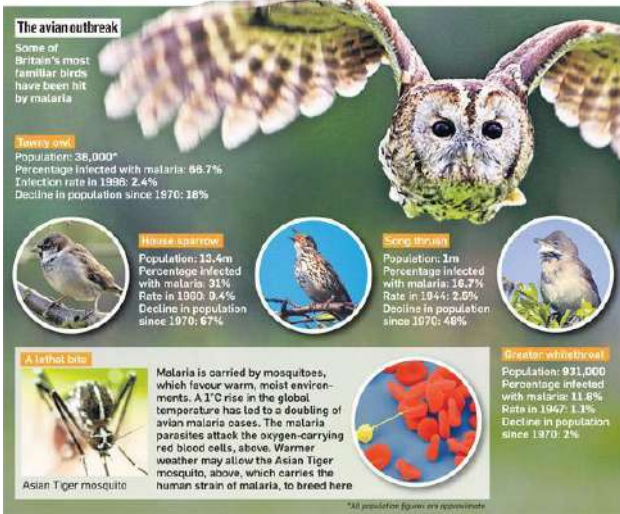
**Lankestrella* (in birds, amphibians)

Invasive/ motile forms of these parasites, in common, possess the **APICAL COMPLEX** consisting of **organelles such as the rhoptries, micronemes, conoid and apical polar rings** visualized through **electron microscopy**

Function of apical complex

- Rhoptries and micronemes are secretory organelles for motility, adhesion to host cells, invasion of host cells and formation of parasitophorous vacuole
- Conoid in invasion of host cells
- Apical polar rings forms MTOC (microtubule organizing center) to support skeletal structure of organism

Malaria Among Animals



P. gallinaceum in chickens



P. berghei in rodents



P. mexicanum in fence lizards in Mexico and western North America



P. agamae in the rainbow lizard (*Agama agama*)



P. aurulentum in neotropical forest gecko (*Thecadactylus rapicaudus*)

Malaria in birds:

P. gallinaceum
P. elongatum
P. relictum
P. cathemerium



Thamnodynastes pallidus, South American snake, host of *P. wenyoni*

Malaria in Monkeys



Macaca fascicularis long-tailed monkeys



Presbytis melalophos banded leaf monkeys



Primates carry:

- *P. cynomolgi*
- *P. inui*
- *P. knowlesi*
- *P. coatneyi*
- *P. fragile*
- Other *Plasmodium* spp.

3.2 Transmission Cycle of Malaria

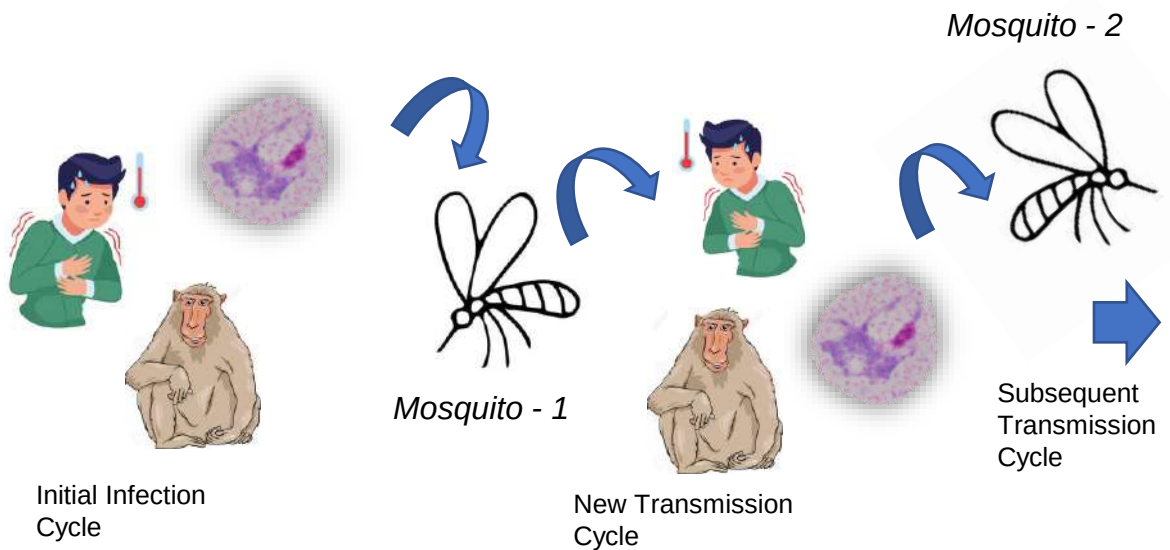
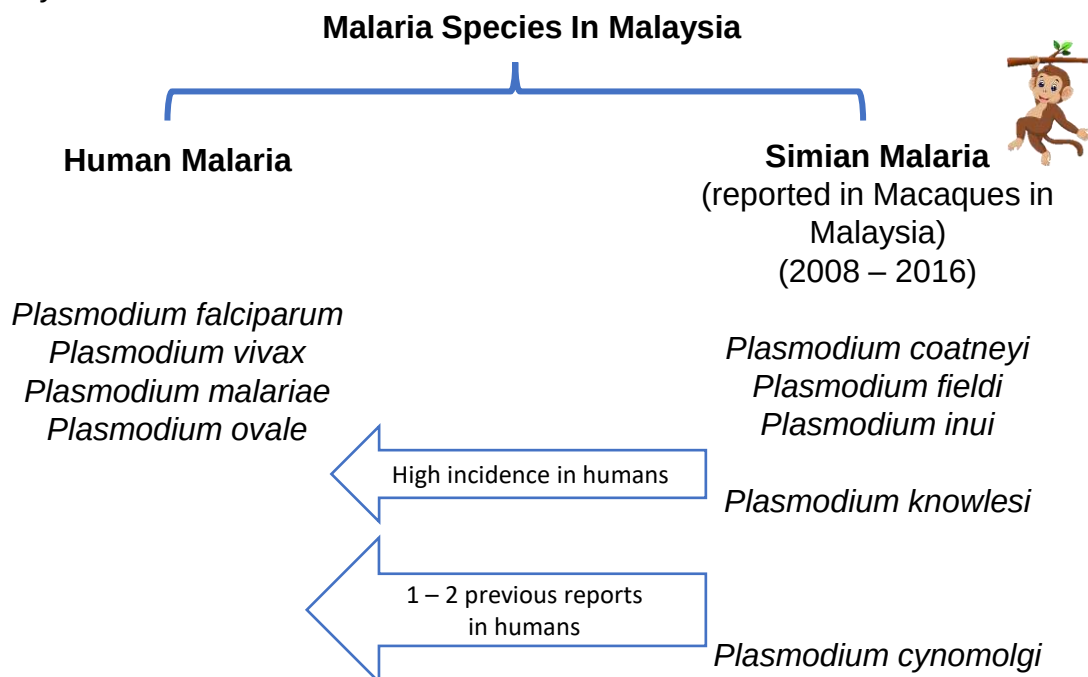


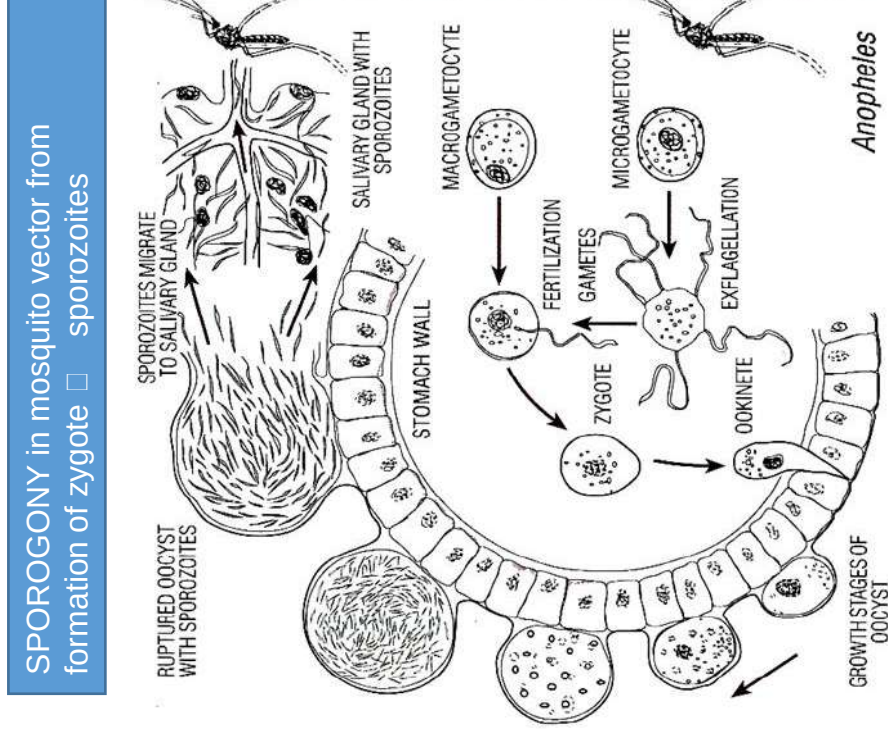
Figure 1.0: The malaria parasites that are present in an infected human / monkey bloodstream introduced into a mosquito vector (Mosquito-1) during a blood meal where the parasites undergo further development into infective forms (sporozoites). In another blood meal, the infective mosquitos inoculate the sporozoites into a new host (human, monkeys) and initiate a new transmission cycle. This cycle is thus repeated in new hosts or get re-infected in the same host again.

3.3 Malaria Species In Man and Primates

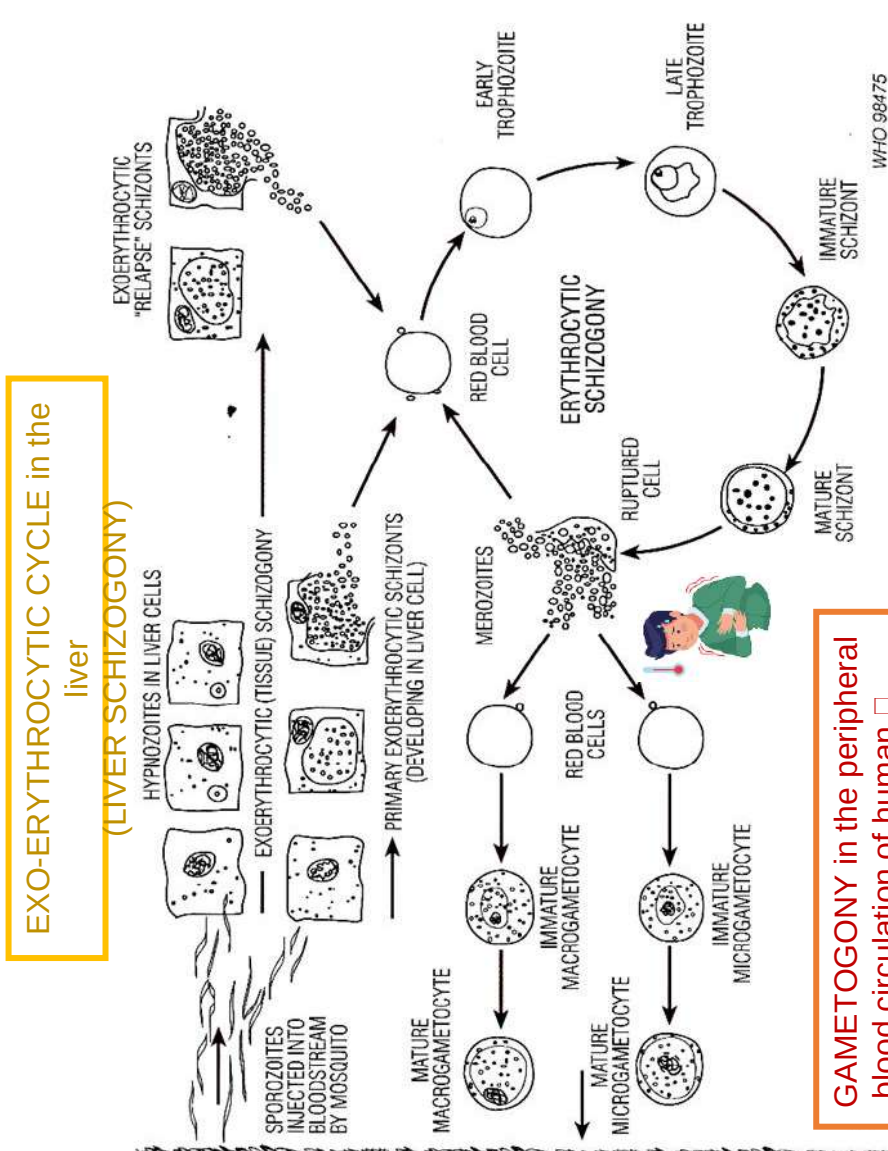
There are 4 malaria species known to infect man and 2 malaria species that originate from monkeys that have been reported to also infect humans in Malaysia.



Understanding The Life Cycle of Malaria Parasites is critical to relate to disease surveillance and monitoring, outbreak investigation, drug actions, new drug discovery, vaccine development, vector control , drafting policies and guidelines to prevent new / introduced / re-emerging malaria



SPOROLOGY in mosquito vector from formation of zygote $\square$ sporozoites



**Ref: WHO Bench Aid, 2000 ,
added content by IMR
Malaysia**

3.4 Relapse and Recrudescence

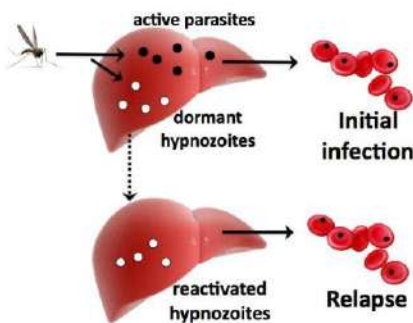
Malaria Case Surveillance and Monitoring

Understanding Significance of *RELAPSE* vs *Recrudescence*

What and when to monitor?? And why??

Note:

- 1⁰ Exo-erythrocytic Schizogony: As soon as sporozoites enter liver cell for first time to produce one and only generation of tissue schizonts. Occurs in all malaria species
- 2⁰ Exo-erythrocytic Schizogony: Dormant hypnozoites reactivated to produce tissue (liver) schizonts. Occurs only in PV and PO causing RELAPSE

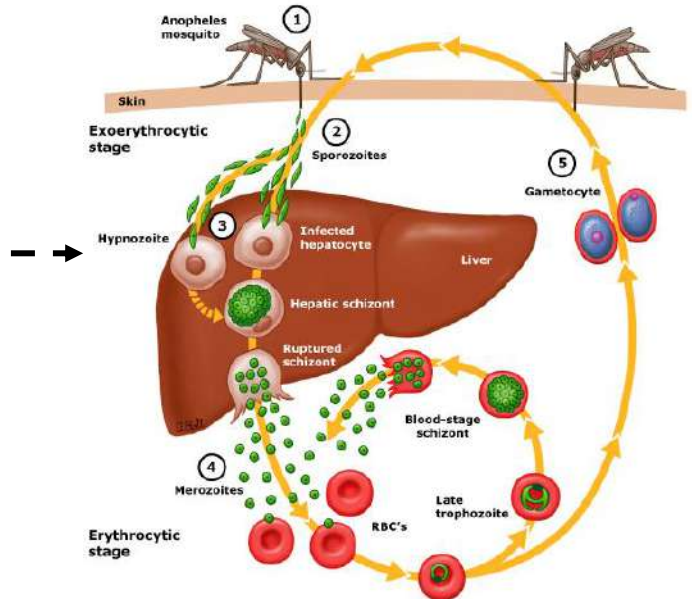


Relapse may appear in 3 – 6 months after patient is “cured”. It may recur for up to 3 years in *P. vivax* and one to 1.5 years for *P. ovale* (however, rare event)

Relapse also will not happen in congenital malaria and blood transfusion malaria because:

- Only erythrocytic stages are passed from infected donor blood into blood recipient,
- Erythrocytic stages of parasites will not re-enter the liver to form hypnozoites and subsequent blood stages

NO RELAPSE IN CONGENITAL / BLOOD TRANSFUSION MALARIA



RECRUDESCENCE vs RELAPSE

Recurrence of malaria infection could be due to:

1) Recrudescence When symptoms return after a symptom free period. Due to parasites surviving in the blood as a result of inadequate or ineffective treatment. All malaria species could exhibit recrudescence. May occur after 1-4 weeks post- symptom free.

In *P. malariae* infection, recrudescence due to low level of blood stage parasites persist for extremely long period in the human blood over 20 – 50 years, even lifetime. In areas of malaria endemicity, often PM mixed infection with PF. PM infection can cause Nephrotic Syndrome (renal disease) due to formation of immune complexes in the kidneys.

2) Relapse When symptoms appear after the parasites have been eliminated from the blood but persist as dormant hypnozoites in the liver cells. Reactivation of hypnozoites releases new merozoite generation in the blood. Occurs only in Pv, Po.

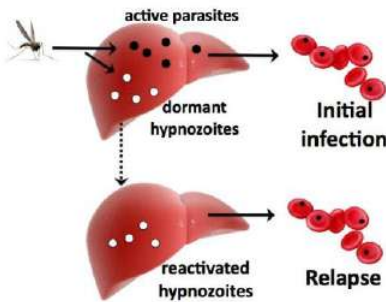
Reinfection A new infection that follows an initial infection. Due to different parasite strain or population that causes the initial infection (but sometimes could be of same parasite genotype)

DIFFICULT TO DIFFERENTIATE RECRUDESCENCE FROM REINFECTION!

Malaria Case Surveillance and Monitoring

Understanding Significance of *RELAPSE* vs *Recrudescence*

What and when to monitor?? And why??

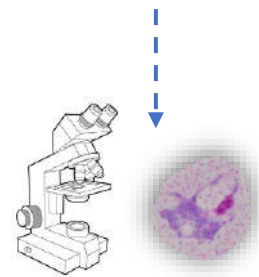
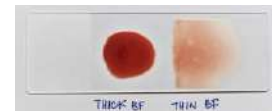


Summary:

- All malaria species exhibit recrudescence
- Relapse event is seen in:
In humans - *Plasmodium vivax*, *Plasmodium malariae*
In monkeys - *Plasmodium cynomolgi*, *Plasmodium fieldi*, *Plasmodium simiovale* and *Plasmodium schwetzi*
- Length of dormancy of hypnozoites varies between strains of malaria species

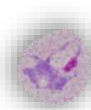
Recrudescence

- Is an event where parasitemia falls below the detectable level (usually by microscopy) and later increases to a patent (visible) parasitemia
- Occurs in all species of malaria parasites
- May predispose to misdiagnosis □ untreated malaria
- Max Duration of Infection (if untreated):
PF 6 – 17 months PV 5 - 8 years
PO 12 - 20 months PM 20 - 50+ years



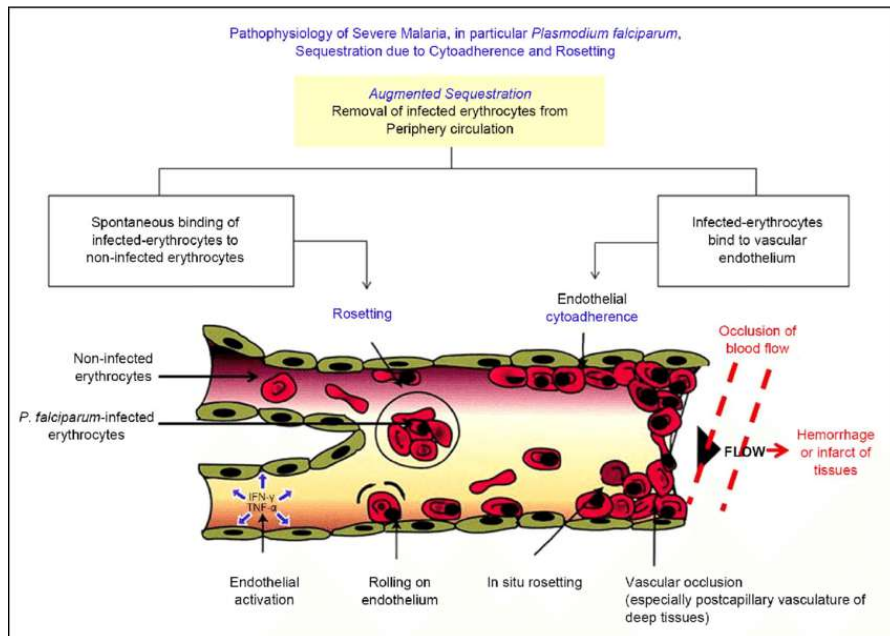
Visible parasitemia in microscopy

When do you expect to see malaria parasites in microscopy?



Erythrocytic Phase	<i>P. falciparum</i>	<i>P. vivax</i>	<i>P. ovale</i>	<i>P. malariae</i>
Pre-patent period: inoculation of sporozoites from mosquito bite to first appearance of parasites in the blood (may absence of clinical symptoms)	6-9 days	8-12 days	10-14 days	15-18 days
Incubation period: Time between inoculation of sporozoites until the onset of malaria symptoms, i.e fever	7-14 days	12-17 days	16-18 days	18-40 days

Why only ring forms (early trophozoites) are frequently seen in microscopy and not mature trophozoites and schizonts in malaria due to *P. falciparum*?



Cytoadherence, Rosetting and Sequestration in *P. falciparum* malaria (also seen in *P. vivax* and *P. knowlesi*)

Reference:

<https://www.malariasite.com/pathophysiology/>

Structural changes occur in the infected red cells resulting in increased rigidity and adhesiveness (sticky). The red cells infected with late stages of *P. falciparum* (during the second half of the 48 hour life cycle) adhere to the capillary and postcapillary venular endothelium in the deep microvasculature (**cytoadherence**).

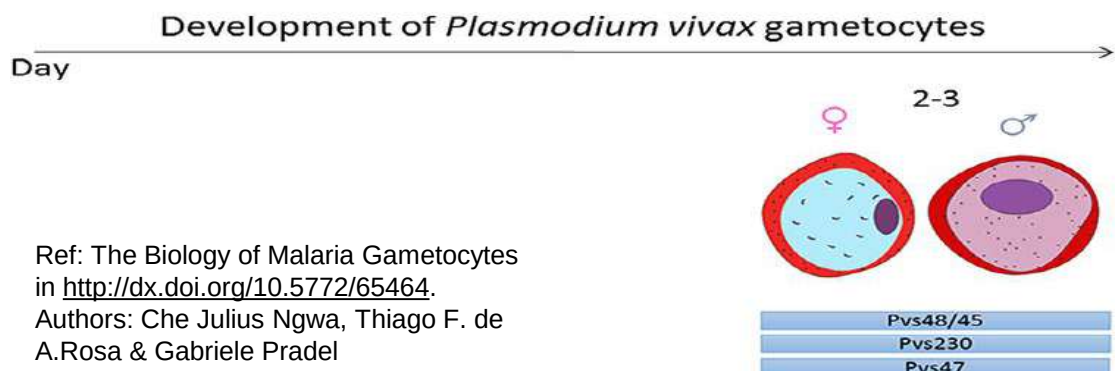
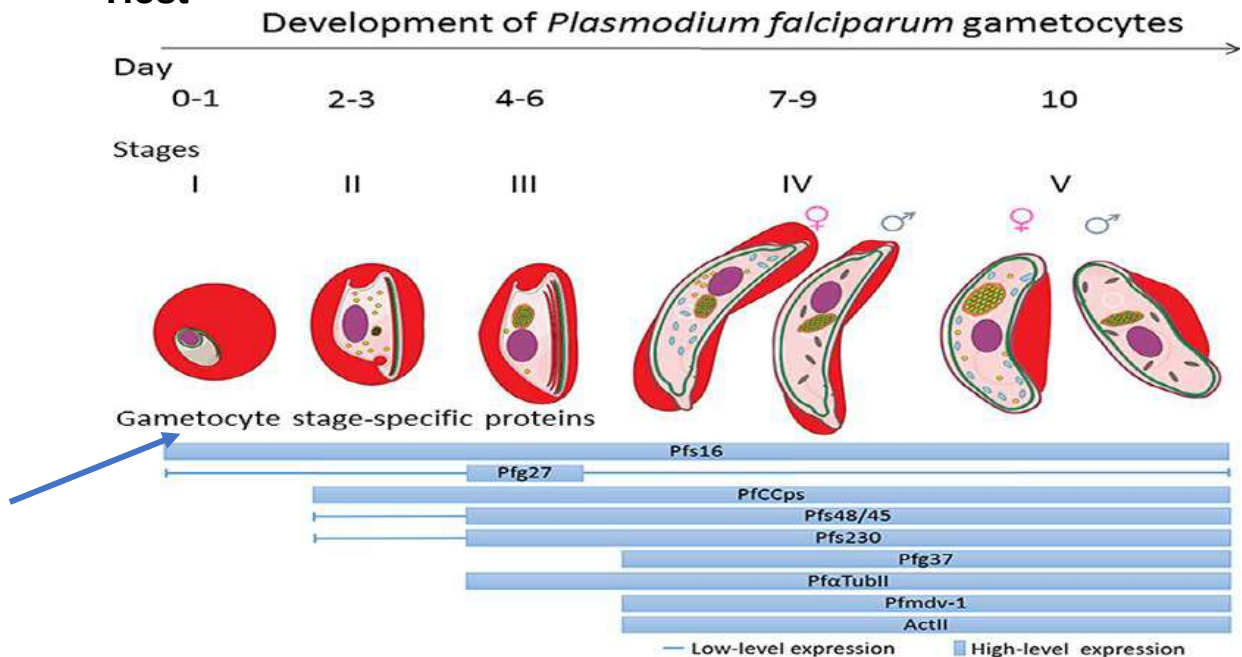
The infected red cells also adhere to the uninfected red cells, resulting in the formation of red cell rosettes (**rosetting**).

Cytoadherence leads to **sequestration** of the parasites in various organs such as the heart, lung, brain, liver, kidney, intestines, adipose tissue, subcutaneous tissues, and placenta.

Effects of sequestration:

- *P. falciparum* parasites continue to grow, multiply and mature within the deeper tissues microvasculature, increasing the number of parasites
- Adhesion of the schizonts to the endothelium of blood vessels allows them to escape clearance by the spleen and to hide from the immune system.
- Therefore more mature trophozoites and schizonts are bound in the deep microvasculature and seldom seen in the peripheral blood circulation
- Only ring forms of trophozoites are commonly seen in *P. falciparum* infection
- If sequestration goes on uninhibited, it ultimately blocks blood flow, limits the local oxygen supply, hampers mitochondrial ATP synthesis, and stimulates cytokine production – all these factors contributing to the development of severe disease.

3.5 Role of Gametocytes in Transmission of Malaria To New Host



Ref: The Biology of Malaria Gametocytes
in <http://dx.doi.org/10.5772/65464>.
Authors: Che Julius Ngwa, Thiago F. de
A.Rosa & Gabriele Pradel

Regarding gametocytes:

- Gametocytes develop in the bone marrow before appearing as mature forms (Stage V) in the blood stream ready for mosquito blood meal
- Crucial for transmission of parasite from human / animal host to the arthropod vector, *Anopheles spp.* causing MALARIA OUTBREAK
- Gametocytes produce specific proteins during its growth that are useful for the survival of the parasites
- Gametocytes do not cause any clinical pathology, unlike the blood stages parasites
- Appearance of gametocytes (Stage IV) in peripheral blood circulation:

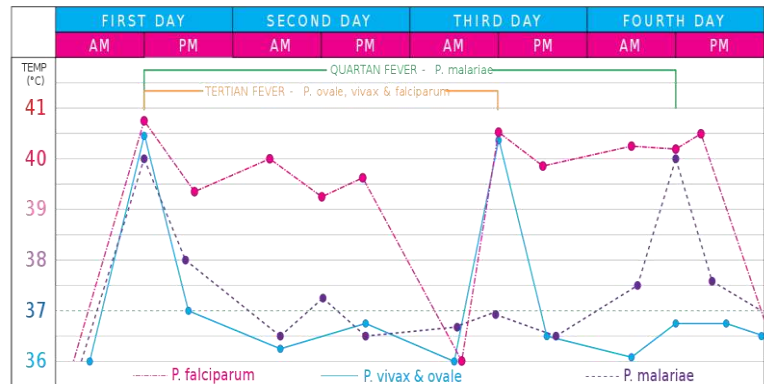
<i>P. falciparum</i>	8 – 11 days
<i>P. vivax</i>	3 – 5 days (earliest to appear)
<i>P. ovale</i>	5 – 6 days
<i>P. malariae</i>	11 – 14 days

Note: Gametocytes are able to survive for at least 3 weeks before their natural death

3.6 Fever in Malaria

Typical Malaria

- Not commonly seen
- Characterised by:
 - **Cold Stage** – shaking, body chills that lasts 15 min – 1 hr
 - **Hot Stage** – high grade fever, dry burning skin, throbbing headache for about 2 -6 hrs
 - **Sweating Stage** – profuse sweating, fever gradually subsides



Source: malaria fever.svg

Typical Malaria Pattern

- Occurs in non-immune and low transmission areas
- Fever paroxysms coincide with rupture of merozoites from infected RBCs
- Typical features are often seen after infection is established over 1 week or so
- Associated with headaches, vomiting, delirium, anxiety, restlessness that disappears when fever subsides
- May be complicated with side effects of antimalarial drugs and secondary infections (pneumonia, urinary tract infection)

Atypical Malaria

- ☐ Often seen in endemic areas
- ☐ Associated with:
 - falciparum malaria
 - early infection
 - patients at extreme age (children / elderly)
 - immunocompromised persons
 - patients on chemoprophylaxis
 - patients who had recurrent attacks of malaria
 - patients with end stage organ failure
 - pregnancy

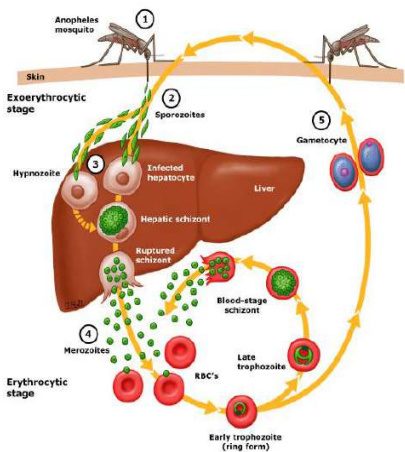
Malaria could be confused with other differential diagnosis:

- All other causes of fever
- Migraine
- Sinusitis
- Pharyngitis, bronchitis, pneumonia, pleurisy
- Acute myocardial infarct, cardiogenic shock, heart failure
- Hepatitis, liver/splenic abscess, acute abdomen, amebiasis
- Acute encephalitis, meningitis, stroke
- Acute confusional disorder / psychosis, mood disorders
- Acute nephritis, nephrotic syndrome, hemoglobinopathies
- Hemolytic anemia, bleeding disorders

FEVER PATTERNS DIFFER BETWEEN MALARIA SPECIES

Note: Fever attacks become apparent only after few cycles of invasion of RBCs (erythrocytic cycles) where parasite density builds up and get synchronised

MALARIA SPECIES	INCUBATION PERIOD	FEVER PERIODICITY
<i>P. falciparum</i>	12 days (9-14 days)	36-48H (malignant tertian)
<i>P. vivax</i>	14 days (8-17 days)	36-48H (benign tertian)
<i>P. ovale</i>	17 days (16-18 days)	36-48H (benign tertian)
<i>P. malariae</i>	28 days (18-40 days)	72H (quartan)
<i>P. knowlesi</i>	9-12 days	24H (quotidian)



Falciparum malaria is termed as **‘malignant’ tertian** malaria because:

- 1) Its preference for all age of RBCs
- 2) Higher numbers of parasites infecting the blood
- 3) Causes severe anemia

Table below shows the types of red blood cells (RBCs) that are invaded by malaria parasites and levels of parasitemia in infected RBCs

Feature	<i>P. falciparum</i>	<i>P. vivax</i>	<i>P. ovale</i>	<i>P. malariae</i>
Preference for RBCs	All age	reticulocyte	reticulocyte	Older RBCs
Average parasitemia / ul blood	20-500,000	20,000	6,000	9,000
Maximum parasitemia / ul blood	2 million	50,000	20,000	30,000
Anemia	++++	++	++	+



CHAPTER 4 : Malaria Vectors In Malaysia

4.1 The Anopheline mosquitos

Anopheles is a genus of mosquito under family Culicidae. Anopheline mosquito is distinguished from other Culicines by its both adult and larvae own characteristic and resting attitude. When resting, adult Anopheline usually form an angle with the surface, not like Culicines and Toxorhynchitines which are mostly humpbacked, resting roughly parallel to the surface. Anopheline larvae, in the other hand lie parallel to and attached to the surface film of the water².

Majority of Anopheline larvae are found in ground water, whether still or flowing, sunlit or shaded. Few species are found to breed in tree holes, bamboos and leaf axils².



4.2 Anophelines as Vectors of Malaria in Malaysia

There are about 460 species of Anophelines has been recognised all around the world, but only 30–40 of it is known as vectors that capable in transmitting Plasmodium, pathogen which cause malaria in humans in endemic areas. To date, 75 species of Anopheles have been recorded in Malaysia, belonging to two main subgenus, Anopheles (anopheles) and Anopheles (cellia). Only nine species have been identified as Malaria vectors¹.

Anopheles maculatus and An. cracens posed as main vectors of Malaria in Peninsular Malaysia. An. maculatus is a vector for human Malaria infection while An. cracens is a vector for zoonotic malaria caused by P. Knowlesi (2014). Meanwhile for Sabah and Sarawak, the main species of the Malaria vector are Anopheles balabacensis and Anopheles letifer respectively. Apart from the aforementioned Anopheles, other species which also identified as Malaria vectors in Malaysia whether human or zoonotic malaria are Anopheles campestris, Anopheles epiroticus, Anopheles sundaicus, Anopheles donaldi, Anopheles flavirostris, and Anopheles latens¹.

4.3 Integrated Vector Control Strategies in Malaysia

Malaysia adapts integrated vector management - a rational decision-making process to optimize the use of resources in Malaria vector control strategy. The aim is to ensure Malaria control to be more effective, efficient, ecologically sound and sustainable. These goals can be achieved by basing decisions increasingly on local evidence, addressing several other vector-borne diseases apart from Malaria using existing systems and local human resources³.

The principle of malaria vector control is to protect the population at risk from Anopheles mosquito bites that could cause malaria infection. This could be achieved through reduction of the vector population or using barriers protection from mosquito bites. Malaysia has implemented indoor residual spray and using of insecticide-treated net as core interventions to control Malaria, whilst other interventions such as larviciding, space spraying, and community engagement has been adapted as complementary interventions³.

References :

1. Spatial distribution of malaria in Peninsular Malaysia from 2000 to 2009. Parasites Vectors 7, 186 (2014). <https://doi.org/10.1186/1756-3305-7-186>, Alias, H., Surin, J., Mahmud, R., Shafie, A., Zin, M.J., Nor, M.M., Ibrahim, A.S., & C. Rundi. 2.
2. Anopheline Mosquitoes of Malaya and Borneo, Reid, J.A.(1968).
3. Handbook for Integrated Vector Management. WHO Geneva (2012).



CHAPTER 5 :

Human and Simian Hosts of Malaria in Malaysia

Important notes

- ✚ Groups at risk of malaria infection in Malaysia are closely related to risky activities carried out, especially forest-related activities.
- ✚ Naturally acquired immunity to malaria is achieved with ongoing exposure to infections and subsequent acquisition of anti-malarial antibodies.
- ✚ The natural hosts of *P. knowlesi* that were initially identified were long-tailed (*Macaca fascicularis*) and pig-tailed (*Macaca nemestrina*) macaques

5.1 Risk Groups and Risk Activities to Acquire Malaria

High Risk Groups

- Indigenous people (Orang Asli, Penan etc);
- Residents living on the outskirts of the forest/ forest area,
- Citizens/non-citizens from endemic countries/endemic areas;
- Individuals involved in activities at risk of malaria infection such as logging, agricultural activities, recreation, security forces and hunters; and
- Close contact individual with confirmed malaria cases

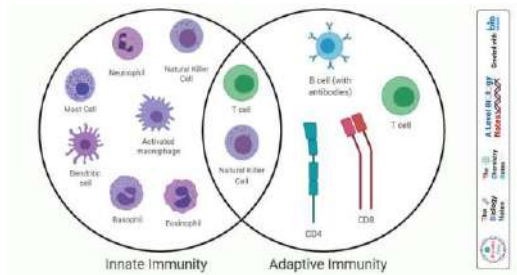
High Risk Activities

- Forest-related activities (Border control, roads & highways building, dam construction, logging, mining, forest ranger patrol, forest related research etc.);
- Hunting;
- Farming (palm oil plantation, rubber estate, fruit orchard etc.);
- Recreational activities (camping, hiking, bird watching etc.); and
- Travelling abroad to malaria endemic countries either for tours or works.

5.2 Innate vs Adaptive Immunity in Malaria

The host immune response to malaria infection is divided into:

1) innate immunity, and 2) adaptive immunity.



INNATE IMMUNITY

- Is naturally present in the host, independent of previous infection
- Inherited disorders of Hemoglobin, red cell polymorphisms and HLA genes confers protection against malaria
- Immediate, non specific immune response is triggered by cells such as neutrophils, monocytes, macrophages, and NK cells
- Respond immediately when triggered by antigens i.e. malaria parasites to provide a first line of defense
- E.g. NK cells in peripheral blood produce IFN- γ in response to infected RBC that activates phagocytosis of parasites by macrophages. The innate immune response then initiate the subsequent adaptive immune response by secreting chemokines and cytokines to recruit and activate additional cells

ADAPTIVE IMMUNITY

- Cells involved in the adaptive response are the B and T cells which produce antibodies and the cytokines.
- Each new T and B cell produced by the host expresses a distinct receptor with different binding properties that allows each cell to recognize a specific antigen (strain) and stages of parasites and amplify population of specific antibodies = Species and stage - specific immune response
- Build up of antibodies takes time, therefore infected human needs to protect against infection by production of maternal antibodies against *P.falciparum* or using effective chemotherapy.
- Repeated malaria infections slowly leads to development of anti-parasite immunity that results in very low or undetectable parasitemia
- Adaptive immunity in malaria does not completely protect the individual, but limit density of parasitemia, thereby reducing overall morbidity and mortality
- ❑ Cytokines produced: Interleukin-12, IFN- γ , IL-10, TGF- β
- ❑ Anti-malarial Antibodies produced: IgM, IgG, IgE that will inhibit merozoite invasion of RBCs and intraerythrocytic growth, enhance clearance of iRBC from the circulation by binding Ab to the iRBC
- ❑ Acquired immunity is not long lasting.
- ❑ It may last for 6 months to 1 year in the absence of re-infection then the individual becomes susceptible to full blown malaria again
- ❑ Also less effective during pregnancy (1 – 2nd trimester) due to physiological immunosuppression and cytoadherence of RBCs to CSA (chondroitin sulfate A) receptors on the placenta that responsible for sequestration in the placental vasculature

Intrinsic Factors (present at birth) could also provide innate immunity:

G6PD Deficiency	Protective against falciparum malaria. Protective mechanism maybe due to impaired parasite growth and enhanced phagocytosis by macrophages. Glucose-6-phosphate dehydrogenase (G6PD) is an enzyme involved in the pentose monophosphate pathway (to make DNA, RNA). Deficiency of this enzyme leads to free radical-mediated oxidative damage to red blood cells, and in turn causes haemolysis.
Absence of Duffy blood group	Duffy antigen present in blood is a receptor for vivax malaria. Absence of Duffy Blood Group found in West African population protects against vivax malaria
Sickle Cell anemia	People with Sickle Cell Disease have shorter life span. However people who are carriers of Sickle Disease (having one sickle gene and one normal Hb gene) confer protection against malaria. In areas with intense malaria transmission, sickle cell trait may confer 60% protection against overall mortality and protection occurs between 2-16 months of life (Study in Western Kenya). Arrested growth of parasites in infected RBCs
Other Structural Defects of Hemoglobin	HbS in sub-Saharan Africa, Middle East and parts of Indian subcontinent; HbC in West and North Africa; HbE in eastern part of Indian subcontinent, SEA; alpha and beta thalassemias
Severe malnutrition and Iron Deficiency	Lack of iron and food not supportive to parasite growth
Function of the spleen	The spleen contains B cells and T cells. The spleen filters abnormal RBCs (deformed / rigid infected RBC) and destroy them within the spleen biomass.
Presence of Pf-maternal antibodies	Protects infants in their first year of life due to passive immunity from mothers in malaria endemic areas
HLA Class 1 B53 and Class II DRB1*1352 alleles	Provide resistance against severe falciparum malaria ie. Anemia, cerebral malaria

5.3 Primates as Natural Host of Zoonotic Malaria

The natural hosts of *P. knowlesi* that were initially identified were long-tailed (*Macaca fascicularis*) and pig-tailed (*Macaca nemestrina*) macaques. The long-tailed macaque is the most abundant and widespread non-human primate in Southeast Asia. It is common to see them in urban areas in most cities, as well as in recreational parks, and suburban and agricultural areas. It has a broad habitat range from coastal mangroves to inland hill forests (home range: ca 1.2 square kilometres; daily movement: ca 1900 square metres). It has the highest occurrence of human–animal conflict in Malaysia and is considered an invasive wildlife species in many countries. *Macaca fascicularis* has at least 10 recognized subspecies and is semi-terrestrial, diurnal and omnivorous. It can live up to 30 years in captivity.



**Macaca
fascicularis**



**Macaca
nemestrina**

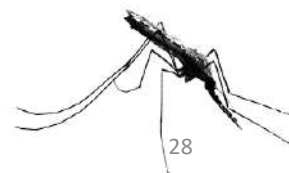
References:

1. National Strategic Planning for Prevention of Malaria in Malaysia (NSFPOR) 2021-2026.
2. World Health Organization. Guidelines for the treatment of malaria Second edition 2010.
3. Expert Consultation On Plasmodium Knowlesi Malaria To Guide Malaria Elimination Strategies, WHO, 2017)

Author:

Dr. Shamilah Hisam, MD, MScPHDC, DAPE. Parasitology Unit, Infectious Diseases Research Center, Institute for Medical Research, NIH Setia Alam, Selangor, Malaysia

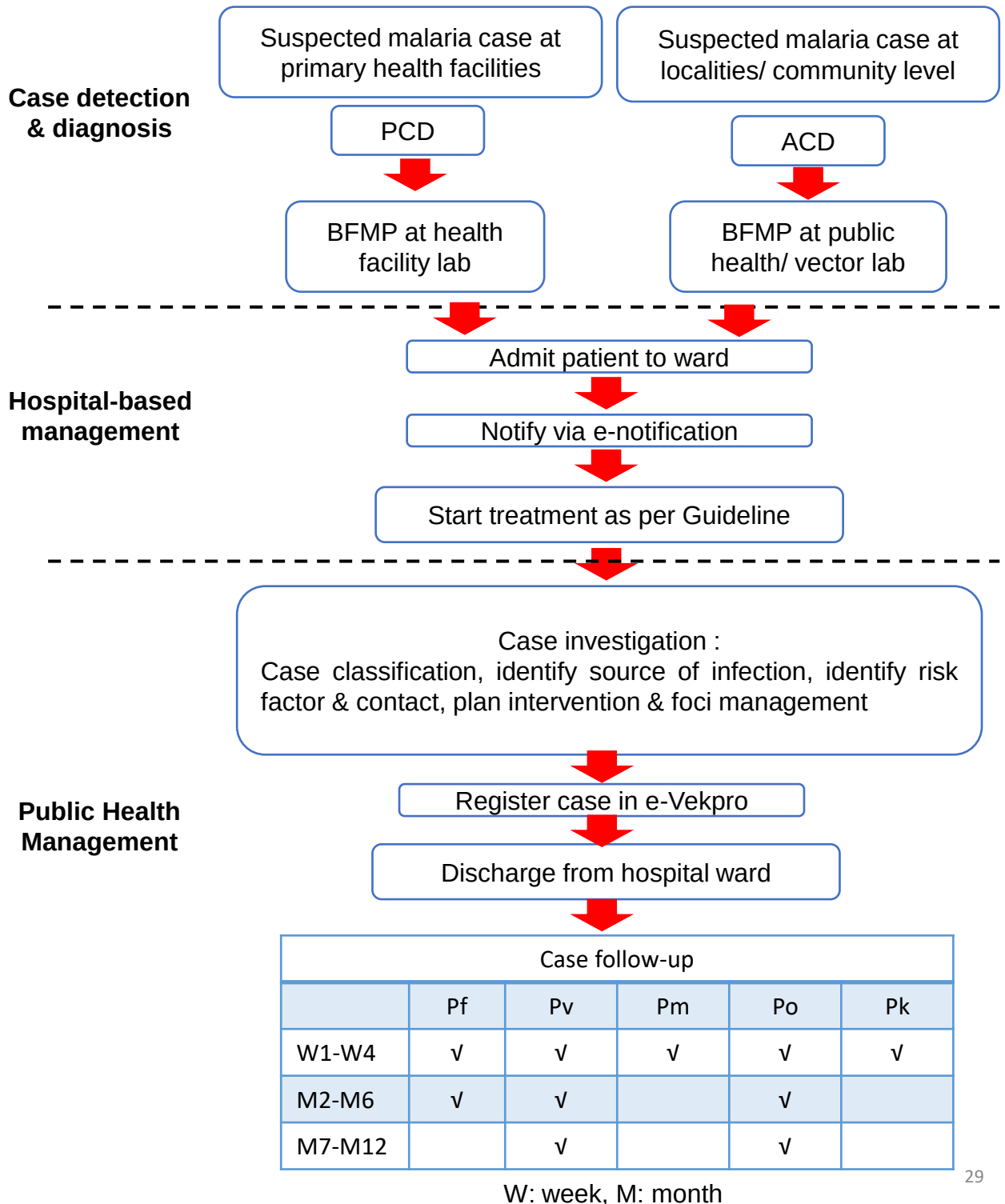
Dr. Mohd Hafizi Abdul Hamid, MD, MPH. Malaria Unit, Vector Borne Disease Sector, Disease Control Division, Ministry of Health, Malaysia



CHAPTER 6 : Management of Malaria In Malaysia

Management of malaria in Malaysia is guided by broad categories of strategies that include enhancing and optimizing case detection and case management, foci stratification by malaria transmission intensity and incorporating routine data collection in a timely and reliable manner.

6.1 Individual Case management



6.2 Malaria Suspected Definition

Malaria has a non-specific signs and symptoms. It is suspected clinically primarily on the basis of fever or history of fever. Besides, its crucial to get a comprehensive history taking to elicit malaria risk to categorise patient as suspected malaria patient.

Category 1

Symptomatic patient with present of fever or history of fever coming from malaria high risk locality (active foci, residual non-active foci, high and medium RV cleared foci).

Category 2

Asymptomatic individual with previous history of travelling or originate from malaria high risk locality or malaria endemic country; and related to malaria high risk activities/occupation. Risk groups in Malaysia consist of:

- i. Orang Asli in Peninsular Malaysia
- ii. Natives/aborigines people in rural Sabah and Sarawak,
- iii. Villagers living within or adjacent to the forests,
- iv. Individuals with history of living or recent travel from and malaria endemic locality/country*
- v. Individuals with malaria-related high-risk forest activities such as logging, plantation/agriculture/recreation, security forces, and forest goers**,
- vi. Malaria close contacts**

Notes :

*Previously reported country origin of malaria imported cases are Myanmar, Indonesia, Thailand, Philippines, African countries (Equatorial Guinea, Nigeria, Liberia, etc), Nepal, India, Pakistan, Bangladesh, Solomon Islands and Papua New Guinea.

**within 6 weeks

6.3 Case Detection:

Passive case detection (PCD) : Detection of malaria cases among patients who, on their own initiative, go to a health post/facilities for treatment, usually for febrile disease.

PCD triaging counter should be made available at emergency department and primary health clinics. Malaria screening should be done to individuals suspected of malaria infection presented at health facilities. Triaging criteria for malaria suspects are as follows:

- People with recent malaria infection;
- People with history of travelling to malaria endemic country.
- Migrant population
- Aborigine population
- People living in:
 - localities with active malaria transmission (active foci);
 - localities with high risk of malaria re-introduction (Receptive & vulnerable)
(refer Guidelines for Prevention of Malaria Re-introduction-2016)
- People working in agriculture sector:
 - palm oil/ rubber plantation worker
 - farmer
- People working in forest related sector:
 - army personnel
 - loggers
 - road/dam construction

Active Case Detection (ACD) : The detection by health workers of malaria infections at community and household level in population groups that are considered to be at high risk. Active case detection can be conducted as fever screening followed by parasitological examination of all febrile patients or as parasitological examination of the target population without prior fever screening.

RACD (Reactive Case Detection : Case detection following a confirmed case):

- Case contacts screening - Fever screening or Mass Blood Screening (MBS)

PACD (Proactive Case Detection) - should be conducted in the following settings:

Settings	Case detection activities
Active Foci	Fever screening & MBS
Residual Non-active Foci	Fever screening
Cleared Foci with high & medium RV	Fever screening
Special Surveillance -High Risk Localities (e.g. Plantations): -High Risk community (e.g. Aborigines): -Point of Entry :	Fever screening or MBS

6.4 Malaria diagnosis methods:

Every case of suspected malaria should be tested to confirm the infection. Microscopic examination (BFMP) of both thick and thin film remains the gold standard for confirmation of malaria in Malaysia.

Microscopy : Blood Film for Malaria Parasite (BFMP) -

- Health Clinics & Hospital Laboratories
- Public Health Laboratories & Private Labs

Polymerase Chain Reaction (PCR)

- Public Health Laboratories, Institute for Medical Research & Private Labs

Rapid Diagnostic Tests (RDT)

- Private Labs

National Reference Laboratory : National Public Health Laboratories (NPHL)

6.5 Mandatory Hospital Admission

Hospitalization of all malaria patients is mandatory for free malaria treatment. Universal access to treatment is provided for both Malaysia and non-Malaysian citizens infected with any forms of malaria. Malaria is one of few diseases that is exempted from treatment fees including non-Malaysian receiving treatment at government health facilities under the Fee Act 2015.

6.6 Notification:

Malaria is a notifiable disease under the Infectious Disease Prevention and Control Act 1988 which mandates notification within 7 days. **However, to ensure early investigation and institution of control measures, medical practitioners are required to notify malaria cases within 24 hours to the nearest health office.**

- Laboratory confirmed (Blood Film for Malaria Parasite -BFMP) cases must be notified.

- How to notify :

Malaria cases can be notified via the online notification system (e-Notifikasi) or using the paper-based or digital notification form.

- Where to notify :

Online notification will be automatically transmitted to central surveillance system. Whereas, paper-based or digital notification form should be submitted to the nearest Health Office, following which will be entered to central surveillance system by health personnel.

6.7 Malaria treatment

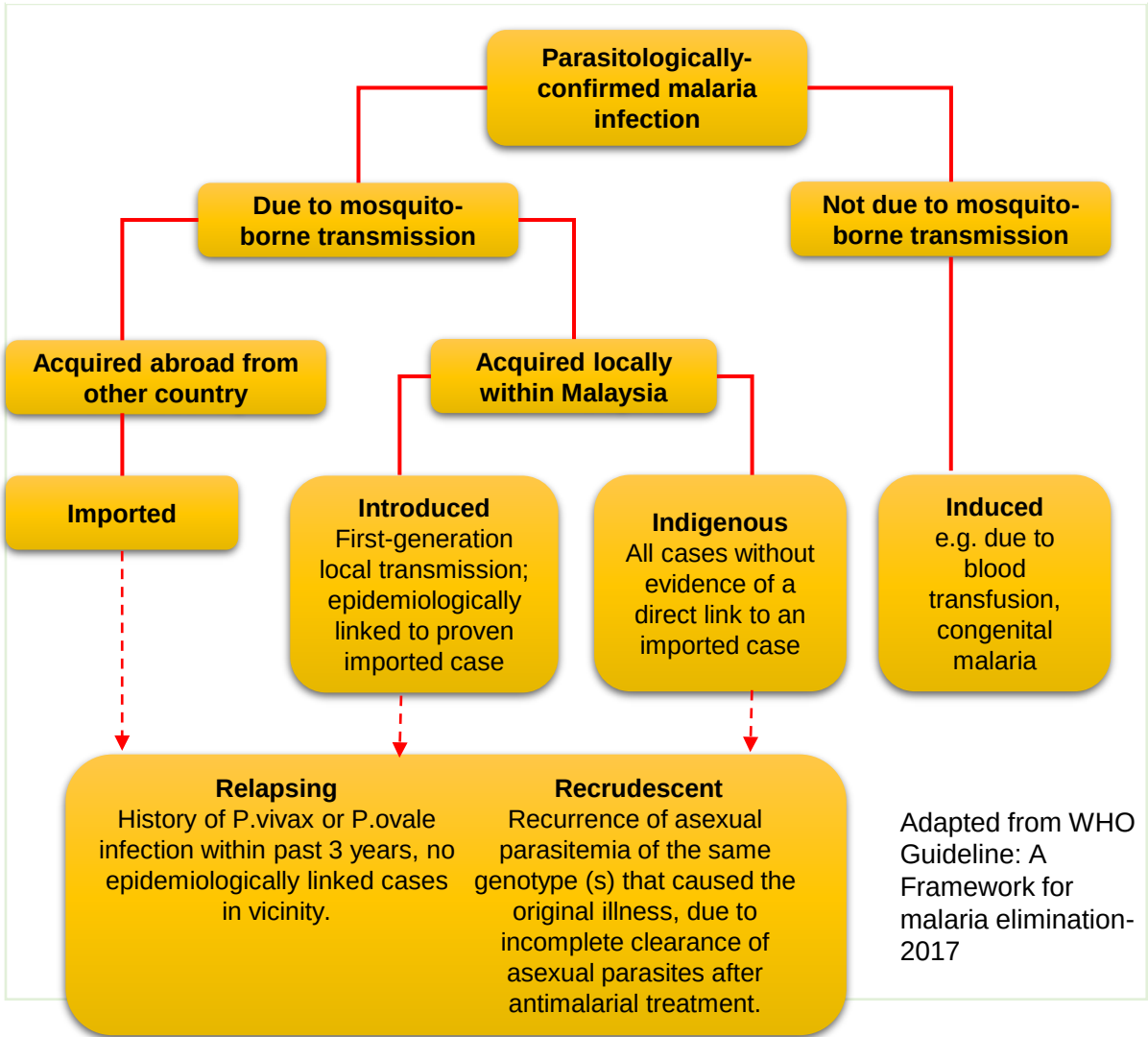
The Malaysian Ministry of Health provides free malaria treatment at all government health clinics free of charge for Malaysian nationals and with a nominal fee for foreign nationals.

Treatment for malaria in Malaysia is based on the Management Guidelines of Malaria in Malaysia published in 2024 which is accessible at Ministry of Health web-page. Hard-copy of the guideline has been produced and distributed to all health facilities in Malaysia.

6.8 Case Investigation

- Initiated within 24 hours after notification and completed within 3 days.
- Investigator: Assistant Environmental Officer (AEHO)/Medical Officer (MO)
- Standard Investigation form: PEM:EPID 9
- Objectives: Case classification and determination of place of infection.
- Case investigation activities include:
 - retesting via microscopy or verification of BFMP result,
 - obtain travel and work history,
 - contact investigation and tracing,
 - epidemiological link analysis,
 - sketching of spot maps, hand drawn or with GIS mapping
 - foci investigation,
 - review of current prevention and control methods,
 - health education,
 - supervision of treatment where applicable.

6.9 Case classification



6.10 Case registration

eVekpro is an online surveillance system for malaria which was developed in 2005 and with revision in 2011.

A confirmed malaria case with positive BFMP result will be investigated and registered into eVekpro. Besides disease registration, eVekpro allows for entry of other surveillance data which includes epidemiological data, drug response surveillance data and vector control data.

6.11 Post discharge follow-up

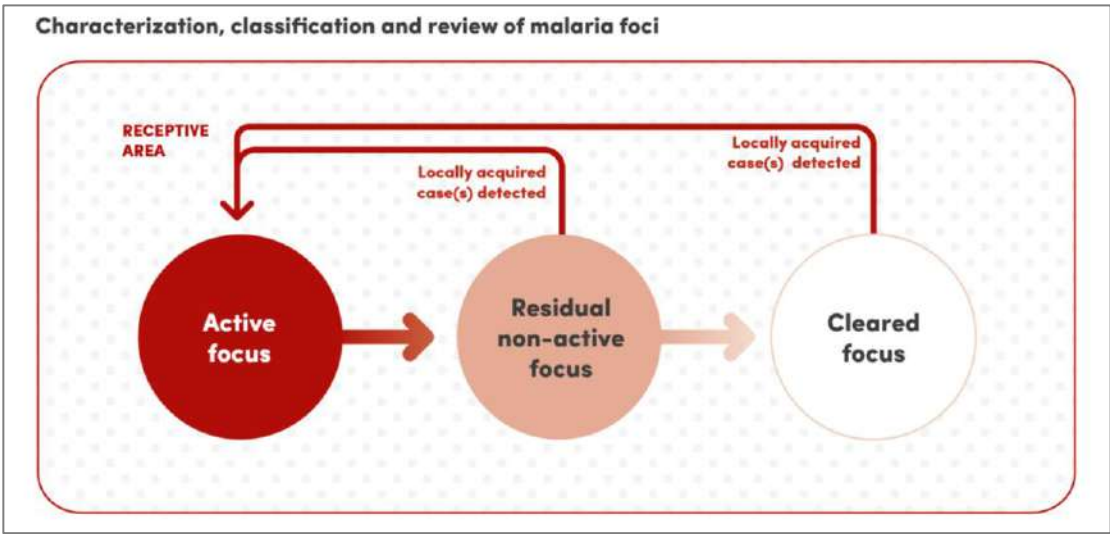
Case follow-up					
	Pf	Pv	Pm	Po	Pk
W1-W4	√	√	√	√	√
M2-M6	√	√		√	
M7-M12		√		√	

W: week, M: month

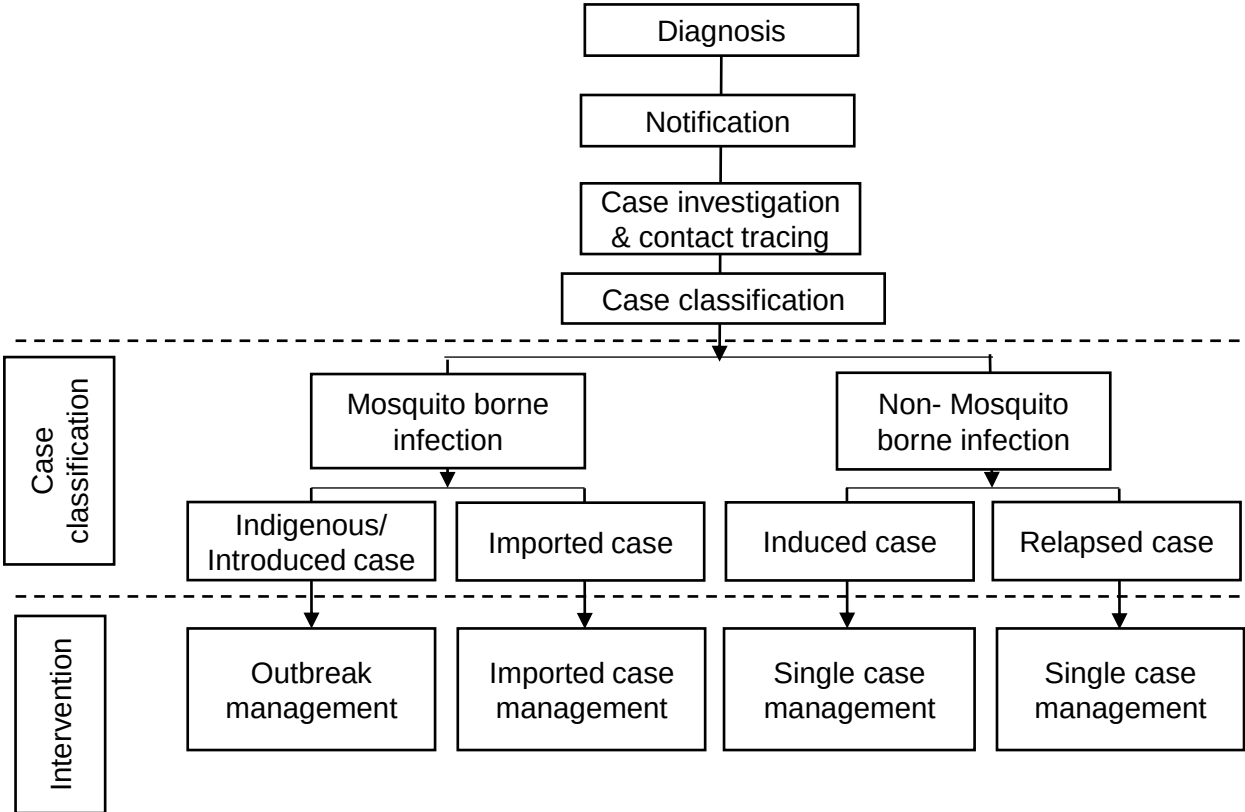
6.12 Focus investigations and response

- Foci investigation is initiated immediately upon confirmation of a locally acquired infection (introduced & indigenous) and foci classification is completed within 7 days.
- Focus investigation will include geography, demography, epidemiology, entomology, receptivity, vulnerability and focus history (previous classification).
- Focus classification is immediately upgraded to ‘active focus’ upon confirmation of a locally acquired malaria.
- Otherwise, malaria focus classification is **reviewed** in the next calendar year.
- Malaysia uses the standard Foci classification recommended by WHO

TYPE OF FOCUS	DEFINITION	OPERATIONAL CRITERIA
Active	A focus with ongoing transmission	Locally acquired case(s) have been detected within the current calendar year. (incl. Introduced)
Residual non-active	Transmission interrupted recently (1-3 years ago)	The last locally acquired case(s) was detected in the previous calendar year or up to 3 years earlier.
Cleared	A focus with no local transmission for more than 3 years	There has been no locally acquired case for more than 3 years, and only imported or/and relapsing or/and recrudescant or/and induced cases may occur during the current calendar year.



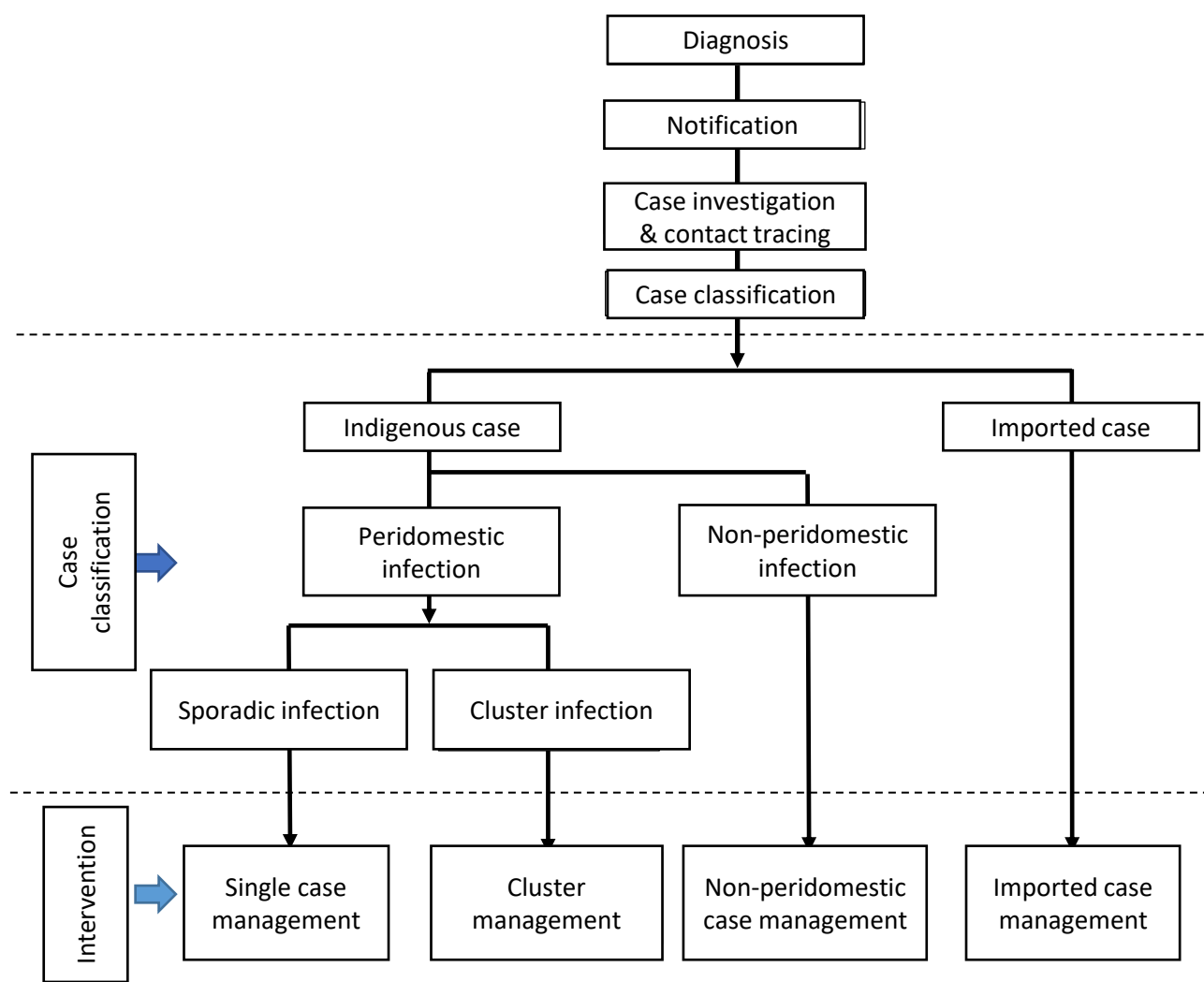
6.13 Human Malaria : Case classification & management



6.14 Intervention matrix for Human Malaria foci management

TYPE OF FOCI	RV INDEX	INTERVENTIONS
Pseudofoci		- Case Management (Imported, Relapse) - PCD
Active Foci		- Epidemiological Investigation - RCD – Reactive Case Detection (contact follow-up 2 years for vivax malaria, 1 year for others) - ACD ± MDA - Outbreak Control - Vector Control(Minimum 6 cycles). - Entomological Investigation (EI) - Bioassay implemented within 2 months after the 1st cycle of IRS - Community Engagement & Mobilization.
Residual Non-active Foci		- Epidemiological Surveillance - Consider ACD - Vector Control (continuation) - ERA after 6 cycles of vector control
Cleared Foci	HIGH	- Epidemiological Surveillance - IVM
	MEDIUM	ERA every 2 cycles of vector control
	LOW	- Case Management (Imported, Relapse) - PCD

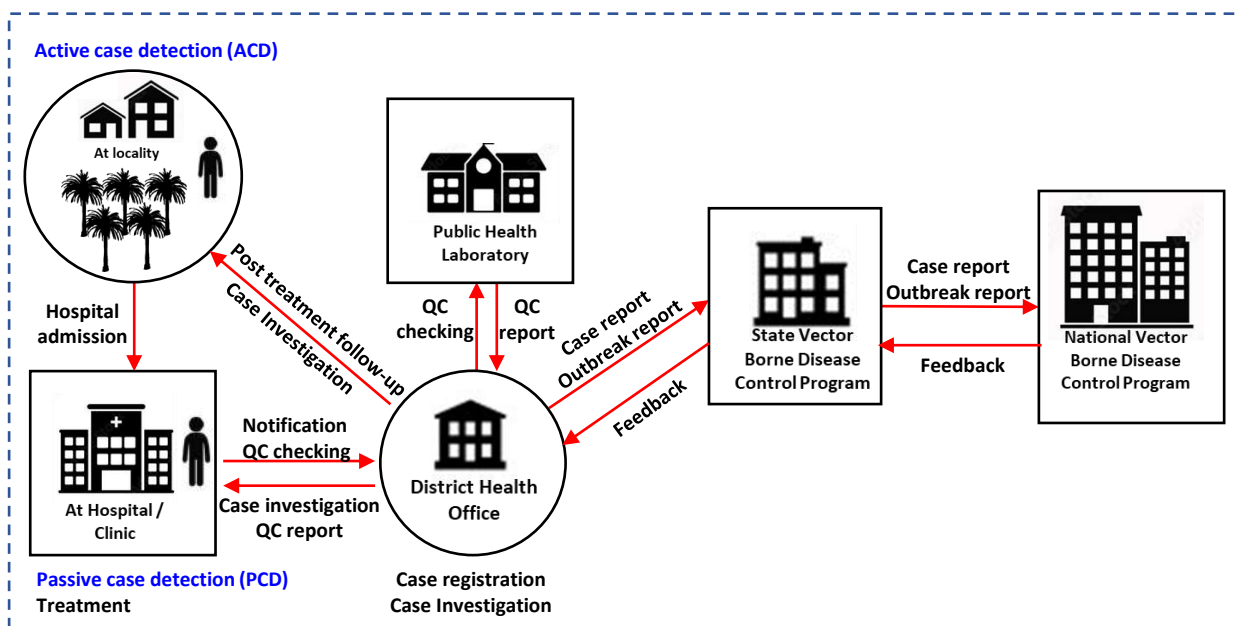
6.15 Zoonotic Malaria : Case classification & management



6.16 Intervention matrix for Zoonotic Malaria management

No	INTERVENTION	INDIGENOUS INFECTION			IMPORTED INFECTION
		Peridomestic Infection		Non Peridomestic infection	
		Sporadic case	Cluster cases		
1	Case investigation	✓	✓	✓	✓
2	Treatment	✓	✓	✓	✓
3	Case follow-up	✓	✓	✓	✓
4	Contact screening	✓	✓	✓	✓
5	ACD	x	✓	x	x
6	ORS	✓*	✓ [#]	x	x
7	LLIN	x	x	x	x
8	Larvaciding	✓	✓	x	x
9	Entomological investigation	✓ [‡]	✓	✓ [‡]	✓ [‡]
10	Bioassay for ORS	x	✓	x	x
11	Health education	✓	✓	✓	✓
12	Macaca population control	x	✓	x	x
13	Inter-agencies collaboration	✓	✓	✓	✓

6.17 Malaria Case Management Data Flow



References:

1. National Strategic Planning for Prevention of Malaria in Malaysia (NSFPOR) 2021-2026.
2. Management of Zoonotic Malaria in Malaysia, 2023.
3. Expert Consultation On Plasmodium Knowlesi Malaria To Guide Malaria Elimination Strategies, WHO, 2017)



CHAPTER 7 : Clinical Features of Malaria

Important notes

- *The incubation period of malaria is variable (refer chapter 2.2); the average being 10–14 days, but may be as short as 7 days or, in exceptional cases, up to 20 years as in P. malariae infection.*
- *Symptoms occur within 6 weeks of the traveler leaving an endemic area in more than 90% of P. falciparum infections, and within 1 year in P. vivax infection.*
- *For P. knowlesi, symptoms occur 9-12 days after a person has visited or worked in a forested or forest-tinge area. There may be a relatively short prodromal period of tiredness and aches.*

7.1 Symptoms of malaria

The signs and symptoms of malaria are non-specific. Malaria is suspected clinically primarily on the basis of fever or a history of fever. In malaria-endemic areas, malaria should be suspected in any patient presenting with a history of fever or temperature $\geq 37.5^{\circ}\text{C}$ and no other obvious cause. There is no combination of signs or symptoms that reliably distinguishes malaria from other causes of fever; diagnosis based only on clinical features has very low specificity and results in overtreatment. Other possible causes of fever and whether alternative or additional treatment is required must always be carefully considered

The early symptoms of malaria are non-specific and similar to the symptoms of a minor systemic viral illness e.g. headache, lassitude, fatigue, abdominal discomfort, muscle and joint aches, usually followed by fever, chills, perspiration, anorexia, vomiting and worsening malaise. The most common presentation of malaria is high fever. The classical paroxysm begins abruptly with an initial 'cold stage', with dramatic rigors during which the patient shakes visibly. This leads to a 'hot stage' in which the patient has a temperature of more than 40°C , may be restless and excitable and may vomit or convulse, and finally a 'sweating stage', when the fever abates and the patient may fall asleep. This paroxysm may last 6–10 hours, and is followed by a prolonged asymptomatic period followed by further rigors in untreated patients. The classic paroxysm is, however, increasingly uncommon in clinical practice.

At early stage of malaria infection, without evidence of vital organ dysfunction, if prompt and effective treatment is given, the patient can experience full rapid recovery. If ineffective medicines are given or if treatment is delayed, especially in *P. knowlesi* and *P. falciparum* malaria, the parasite burden continues to increase every 24 or 48 hours, severe malaria and even death may ensue. This progression may occur within a few hours.

In high-transmission areas, such as in many sub-Saharan Africa, the majority of infections are asymptomatic and infections are acquired repeatedly throughout life. In such areas, symptomatic and sometimes fatal malaria occurs in the first few years of life, but, thereafter, it is increasingly likely to be asymptomatic. This reflects a state of imperfect immunity, where the infection is controlled, usually at levels below those causing symptoms. The rate at which imperfect immunity is acquired depends on the intensity of transmission.¹

The nature of malarial clinical disease depends greatly on the background level of the acquired protective immunity and intensity of malaria transmission in the area of residence. Non-immune travelers to a malaria endemic area are at a higher risk of acquiring malaria, unless protective measures are taken. Immunity is, however, modified in pregnancy, and it is often gradually lost, at least partially, when individuals move out of the endemic areas for long durations (usually many years).

7.2 Definition of Uncomplicated Malaria

A patient who presents with symptoms of malaria and a positive parasitological test (microscopy or RDT) but with no features of severe malaria is defined as having uncomplicated malaria.

7.3 Definition of Severe/ Complicated Malaria

Severe malaria usually manifests with one or more of the following: coma (cerebral malaria), metabolic acidosis, severe anaemia, hypoglycaemia, acute renal failure or acute pulmonary oedema. By this stage of the disease, the case fatality in people receiving treatment is typically 10–20%. However, if left untreated, severe malaria is fatal in the majority of cases. Long-term cognitive impairment is also common in children with cerebral malaria.

In a patient with malaria infection and no other obvious cause of symptoms, the presence of one or more of the following clinical or laboratory findings classifies the patient as suffering from severe malaria.

Table 7.1: Clinical Features, Laboratory Findings and Complications of Severe & Complicated Malaria ¹

Clinical features	● Impaired consciousness or unrousable coma (Glasgow coma score < 11 in adults or a Blantyre coma score < 3 in children) ● Prostration (generalized weakness so that the patient is unable to sit, stand or walk without assistance) ● Multiple convulsion (> 2 episodes within 24 hours) ● Respiratory distress: rapid, deep, laboured breathing (could be manifestation of severe acidosis). ● Circulatory collapse or shock- either compensated / decompensated. In compensated shock, patient may present with prolonged capillary refill ≥ 3s with no hypotension. In decompensated shock, systolic blood pressure < 80 mm Hg (adults) and < 70 mm Hg in children with cool peripheries and prolonged capillary refill. ● Significant bleeding – including recurrent or prolonged bleeding from mucosal area – nose, gums, or venupuncture sites. Hematemesis or malena. ● Pulmonary oedema (radiological or clinical with lungs crepitations and tachypnoeic)
Laboratory findings	● Hypoglycaemia (blood or plasma glucose < 2.2 mmol/l or 40mg/dL) ● Metabolic acidosis (plasma bicarbonate < 15 mmol/l or base deficit of > 8 mEq/L or venous plasma lactate ≥ 5 mmol/l) ● Severe normocytic anaemia (Hb < 7 g/dl, packed cell volume < 20% for adult, Hb ≤ 5 g/ dL or a haematocrit of $\leq 15\%$ in children < 12 years of age) with parasite count > 10 000/ μL ● Hyperparasitaemia (P falciparum – parasitemia > 10%, *P. knowlesi – parasite density > 15,000 / μl alone₇, P vivax – no established threshold) ● Renal impairment - Plasma or serum creatinine > 265 μmol/L (3 mg/dL) or blood urea > 20 mmol/L ● Jaundice: Plasma or serum bilirubin > 50 μmol/L (3 mg/dL) with parasite count > 100 000/ μL)
Complications	● Cerebral malaria (Coma) ● Hyperpyrexia ● Convulsions ● Severe anaemia ● Respiratory distress / Acute Respiratory Distress Syndrome (ARDS) ● Acute renal failure ● Hypoglycaemia ● Circulatory collapse (shock) ● Spontaneous bleeding and coagulopathy

7.4 Definition of severe *P knowlesi* Malaria ²

Laboratory findings – Parasite count	• <i>P. knowlesi</i> hyperparasitaemia – parasite count > 100,000/μL or > 2% of infected red blood cells; OR • Parasitaemia > 15,000 / μl in presence of jaundice (Plasma or serum bilirubin > 50 μmol/L (3 mg/dL)
OR	
Clinical manifestations (one or more of the criteria in the absence of alternative cause)	• Acute kidney injury - Plasma or serum creatinine > 265 μmol/L (3 mg/dL) or blood urea > 20 mmol/L • Respiratory distress: oxygen saturation < 92% with respiratory rate > 30 breaths /minute) • Circulatory collapse or shock- either compensated / decompensated. In compensated shock, patient may present with prolonged capillary refill ≥ 3s with no hypotension. In decompensated shock, systolic blood pressure < 80 mm Hg (adults) and < 70 mm Hg in children with cool peripheries and prolonged capillary refill. • Severe normocytic anaemia (<i>Hb</i> < 7 g/dl, <i>packed cell volume</i> < 20% for adult) • Significant bleeding – including recurrent or prolonged bleeding from mucosal area – nose, gums, or venupuncture sites. Hematemesis or malena. Metabolic acidosis (<i>plasma bicarbonate</i> < 15 mmol/l or <i>venous plasma lactate</i> ≥ 5 mmol/L ; or <i>base deficit</i> of > 8mEq/L)

References:

1. WHO Guidelines for Malaria 16 October 2023 (WHO/UCN/GMP/2023.01 Rev.1)
2. Grigg, M. J., William, T., Barber, B. E., Rajahram, G. S., Menon, J., Schimann, E., Piera K., Wilkes C. S., Patel, K., Chandna. A., Drakeley, C. J., Yeo, T. W. & Anstey, N. M. (2018). Age-related clinical spectrum of Plasmodium knowlesi malaria and predictors of severity. Clinical infectious diseases, 67(3), 350-359.

Author: Dr Lee Heng Gee, Infectious Disease Consultant, Ministry of Health, Malaysia



7.5 Chronic Malaria

Chronic Malaria and its Most Severe Complication, Hyper-Reactive Malarial Splenomegaly (HMS) Syn. Tropical Splenomegaly Syndrome (TSS)

Chronic malaria is usually defined as a long-term malarial infection in semi-immune subjects, usually without fever or other acute symptoms. The untreated infection may evolve to hyper-reactive malarial splenomegaly (HMS), a life-threatening complication.

The following case definition for HMS was used:

- antimalarial antibody titre (IFAT-Biomérieux) $\geq 1/160$, PLUS:
- massive splenomegaly (ultrasound longitudinal diameter, corresponding to spleen length, ≥ 16 cm) (normal value in healthy adults 8–11 cm);
- IgM level ≥ 2 SD above the local mean (or ≥ 2.5 g/L according to the local laboratory);
- no other identifiable cause of splenomegaly or of raised IgM, such as schistosomiasis, cirrhosis, haematological conditions, and others as described previously.



Regarding HMS,

- Seen in older children and adults in malaria hyperendemic areas
- Repeated malaria infections induce exaggerated immune response
- Characterized by anemia, massive splenomegaly and elevated IgM levels
- Malaria parasites are not seen or scanty in peripheral blood circulation i.e. negative BFMP. Therefore speculation that HMS is an immune-mediated mechanism, and not to actual presence and response to the parasites
- Some studies claimed that HMS can be detected using QBC (quantitative buffy coat); RDTs or PCR in BFMP-negative samples
- Response well to prolonged treatment with antimalarial drugs.
In endemic countries, treatment could be given over one month to lifelong while in non-endemic countries treatment could be over 6 weeks to 36 months
- Recovery post treatment could be partial or complete, with reduced spleen size
- Severe complications of HMS: acute infectious diseases, hemolytic anemia especially during pregnancy, splenic rupture
- Fatal if untreated



CHAPTER 8 : Diagnosis Of Malaria

Important notes

- ✚ All case of suspected malaria should have a parasitological test to confirm diagnosis and supported with quality assurance program.
- ✚ Microscopic examination of both thick and thin film remains the gold standard for confirmation of malaria in Malaysia.

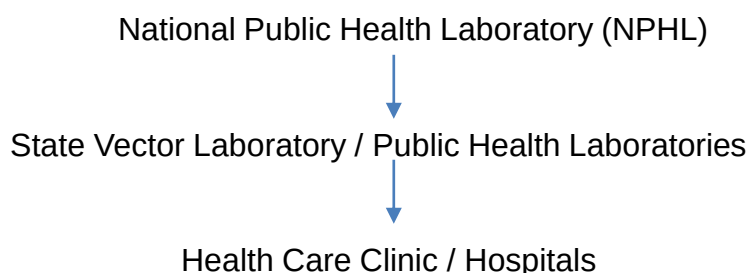
8.1 Malaria Diagnosis System

Prompt and accurate diagnosis of malaria is part of effective disease management. The diagnosis of malaria is basically based on clinical suspicion and on the detection of parasites in the blood (parasitological or confirmatory diagnosis). Several tests are available for laboratory diagnosis of malaria in Malaysia such as Blood Film for Malaria Parasite (BFMP), Polymerase Chain Reaction (PCR) and Rapid Diagnostic Test (RDT).

Table 8.1: Laboratory Networking in Malaysia

Responsible bodies	Roles and Functions
Field sampling/Health Care Clinic /Hospitals	• Performing the screening of blood film malaria parasite • Sending 100% of microscopy-positive slides and 10% of negative slides to State Vector Laboratory.
State Vector Laboratory/ Public Health Laboratories	• Performing rechecking of 100% of positive slides and 10% of negative slides for quality control received. • Sending all the slides to National Public Health Laboratories (NPHL) for quality checking. • All public health laboratories (PHLs) shall provide the PCR services as requested.
National Public Health Laboratory (NPHL)	• Rechecking 100% from positive slides and 2% from negative slides, which are sent from State Vector Laboratory/other Public Health Laboratories for quality control. • Slides with discrepancy or “errors” identified by NPHL will be proceed with PCR. Results shall be reported immediately using the “errors/discrepancy” notification form. • EQA Proficiency Testing (PT) provider for Blood film malaria parasites (BFMP)

8.1.1 Laboratory network in Malaysia



8.1.2 National reference laboratory

National Public Health Laboratory (NPHL) Sungai Buloh is recognized as the National Malaria Microscopy Reference Laboratory. NPHL is responsible for issues related to Malaria Microscopy in Malaysia and to ensure that the Malaria Microscopy Quality Assurance Programme at MOH is implemented in accordance with WHO Malaria Microscopy Quality Assurance Manual, Version 1 (2009) and Version 2 (2016).

Table 8.2: Roles of National Reference Laboratory

Reference Lab	Roles
National Public Health Laboratory (NPHL)	• Reference Laboratory for Quality Checking of Blood Film Malaria Parasite • Proficiency Testing (EQA PT) Provider for Malaria Microscopy • Training Center for Malaria Microscopy • National External Competency Assessment for Malaria Microscopy. • Collaboration with WHO-WPRO for External Competency Assessment of Malaria Microscopit (WHO-ECA) • Collaboration with WHO UK NEQAS for External Quality Assessment for Malaria Nucleic Acid Amplification Techniques (EQA NAAT). • Collaboration with WHO/WPRO-RITM, Philippines for External Quality Assessment (EQA) Malaria Microscopy. • Coordinator for Technical Working Group (Parasitology) for Public Health Laboratories and State Vector Laboratory specifically related to National Malaria Elimination Strategic Plan.

8.2 Malaria Suspected Definition

Malaria has a non-specific signs and symptoms. It is suspected clinically primarily on the basis of fever or history of fever. Besides, its crucial to get a comprehensive history taking to elicit malaria risk to categorise patient as suspected malaria patient.

7.2.1 Category 1

Symptomatic patient with present of fever or history of fever coming from malaria high risk locality (active foci, residual non-active foci, high and medium RV cleared foci).

7.2.2 Category 2

Asymptomatic individual with previous history of travelling or originate from malaria high risk locality or malaria endemic country; and related to malaria high risk activities/occupation. Risk groups in Malaysia consist of:

- i. Foreign workers (commonly lives in receptive areas)
- ii. Malaysians who work in malaria endemic countries
- iii. Aborigines
- iv. Sabah and Sarawak rural residents
- v. Residents of traditional village (forest fringe)
- vi. Activities related risk groups (forest related sector).

8.3 Clinical Diagnosis

The particular clinical diagnostic criteria vary from area to area according to the intensity of transmission, the species of malaria parasite and other prevailing causes of fever³. Malaria must be excluded in all febrile patients ($\geq 37.5^{\circ}\text{C}$) or patient presenting with history of fever and fulfilled with one or more criteria as listed below:

1. living in, or returning from, an endemic locality/country, regardless of whether they have been taking antimalarial drugs;
2. aborigines (Orang Asal or Penan);
3. people from rural locality/forest fringe traditional village or high RV locality;
4. immigrants from malaria endemic countries;
5. malaria risk related activities in jungle (hunting, camping, exploration convoy etc.);
6. malaria risk related occupation (farmers, estate worker, soldier, land surveyor etc.).

In all settings, clinical suspicion of malaria should be confirmed with a parasitological diagnosis. However, in settings where parasitological diagnosis is not possible, the decision to provide antimalarial treatment must be based on the prior probability of the illness being malaria. Malaria should be suspected in the presence of thrombocytopenia, relative lymphopenia, atypical lymphocytes, and an elevated *lactate dehydrogenase* (LDH) level. (Table 5.3) Other possible causes of fever and need for alternative treatment must always be carefully considered.

Table 8.3: Investigating suspected malaria⁸

Investigations	Notes
Blood Film for Malaria Parasite (BFMP)	To be done immediately. There is no need to wait for a peak of fever before examining thick and thin blood films. If the initial film is negative, repeat another 2 samples or more especially at peak of fever if symptoms persist.
Full Blood Count (FBC)	In falciparum malaria, the WBC count is generally normal, with relative lymphopenia. Neutrophilia suggests secondary bacterial infection or severe disease. Thrombocytopenia is present in more than 90% of non-immune patients.
Blood Urea & Serum Electrolytes (BUSE)	Sodium, calcium and albumin are low; these usually resolve spontaneously with treatment.
CRP level	C-reactive protein (CRP) is raised
Liver profile	Bilirubin is often raised; in the context of normal liver function, this is related to haemolysis. Although malaria may cause mild elevations of liver enzymes, jaundice in the presence of liver enzyme abnormalities may suggest that other diagnoses (e.g. viral hepatitis) should be considered.
Blood glucose	Blood glucose can be low in severe cases (with high parasitaemia) and during quinine treatment in adults.

8.4 Laboratory Diagnosis

Prompt and accurate diagnosis of malaria is part of effective disease management. The diagnosis of malaria is based on clinical suspicion and on the detection of parasites in the blood (parasitological or confirmatory diagnosis). Once Plasmodium is detected, the BFMP test should be done daily during hospital stay, weekly tests x 4 during follow-up for all species and additional monthly tests x 11 during follow-up for *P.vivax*.

8.4.1 Tests Available For Laboratory Diagnosis

Several tests are available for laboratory diagnosis of malaria such as Blood Film for Malaria Parasite (BFMP), Polymerase Chain Reaction (PCR) and Rapid Diagnostic Test (RDT) and these will be further elaborated here.

a. Blood Film for Malaria Parasite (BFMP)

Microscopic examination of both thick and thin film remains the gold standard for confirmation of malaria. The thick film is more sensitive for detecting malaria parasites as the blood is more concentrated which allow for a greater volume of blood to be examined. On the other hand, the thin film helps in parasite species identification. (Table 5.3)

Table 8.4: Advantages and limitations of microscopy

Advantages	Limitation
• High sensitivity. It is possible to detect malarial parasites at low densities. • Can quantify the parasite load. • Possible to distinguish the various species of malaria parasite and their different stages.	• Need adequately trained laboratory technician.

i. Specimen

For every suspected malaria patient, specimen collection includes BFMP slides and blood in EDTA tube.

The thick and thin film should be prepared in one slide and sent in a slide box to the laboratory. Stained the thick (unfixed) and thin film (fixed with methanol) with 10% Giemsa stain in buffered solution with pH7.2 for 10 minutes. If BFMP slides are of poor quality, lab technologist should have made the films using the EDTA blood within 2 hours or as soon as possible to minimize morphological changes in the parasites.

Blood taken in EDTA tubes, must be kept in freezer for future use as it may need to be used for further test such as PCR or DNA sequencing.

i. Reporting Format

Quantitative malaria microscopy is more accurate to calculate the number of parasites per micro liter of blood. This method is recommended as it can assist the doctor to assess the disease severity, monitor response to treatment and the patient's progress. The quantitative result should report different stages of parasite and therefore, the laboratory technician has to calculate both the asexual (trophozoites, schizonts) and sexual (gametocytes) stages of the parasite. (Box 5.1 and Box 5.2)

Microscopy results recorded as *P. malariae* should be reported as *P. knowlesi* / *P. malariae* to guide case management. These findings should be confirmed with PCR.

i. Laboratory Turnaround Time

Hospital & health clinic with laboratory technician: Within 2 hours after BFMP slides have reached the laboratory.

Health Care Clinics: Within the same day as taken.

* Sample sent in EDTA tube may take a longer time to process

Box 8.1: Method of calculation

When the thick film is examined and if *P. falciparum* parasites are found, they will be counted in the following way:

Asexual will be calculated against 200 White Blood Cells but in VERY HEAVY infections the number may be 50 WBCs. The higher the number of WBC's counted the more accurate the final calculation.

Sexual will be counted against WBC's as above.

Please note: A number of assumptions are made here which the doctor should be aware of:

The number of WBC's per micro-liter of blood is 8,000.

The number of red blood cells per micro-liter of blood is 5,000,000.

That one parasite occupies one red blood cell.

Formula for calculation:

$$\frac{\text{No. of parasites}}{\text{No. of WBC's}} \times 8,000 = \text{Parasites per } \mu\text{l blood}$$

Box 8.2: Example of Calculation and Reporting

Thick film examined as *P. falciparum*

No. of Asexual	:	100
No of Gametocytes	:	10
No of WBC counted	:	200

$$\begin{aligned}\text{Asexual density} & : \quad \frac{100}{200} \times 8,000 = 100 \times 40 \\ & = 4,000 \text{ parasites per } \mu\text{l blood}\end{aligned}$$

$$\begin{aligned}\text{Sexual density:} & \quad \frac{10}{200} \times 8,000 = 10 \times 40 \\ & = 400 \text{ parasites per } \mu\text{l blood}\end{aligned}$$

⇒ Report : = Pf Asexual / sexual
= Pf 4,000 / 400 per μl blood

a. Polymerase Chain Reaction (PCR)

PCR for confirmation and species determination is indicated in certain situation such as;

- With clinical symptoms of malaria but no malaria parasite seen in BFMP
- Mortality cases
- Cases having microscopic appearance of *P. malariae*^{10,11}
- Determination of cases link in outbreak

PCR test is available at the National Public Health Laboratory (MKAK) at Sungai Buloh.

i. Specimen

Whole blood in EDTA or dried blood spot on filter paper are the preferable specimens for PCR analysis. Whole blood via venipuncture in EDTA should be kept in a clean biohazard plastic bag and transported in a cold chain (4°C) within 3-4 hours or minus 20°C if a longer period is needed for transportation. For dried blood spot prepared on filter papers (Whatman Grade 1), blot approximately 125 μl of blood (about 20 cent size in diameter) on the filter paper and leave to air dry. Label with appropriate ID, i.e. date of sample collection and initials of technician responsible. Dried blood spot should be transported to the laboratory at room temperature in a separate biohazard zip lock plastic bag (if specimens are obtained from more than one patient) within 48 hours. **(Note: A thick and thin film needs to be sent together with whole blood/dried blood spot specimen for microscopy diagnosis.)**

i. Laboratory Turnaround Time

Nested PCR : 7 working days
qPCR (real time) : 3 working days

a. **Rapid Diagnostic Test (RDT)**

Rapid Diagnostic Tests are based on the detection of circulating parasite antigens. Rapid antigen detection tests (immunochromatographic tests) cannot replace microscopy but can be supplementary when malaria diagnosis is being performed by relatively inexperienced staff (e.g., in low prevalence areas and outside normal working hours). Result of RDTs should be always interpreted together with the results of microscopy, read by an experienced laboratory technician.

In theory malaria RDTs should have high sensitivity (>95%) and should be highly stable in all situations. Histidine-rich protein-2 is a water-soluble protein produced by trophozoites and young (but not mature) gametocytes of *P. falciparum*. The RDTs that use HRP2 detect *P. falciparum* only. These tests should be used only for diagnosis of acute malaria infections, and not for follow-up, as they may remain positive for several weeks even after successful treatment. *Plasmodium lactate dehydrogenase* (pLDH) and the pan-specific *aldolase* are produced by both asexual and sexual stages (gametocytes) of malaria parasites. There is considerable variability in performance of the pLDH and aldolase-based tests for the non-falciparum species.⁹ Because none of the aldolase-based tests are sufficiently sensitive for the non-falciparum species (including *P. vivax* and *P. knowlesi*), the use of pLDH-based RDTs with high sensitivity is recommended. No commercially available RDTs have been sufficiently sensitive overall for *P. knowlesi*, thus a negative pLDH-RDT or aldolase-based RDT does not exclude the diagnosis of knowlesi malaria. Several types of RDTs are available and information on this is available at the WHO website (<http://www.wpro.who.int/sites/rdt>).

References:

1. Marsh K *et al.* Clinical algorithm for malaria in Africa. *Lancet*, 1996, 347:1327–1329.
2. Redd SC *et al.* Clinical algorithm for treatment of *Plasmodium falciparum* in children. *Lancet*, 347:223–227.
3. WHO Expert Committee on Malaria. Twentieth report. Geneva, World Health Organization, 2000 (WHO Technical Report Series, No. 892).
4. Nwanyanwu OC *et al.* Malaria and human immunodeficiency virus infection among male employees of a sugar estate in Malawi. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, 1997, 91:567–569.
5. WHO Expert Committee on Malaria. Twentieth report. Geneva, World Health Organization, 2000 in WHO Technical Report Series, No. 892.
6. Ng OT, Ooi EE, Lee CC, Lee PJ, Ng LC, Wong PS, *et al.* Naturally acquired *Plasmodium knowlesi* infection, Singapore. *Emerg Infect Dis* 2008;14:814e6.
7. World Health Organization. Guidelines for the treatment of malaria Second edition 2010.
8. Geoffrey P. Malaria. *Protozoal tropical infections* 2005; *Medicine* 33:8:39-43
9. Barber *et al.*, Evaluation of the sensitivity of a pLDH-based and an aldolase-based Rapid Diagnostic Test for the diagnosis of uncomplicated and severe malaria caused by PCR-confirmed *Plasmodium knowlesi*, *P.falciparum* and *P. vivax*. *J Clin Micro* 2013;4:1118-1123
10. Informal Consultation on the Public Health Importance of *Plasmodium knowlesi*. Kuching, Sarawak, Malaysia 22 to 24 February 2011, World Health Organization.
11. Barber *et al.* Limitations of microscopy to differentiate *Plasmodium* species in a region co-endemic for *Plasmodium falciparum*, *Plasmodium vivax* and *Plasmodium knowlesi*. *Malar J.* 2013, 812:8
12. Singh, B., and C. Daneshvar. Human infections and detections of *Plasmodium knowlesi*. *Clin. Microbiol.Rev.* 2013, 26(2):165.
13. Barber *et al.* A prospective comparative study of knowlesi in Spain by Real Time PCR in a traveler from Southeast Asia. *Clin Infect Dis*, 2013; 56(3):383-97



CHAPTER 9 :

Malaria Treatment

9.1 : Treatment of uncomplicated malaria

Important notes

- *All malaria patient should be admitted and treated in hospital.*
- *All confirmed uncomplicated malaria should be treated with registered ACTs in Malaysia :*
 - *First line : artemether + lumefantrine*
 - *Alternative : artesunate + mefloquine*
 - *Duration of ACT regimen treatment: 3 days treatment*
 - *For pregnant women (All trimester), the recommended ACTs is arthemether + lumefantrine.*
- *To reduce transmissibility of treated *P.falciparum* infection: Give a single dose of 0.25 mg/kg bw primaquine with ACT on the first day of treatment to patients with *P. falciparum* malaria (except pregnant women, infants aged < 6 months and women breastfeeding infants aged < 6 months) to reduce transmission. G6PD testing is not required.*
- *To prevent relapse in *P.vivax* & *P.ovale* infection: Primaquine 0.5mg/kg to be given once daily for 14 days of treatment in addition to artemisinin regime and quinine (except for pregnant women and infant < 1 year of age).*

Therapeutic objectives

The clinical objectives of treating uncomplicated malaria are to cure the infection as rapidly as possible and to prevent progression to severe disease. “Cure” is defined as elimination of all parasites from the body. The public health objectives of treatment are to prevent onward transmission of the infection to others and to prevent the emergence and spread of resistance to antimalarial drugs.

WHO recommends the use of Artemisinin Combination Therapy (ACT) as the standard treatment for malaria and discourages the prescription of monotherapy or sub-standard ACT as this will promote resistant plasmodium.

Treat adults with uncomplicated *P. falciparum* malaria (except pregnant women in their first trimester) with one of the following ACTs. The treatment regimen and dosing are as shown in Treatment Table (Table 9.1).

9.1.1 : Patient Who Are Unable To Tolerate Orally

Some patients cannot tolerate oral treatment and will require parenteral administration for 1–2 days, until they can swallow and retain oral medication reliably. Although such patients do not show other signs of severity, they should receive the same initial

antimalarial treatments recommended for severe malaria. Initial parenteral treatment must always be followed by a full 3-day course of ACT.

9.1.2 : Recurrent falciparum malaria

Recurrence of *P. falciparum* malaria can result from re-infection or recrudescence (treatment failure). Treatment failure may result from drug resistance or inadequate exposure to the drug due to sub-optimal dosing, poor adherence, vomiting, unusual pharmacokinetics in an individual, or substandard medicines. It is important to determine from the patient's history whether he or she vomited the previous treatment or did not complete a full course of treatment. When possible, treatment failure must be confirmed parasitologically.

Failure within 28 days

The recommended second-line treatment is an alternative effective ACT.

Failure after 28 days

Recurrence of fever and parasitaemia more than 4 weeks after treatment may be due to either recrudescence or a new infection. As PCR genotyping is not routinely used in patient management, all presumed treatment failures after 4 weeks of initial treatment should be considered new infections and be treated with the first-line ACT. However, reuse of mefloquine within 60 days of first treatment is associated with an increased risk for neuropsychiatric reactions, and an alternative ACT should be used.

9.2 : Treatment of severe and complicated malaria

Important notes

- *All malaria patients should be admitted and treated in hospital.*
- *All patients with severe/ complicated malaria, immuno-compromised patients and pregnant mothers should be referred to hospitals with specialists for further management.*
- *Antimalarial treatment should be given to patients prior to transfer to referral hospital. Give IV Artesunate (2.4 mg/kg) stat. If this is not possible, give it IM. Administer IV Quinine if IV Artesunate is not available (and IM, if no IV access).*

Severe malaria is caused by the presence of a high density of asexual forms of *Plasmodium sp.* in the patient's body. Such a patient needs urgent management at secondary or tertiary hospitals. However, in areas of high malarial endemicity or in outbreak situation, clinical suspicion alone should prompt a therapeutic trial with an appropriate antimalarial drug even if blood smears are negative.

Severe malaria is a medical emergency with high mortality if untreated. Mortality is estimated to be around 100% in untreated cerebral malaria. The overall mortality rate falls to 10–20% with prompt and effective antimalarial treatment as well as good supportive care.

The mortality risk also depends on the species of infecting malaria parasite, the number of organs affected and the degree of dysfunction, age, underlying pre-morbid, concomitant medical problems, previous immunity as well as access to appropriate treatment.

9.2.1 Hospital Admission For Malaria Management

All malaria patients should be admitted. Admission is **mandatory** with the following :

- a. Patients with manifestations of severe/ complicated malaria (Chapter 6)
- b. Patients with high parasitaemia ($>100,000/\mu\text{l}$) [$>15,000/\mu\text{l}$ for **P. knowlesi*]
- c. Patients with severe malnutrition
- d. Patients who cannot tolerate orally
- e. All pregnant mothers with acute malaria
- f. Children
- g. Patients with G6PD deficiency

*Early diagnosis and aggressive management of *P. knowlesi* infection is important as severe *P. knowlesi* malaria develops at lower parasite densities.

In clinics where there is no medical officers, assistant medical officers are allowed to initiate anti-malaria treatment after consultation with medical officers and refer to the nearest hospitals.

9.2.2 Referral To Hospital With Specialist

These patients should be referred to hospitals with specialists for further management:

- a. All patients with severe and complicated malaria
- b. Immuno-compromised patients with significant co-morbidities e.g. hepatic or renal dysfunction
- c. Pregnant mothers

Steps for referral

When referring patients, ensure

- Consult physician at receiving hospital
- Referral letter with patient's summary and BFMP slide of patient are sent along
- Antimalarial treatment is commenced prior to transfer
 - Give IV Artesunate (2.4 mg/kg) stat. If this is not possible, give it IM. Administer IV Quinine if IV Artesunate is not available (and IM, if no IV access). **(IM Artesunate > IM Athemether > IM Quinine)**

(Children <6 y.o: If IM Artesunate not available, give rectal Artesunate 10mg/kg bw)

If admission is not possible

If admission to hospital with specialist is not possible, the patient's plan of management **must be** discussed with the Physicians/ Infectious Disease Specialists.

9.2.3 Criteria for referral to the High Dependency Unit (HDU) or Intensive Care Unit (ICU)

- If a patient has any of the features of severe malaria, the patient should be referred to a hospital with a HDU or ICU.
- An early referral to intensivist should be made for further evaluation and management.
- If the patient cannot be transferred, then the patient should be monitored closely in the district hospital in close consultation with the physician.
- All patients with complicated malaria should be referred to the physician on call.

9.2.4 General Management for Severe Malaria

1. Take a complete history and physical examination.
2. Examine the Glasgow Coma Scale (GCS), presence of neck stiffness, blood pressure, pulse rate, respiratory rate. Examine the fundi as well if possible.
3. Secure an open airway in unconscious patients and assess the breathing and circulation. Insert an intravenous cannula.
4. Measure weight or estimate patient's body weight; so that medicines, including antimalarial drugs and fluids, can be given appropriately and promptly.
5. Send the following investigations.
6. Admit the patient to the ICU or the High Dependency Unit
7. The patient should be referred to the Physician or the Intensivist.
8. Give IV Artesunate (2.4 mg/kg) stat. If this is not possible, give it IM. Administer IV Quinine if IV Artesunate is not available (and IM, if no IV access). For complete dosing, please look at the section on antimalarial treatment.
9. Administer the intravenous antimalarial for at least 24 hours before switching to an oral formulation if the patient improves and can tolerate orally.
10. Close observation of the vital signs, GCS, by the nurses and doctor is of ~~vital~~ importance.
11. **Pulmonary oedema is a grave complication of severe *P. knowlesi* and *falciparum* infection.** It has a high mortality (over 80% in falciparum malaria); the prognosis is better in vivax malaria. Pulmonary oedema may develop several days after chemotherapy has been started, at a time when the

patient's general condition is improving and the peripheral parasitaemia is falling.

- a. Close input output monitoring is very important as the patient can develop pulmonary oedema easily if too much fluids is given. Give fluids just enough to maintain the circulation and correct the dehydration.
 - b. Patients who are unable to tolerate orally should be given maintenance fluids – for children at 3-4ml/kg/hour, and adults at 1-2ml/kg body weight per hour, until the patient is able to take and retain oral fluids.
 - c. Rapid fluid boluses are contraindicated in severe malaria resuscitation. Dehydration should be managed cautiously and ideally guided by urine output (with a urine output goal of > 1ml/kg body weight per hour), unless the patient has anuric renal failure or pulmonary oedema, for which fluid management should be tailored to the needs of the patient and reassessed frequently.
12. If the patient's GCS is reduced, look for other causes such as meningitis. A lumbar puncture may then be indicated and antibiotics instituted.
 13. There is considerable clinical overlap between septicaemia, pneumonia and severe malaria, and these conditions may coexist. If secondary infections are suspected, then start the patient on a broad spectrum IV antibiotic such as Ceftriaxone 2g OD (after a septic workup is done).
 14. Monitor for clinical and parasitological therapeutic response.
 15. High fever can be managed by tepid sponging and paracetamol.

9.2.5 Management of malaria complications

Malaria, due to *P. falciparum* or *P. knowlesi* is a medical emergency because clinical deterioration can occur rapidly and unpredictably. Complicated malaria without prompt and appropriate treatment may lead to death. Most cases of severe malaria in Malaysia are caused by *P. knowlesi* and *P. falciparum* infection.

Immediate clinical management of severe manifestations and complications of malaria

Manifestations/ Complications	Immediate Management
Coma (cerebral malaria)	Maintain airways, place patient on his or her side and for ventilator support if necessary; exclude other treatable causes of coma (e.g. hypoglycaemia, bacterial meningitis); avoid harmful ancillary treatment, such as corticosteroids, heparin and adrenaline.

Convulsions	• Maintain airways; treat promptly with intravenous or rectal diazepam and/ or other anticonvulsants. Check blood glucose, blood urea and serum electrolytes. • Prophylactic anticonvulsants are not recommended.
Hypoglycaemia	• Correct hypoglycaemia and maintain with glucose containing infusion.
Acute renal failure	• Exclude pre-renal causes, check fluid balance and urinary sodium; in established renal failure initiate haemofiltration or haemodialysis, or peritoneal dialysis.
Severe Anaemia	• Blood transfusion with screened fresh whole blood where indicated.
Acute pulmonary oedema	• Prop patient up at an angle of 45°, give oxygen and other respiratory support as indicated. Adjust fluid therapy appropriately. Diuretics if necessary.
Shock	• Assess hydration status. Give appropriate fluid therapy. Correct haemodynamic disturbances. Suspect septicaemia (bacterial in origin) in unexplained clinical deterioration. Take blood cultures; give parenteral broad-spectrum antimicrobials and de-escalate according to culture results.

9.2.6 Adjunct therapy

In an attempt to reduce the unacceptably high mortality of severe malaria, besides appropriate and effective antimalarial drugs, various adjunctive treatments for the complications of malaria have been evaluated in clinical trials. Adjunctive therapy is defined as any additional therapy that modifies physiologic processes caused by malaria. These therapies may act directly on specific biologic pathways altered by malaria or more generally on end-stage factors produced in malaria by a number of different specific processes.

Exchange Blood Transfusion (EBT)

There is no clear evidence of benefit due to absence of randomized clinical trial (RCT) on EBT and potential risk in patients who are hemodynamically unstable³ The procedure requires expertise and intense monitoring, which limits its suitability in resource-limited settings. The rationale for EBT has been variously proposed as:

- Removing infected red blood cells from the circulation and, therefore, lowering the parasite burden;
- Reducing rapidly both the antigen load and the burden of parasite-derived toxins and toxic mediators; and
- Replacing the rigid unparasitized and parasitized red cells by more deformable cells and, therefore, reducing microcirculatory obstruction.

Adjunctive Treatments which are not recommended

Other supportive strategies and interventions have been used in severe malaria patients in an effort to further reduce the mortality, but very few are supported by evidence of benefit and this includes the use of these listed drugs.^{1,2,4}

- Modulate the immune response to *P. falciparum* (dexamethasone, intravenous immunoglobulin, monoclonal antibodies to TNF- α , pentoxifylline and curdlan sulfate)*
- Reduce iron burden (iron chelation with desferrioxamine or deferipone)
- Reduce oxidative stress (*N*-acetylcysteine [NAC])
- Counteract the prothrombotic state and prevents the formation of rosettes of infected red blood cells (RBCs) (heparin and aspirin)
- Reduce parasitemia (EBT)
- Expand volume and potentially decrease acidosis (albumin)**
- Decrease intracranial pressure and cerebral oedema (mannitol and dexamethasone)
- Prevent seizure activity (prophylactic phenobarbital)

9.3 : Treatment in special risk groups

9.3.1 : Young children and infants

Important notes:

1. The treatment regime for paediatric patients is the same as for adults except that the drug dose is adjusted according to the weight of the patient.
2. Sulphadoxine-pyrimethamine, SP (with artesunate) should be avoided in the first week of life as it could aggravate neonatal hyperbilirubinemia.
3. Tetracyclines such as Doxycycline is contraindicated in children less than eight years old.
4. Primaquine should be avoided in the first 6 months of life.
5. Primaquine *should be given to paediatric patients only after they have been screened for G6PD deficiency.*
 - i. In children with mild to moderate G6PD deficiency, Primaquine should be given in a dose of 0.75 mg base/kg body weight once a week for 8 weeks, with close medical supervision for potential Primaquine- induced haemolysis. If significant haemolysis occurs on treatment, then Primaquine should be stopped immediately.
 - ii. In severe G6PD deficiency, Primaquine is contraindicated and should not be used.

6. Infant weighing less than 5kg with uncomplicated *P. falciparum* malaria should be treated with ACT. The weight-adjusted dose should achieve the same mg/kg body weight target dose as for children weighing 5kg.
7. It is important to ensure accurate dosing and retention of the ACT in young children. Mothers often need to be taught on techniques of drug administration and to repeat drug administration if it is regurgitated **within 1 hour** of administration.
8. Refer table 9.2 for treatment regime for children.

Congenital malaria

Congenital malaria is defined as presence of asexual form of malaria parasites in newborn's cord blood and/or peripheral blood during first week of life, irrespective of clinical symptoms. It can be acquired by transmission of parasites from mother to child during pregnancy or perinatally during labour. Prevalence of congenital malaria varies from 0% to 54.2% depending on the geographic area, endemicity, and method of detection. Congenital malaria from *P. vivax* is more commonly reported in Asia whereas infection from *P. falciparum* is mainly described in African countries. Most babies present with symptoms between 10 and 30 days of age (range: 14hr to several months of age). The clinical features of neonatal malaria include anaemia (77%), fever (74%), hepatosplenomegaly (68%), poor feeding/ lethargy/ irritability, jaundice and severe thrombocytopaenia.

Congenital malaria may mimic neonatal sepsis and should be considered in the differential diagnosis of neonatal sepsis. All newborn babies of mother with malaria should be screened for congenital malaria.

Primaquine is not required for treatment as the tissue/ exo- erythrocytic phase is absent in congenital malaria.

Treatment for ***P. falciparum* infection**: Quinine 10mg/kg 8 hourly for 1 week.

Mixed Malaria infections

Mixed malaria infections are common. ACTs are effective against all malaria species and are the treatment of choice. Treatment with Primaquine should be given to patients with confirmed *P. vivax* infection.

9.3.2: Pregnant and lactating women

Malaria in pregnancy have an association with fetal growth restriction, low- birth- weight babies, maternal anaemia and may increase the risks for severe malaria, miscarriage and both maternal and fetal mortality especially in low-transmission areas.

In high- transmission settings, the cases are usually only mild, or asymptomatic despite having increased risk for fetal growth restriction.

The risk of malaria infection is reported to be highest in the first and second trimesters of pregnancy.

9.3.2.1 First trimester

Based on latest updated WHO Malaria Guidelines, artemether- lumefantrine (AL) is strongly recommended as the preferred treatment of uncomplicated *P falciparum* malaria during the first trimester of pregnancy, despite the low certainty of evidence because:

- there was huge beneficial effect of treatment on efficacy (demonstrated in the second and third trimesters of pregnancy), specifically a six-fold reduction in treatment failures following AL, as compared to the quinine-based therapies
- AL was associated with trivial adverse events and significantly lower risk for adverse pregnancy outcomes in the first trimester of pregnancy
- AL was better tolerated as compared to quinine-based therapies
- there is probably increased equity, acceptability and feasibility, resulting from better access to AL and more efficient implementation of ACTs compared to quinine-based treatments.

Due to limited exposures to other ACTs (artesunate-amodiaquine, artesunate-mefloquine and dihydroartemisinin piperazine), current evidence is insufficient to make a recommendation for routine use of these other ACTs in the first trimester of pregnancy. However, these other ACTs may be considered for use where AL is not available, given the demonstrated poorer outcomes of quinine treatment.

In any cases especially those with severe form of malaria infection, the benefit of the treatment must always be weighed against the potential harm to both the mother and the growing foetus. It is always recommended that the care decision made through a multidisciplinary care approach under the supervision of a dedicated obstetrician or maternal foetal medicine specialist.

9.3.2.2 Second and third trimesters

Treating Malaria with Artemisinin derivatives in the second and third trimesters have been shown safe to both mother and foetus based on over 4000 documented pregnancies.

The current assessment of risk–benefit suggests that ACTs should be used to treat uncomplicated falciparum malaria in the second and third trimesters of pregnancy.

Mefloquine is considered safe for the treatment of malaria during the second and third trimesters; however, it should be given only in combination with an Artemisinin derivative.

Quinine may cause hypoglycaemia in late pregnancy, and it should be used (with Clindamycin) only if effective alternatives are not available.

Primaquine and Tetracyclines should not be used in pregnancy due to limited safety data and potential harm to both mother and foetus.

9.3.2.3 Dose modification during pregnancy

Data on the pharmacokinetics of antimalarial agents used during pregnancy are limited. Those available indicate that pharmacokinetic properties may be altered during pregnancy but that the alterations are insufficient to warrant dose modifications. No significant differences in exposure have been seen with Quinine as treatment during pregnancy.

9.3.2.4 Lactation

The amounts of antimalarial drugs that enter breast milk and are consumed by breastfeeding infants are relatively small. Tetracycline is contraindicated in breastfeeding mothers because of its potential yellow discolouration effect on infants' bones and teeth.

Pending further information on excretion in breast milk, Primaquine should not be used for nursing women, unless the breastfed infant has been checked for G6PD deficiency. Primaquine potentially may cause hemolytic anemia in a glucose-6-phosphate dehydrogenase (G6PD)- deficient foetus.

9.3.2.5 Intermittent preventive treatment of malaria in pregnancy

All pregnant women staying or travelling in malaria-endemic areas should be given antimalarial medicine at certain intervals to reduce disease burden in pregnancy and adverse pregnancy and birth outcomes including maternal anemia, placental and maternal malaria infection at delivery and congenital malaria.

Sulfadoxine- pyrimethamine (SP) has been widely used for malaria chemoprevention during pregnancy and is effective in improving key pregnancy outcomes.

SP best given under directly observed therapy (DOT) method with three tablets of SP (each tablet containing 500 mg/25 mg SP), for the total required dosage of 1500 mg/75 mg SP.

It should start as early as possible in the second trimester and not before week 13 of pregnancy, with doses should be given :

- at least one month apart
- at least three doses are received

General precautions such as rescheduling the travel plan to endemic areas, use of bed nets and also the use of malaria chemoprevention drugs should be advised to the pregnant women at risk.

9.3.3 : Other risk group

9.3.3.1 : Patients co-infected with HIV

In HIV infected individuals, the risk of severe malaria and malaria-related deaths are influenced by the immune-compromised state of HIV and whether they are partially immune to malaria due to constant contact in areas where malaria transmission is stable.

There are however limited information on how HIV infection modifies therapeutic responses to ACTs.

Early studies suggested that individuals with more advance HIV will have decreased treatment response to antimalarial drugs.

The treatment recommendation will be as shown in Table 9.1 as there is presently insufficient information to modify the general malaria treatment recommendations for patients with HIV/AIDS except if they are on particular ACTs that will interact with the anti-retroviral therapy.

In uncomplicated falciparum malaria, avoid Artesunate + SP if the HIV/AIDS patients are being treated with Co-trimoxazole, and avoid Artesunate + Amodiaquine if they are being treated with Efavirenz or Zidovudine.

9.3.3.2 : Non-Immune travellers

Most travellers who acquire malaria are those who are non-immune against malaria, lived in cities in an endemic countries with little or no transmission or are visitors from non-endemic countries travelling to areas with malaria transmission.

They are at higher risk for acquiring severe malaria. Therefore, they should be treated according to the treatment recommendation as shown in Table 9.1.

Table 9.1: Treatment Regimen for Malaria

DIAGNOSIS	FIRST LINE	ALTERNATIVE	COMMENTS																	
PLASMODIUM FALCIPARUM																				
<i>Plasmodium falciparum</i> A) Non Complicated (i) New Infection	ADULT Riamet* (1 tablet: 20 mg artemether/ 120 mg lumefantrine) A 3-day treatment schedule with a total of 6 doses is recommended The patient should receive an initial dose followed by the 2 nd dose 8 hours later, then 1 dose BD for the following 2 days	Artesunate /Mefloquine (AS+MQ FDC Tablet)* 2 FDC strength – 25mg AS /55 MQ and 100mg AS / 220mg MQ A 3-day treatment schedule, once daily regime	The choice of drug should be governed by drug availability and safety. Riamet should be taken with a fatty meal eg milk to increase bioavailability Give a single dose of 0.25 mg/kg bw primaquine with ACT on the first day of treatment to patients with <i>P. falciparum</i> malaria (except pregnant women, infants aged < 6 months and women breastfeeding infants aged < 6 months) to reduce transmission. G6PD testing is not required.																	
	CHILDREN Riamet* (1 tablet: 20 mg artemether/120 mg lumefantrine) Dosing regimen: An initial dose followed by the 2 nd dose 8 hours later, then 1 dose BD for the following 2 days 5 -<15kg : 1 tablet per dose 15 -<25kg : 2 tablets per dose 25 -<35kg : 3 tablets per dose ≥35kg : 4 tablets per dose	Artesunate /Mefloquine (AS+MQ FDC Tablet)* <table><tr><td>Weight (kg)</td><td>Dose (once daily regimen)</td></tr><tr><td>5 – 8</td><td>1 tablet of 25mg AS/55mg MQ</td></tr><tr><td>9 - 17kg</td><td>2 tablets of 25mg AS/55mg MQ</td></tr><tr><td>18 - 29kg</td><td>1 tablet of 100mg AS/220mg MQ</td></tr><tr><td>≥30kg</td><td>2 tablets of 100mg AS/220mg MQ</td></tr></table>		Weight (kg)	Dose (once daily regimen)	5 – 8	1 tablet of 25mg AS/55mg MQ	9 - 17kg	2 tablets of 25mg AS/55mg MQ	18 - 29kg	1 tablet of 100mg AS/220mg MQ	≥30kg	2 tablets of 100mg AS/220mg MQ	<table><tr><td>Body weight (kg)</td><td>Single dose of primaquine (mg base)</td></tr><tr><td>10 to < 25</td><td>3.75</td></tr><tr><td>25 to < 50</td><td>7.5</td></tr><tr><td>50 to 100</td><td>15</td></tr></table> *In pregnancy: 1 st trimester : The recommended ACT treatment is arthemether-lumefantrine (AL	Body weight (kg)	Single dose of primaquine (mg base)	10 to < 25	3.75	25 to < 50	7.5
Weight (kg)	Dose (once daily regimen)																			
5 – 8	1 tablet of 25mg AS/55mg MQ																			
9 - 17kg	2 tablets of 25mg AS/55mg MQ																			
18 - 29kg	1 tablet of 100mg AS/220mg MQ																			
≥30kg	2 tablets of 100mg AS/220mg MQ																			
Body weight (kg)	Single dose of primaquine (mg base)																			
10 to < 25	3.75																			
25 to < 50	7.5																			
50 to 100	15																			

***In pregnancy:**

1st trimester : The recommended ACT treatment is arthemether-lumefantrine (AL

DIAGNOSIS	FIRST LINE	ALTERNATIVE	COMMENTS
PLASMODIUM FALCIPARUM			
			Exposure of artemisinin derivatives during 2nd and 3rd trimester have shown no adverse effects on the mother or fetus. Lactating woman Do not use primaquine and tetracycline *AS + MQ • Avoid its use in children with epilepsy as it may cause seizures • Interact with Quinine, Chloroquine and Halofantrine and may cause arrhythmia • Do not repeat AS-MQ regime within 60 days. (Reuse of mefloquine within 60 days of first treatment is associated with increased risk of neuropsychiatric reactions.)

PLASMODIUM FALCIPARUM			
Plasmodium falciparum		Oral Quinine (10mg/kg) 8 hourly with	Give a single dose of 0.25 mg/kg bw primaquine with ACT on the first day of treatment to patients with <i>P.</i>

	*For all trimester stages of pregnancy: IV Artesunate as for normal adults³ *CHILDREN - treatment regimen same as for normal adults	Oral Doxycycline 100mg BD for 7days (adult). For Children, Doxycycline (>8yrs) (3.5 mg/kg OD) OR Clindamycin (<8yrs) (10 mg/kg/dose bd) given for 7 days (*Doxycycline use if ACT not available : artesunate+doxy / quinine+doxy)	be reduced by one third after 48 h, to 10 mg salt/kg bw every 12 h. No dosage adjustments are necessary if patients are receiving haemodialysis Give a single dose of 0.25 mg/kg bw primaquine with ACT on the first day of treatment to patients with <i>P. falciparum</i> malaria (except pregnant women, infants aged < 6 months and women breastfeeding infants aged < 6 months) to reduce transmission. G6PD testing is not required
PLASMODIUM VIVAX / OVALE			
<i>P. vivax, P. ovale</i> A)New Infection ** Treatment for <i>Plasmodium vivax/ovale</i> malaria for children is the same regimen as <i>Plasmodium falciparum</i> malaria in children.	Riamet* (1 tablet: 20 mg artemether/ 120 mg lumefantrine) A 3-day treatment schedule with a total of 6 doses is recommended The patient should receive an initial dose followed by the 2nd dose 8 hours later, then 1 dose BD for the following 2 days	ACT (AS+MQ) for 3 days OR Oral Quinine 10mg/kg 8 hourly for 7 days AND Primaquine 0.5 mg/kg (max 30mg) OD for 14 days	The objectives of treatment of vivax malaria are twofold: to cure the acute blood stage infection and to clear hypnozoites from the liver to prevent future relapses. This is known as “radical cure” G6PD: In patients with mild-moderate variants of G6PD deficiency, primaquine should be given in a dose of 0.75 mg base/kg body weight once a week for 8 weeks, with close medical supervision for potential primaquine-induced haemolysis. If significant

B)Treatment Failure or suspected chloroquine resistant	5 - <15kg : 1 tablet per dose 15 - <25kg : 2 tablets per dose 25 - <35kg : 3 tablets per dose ≥35kg : 4 tablets per dose AND Primaquine 0.5 mg/kg (max 30mg) OD for 14 days Repeat course: ACT (Riamet) for 3 days AND Primaquine 0.5 mg/kg (max 30mg) OD for 14 days		haemolysis occurs on treatment, then primaquine should be stopped. In severe <i>P. vivax</i>, treatment is as severe <i>P. falciparum</i> For adults > 70kg, continue 30 mg daily beyond 14 days until a total cumulative dose of 6 mg/kg is reached In pregnancy and breastfeeding, the full course of Chloroquine will be given, then followed by 300mg weekly chemoprophylaxis until delivery and breastfeeding is completed. The full course of Primaquine will only be given after that on the basis of G6PD status to prevent future relapse.
PLASMODIUM MALARIAE/KNOWLESII**			
<i>P.malariae</i>, <i>P.knowlesi</i> A) New Infection	Riamet (1 tablet: 20mg artemether/120mg lumefantrine) A 3-day treatment schedule with a total of 6 doses is recommended.	ACT (AS+MQ) for 3 days OR Chloroquine Total dose of 25 mg base/kg administered as 10 mg base/kg orally on day 1, followed by 10 mg base/kg	** <i>P. knowlesi</i> and <i>P. malariae</i> cannot be distinguished by microscopy. In Sabah and Sarawak, <i>P. malariae</i> is rare; nearly all reports of <i>P. malariae</i> are in fact <i>P. knowlesi</i> when tested by PCR. Patient diagnosed by microscopy as <i>P.malariae</i>

B) Severe	The patient should receive initial dose followed by the 2nd dose 8 hours later, then 1 dose BD for the following 2 days. 5 - <15kg : 1 tablet per dose 15 - <25kg: 2 tablets per dose 25 - <35kg: 3 tablets per dose ≥35kg : 4 tablets per dose Treat as severe <i>P falciparum</i> infection with IV Artesunate	orally on day 2, and 5 mg base/kg on day 3 *Chloroquine can be used in places where ACT is not available and with close monitoring of treatment response.	should be presumed to have <i>P.knowlesi</i> infection. It does not require radical cure with primaquine, as no hypnozoites are formed in infection with this species. ACTs clear parasites more quickly than chloroquine. ACTs with long half-lives provide a longer period of suppressive post-treatment prophylaxis against relapses and new infections ** Treatment for <i>Plasmodium malariae/knowlesi</i> malaria for children is the same regimen as <i>Plasmodium falciparum</i> malaria in children.
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9.4 Field Treatment

Field treatment for malaria is not advisable as a routine mode of treatment. However, it is still practiced in certain remote areas in Sarawak and probably Sabah due to unforeseen logistic situation. However, this is limited to interior areas where it is only reachable by flying doctor services, the patient with clinically uncomplicated malaria. The treatment should follow treatment guidelines accordingly. However, all malaria cases must be encouraged to be hospitalized.

*As testing for G6PD deficiency is not readily available at field level, patient need to be educated on signs and symptoms of hemolysis i.e. dark colour urine. The first sign of primaquine associated hemolysis tend to become apparent within 2 to 3 days of starting treatment.²² Patient needs to be advised to stop Primaquine immediately if such signs and symptoms are present and seek immediate medical attention.

History of previous anti-malarial exposure especially to Primaquine can be elicited. If there is no history of adverse reaction to Primaquine, then it may be reasonable to assume that there is no G6PD deficiency.

9.5 Presumptive Treatment

Presumptive treatment is given for those patients whom a blood film was taken but the results would not be available at that moment due to logistic reason, like those in very remote areas and risks of defaulting follow- up. This practice should be discouraged and is only done under extreme circumstances. This mode of treatment is given to those in endemic areas with classical symptoms of malaria i.e. fever for more than 3 days with chills and rigors. The treatment should be based on the predominant species in that area. The safety of the drug should be monitored and supervised as required. However, with availability of reliable and accurate rapid diagnostic onsite, the need to give presumptive treatment will be overcome and phased out in near future.

9.6 Discharge and Follow-up

All patients diagnosed with malaria must be admitted for the duration of treatment to ensure completion. In high transmission settings, it is important to keep the patients warded until they are cleared of gametocytes which play a crucial role in the spread of malaria and thus of public health concern. Otherwise, the patients can be discharged once they are clinically well with two negative BFMP slides within 48 hours.

After completed the treatment and discharge from ward, patients should be followed up weekly for a month (Day 7, 14, 21, 28). Thereafter, follow up for those infected with *P. vivax*/ *P. ovale* infection should be followed up monthly for one year extension. However, if antimalarial drug resistance is suspected, follow up should be extended; for example, in the case of suspected *P. falciparum* drug resistance, beyond two months. Such cases should be notified to the state and ministerial level for further action.

References:

1. WHO Guidelines for Malaria 30 November 2024.
2. G Watt, K Jongsakul and R Ruangvirayuth. A pilot study of N Acetylcystein as adjunctive therapy for severe malaria. *Q J Med* 2002 95: 285-290
3. Riddle MS, Jackson JL, Sanders JW, Blazes DL. Exchange trans fusion as an adjunct therapy in severe *Plasmodium falciparum* malaria: a meta-analysis. *Clin Infect Dis* 2002 May 1; 34(9):1192e8.
4. John C. C., Kutamba E., Mugarura K., Opoka R. O.. Adjunctive therapy for cerebral malaria and other severe forms of *Plasmodium falciparum* malaria. Expert review of antiinfective therapy. 2010, 8(9):997-1008
5. Maitland K, Nadel S, Pollard AJ, Williams TN, Newton CR, Levin M. Management of severe malaria in children: proposed guidelines for the United Kingdom. *Br Med J*. 2005; 331(7512):337–343.
6. Barber BE *et al*, A prospective comparative study of knowlesi, falciparum, and vivax malaria in Sabah, Malaysia: high proportion with severe disease from *Plasmodium knowlesi* and *Plasmodium vivax* but no mortality with early referral and artesunate therapy. *Clin. Infect Dis*. 2013; 56(3):383-97.
7. Management of Severe Malaria: A Practical Handbook. WHO Third Edition 2012
8. William T *et al*. Severe *Plasmodium knowlesi* malaria in a tertiary care hospital, Sabah, Malaysia. *Emerg Infect Dis*. 2011,17(7):1248-55
9. Barber BE *et al*, A prospective comparative study of knowlesi, falciparum, and vivax malaria in Sabah, Malaysia: high proportion with severe disease from *Plasmodium knowlesi* and *Plasmodium vivax* but no mortality with early referral and artesunate therapy. *Clin. Infect Dis*. 2013; 56(3):383-97
10. Expert Consultation On Plasmodium Knowlesi Malaria To Guide Malaria Elimination Strategies [RS-2017-GE-05-MYS-eng.pdf \(who.int\)](#)
11. Grigg, M. J., William, T., Barber, B. E., Rajahram, G. S., Menon, J., Schimann, E., Piera K., Wilkes C. S., Patel, K., Chandna. A., Drakeley, C. J., Yeo, T. W. & Anstey, N. M. (2018). Age-related clinical spectrum of *Plasmodium knowlesi* malaria and predictors of severity. *Clinical infectious diseases*, 67(3), 350-359.
12. Uneke CJ. Congenital *Plasmodium falciparum* malaria in sub-Saharan Africa: a rarity or frequent occurrence?. *Parasitol Res*. 2007;101(4):835-842.
13. Natama HM, Ouedraogo DF, Sorgho H, et al. Diagnosing congenital malaria in a high-transmission setting: clinical relevance and usefulness of *P. falciparum* HRP2-based testing. *Sci Rep*. 2017;7(1):2080.
14. Dondorp A, *et al*. Artesunate versus quinine in the treatment of severe falciparum malaria in African children (AQUAMAT): on open-label, randomized trial; *Lancet* 2010 Nov 13;376(9753): 1647-1657.
15. UK Malaria guidelines 2007, Metha PN 2004

16. Neena V. et al. Congenital malaria with atypical presentation: a case report from low transmission area in India. *Malaria Journal* 2007, 6:43
17. Del Punta et al. Congenital *Plasmodium vivax* malaria mimicking neonatal sepsis: a case report. *Malaria Journal* 2010, 9:63
18. Red Book 2009
19. Tarning J, Lindeganullrdh NH, Gilder ME, et al. "A study of the pharmacokinetics of primaquine in lactating women and breastfed infants for the radical treatment of uncomplicated maternal *plasmodium vivax*." *Am J Trop Med Hyg* 91 (2014): 587
20. Irvine MH, Einarson A, Bozzo P. Prophylactic use of antimalarials during pregnancy. *Can Fam Physician*. 2011;57(11):1279-1281.
21. Rai P, Majumdar K, Sharma S, Chauhan R, Chandra J. Congenital malaria in a neonate: case report with a comprehensive review on differential diagnosis, treatment and prevention in Indian perspective. *J Parasit Dis*. 2015;39(2):345-348. doi:10.1007/s12639-013-0342-1
22. Yilma D, Groves ES, Brito-Sousa JD, Monteiro WM, Chu C, Thriemer K, Commons RJ, Lacerda MVG, Price RN, Douglas NM. Severe Hemolysis during Primaquine Radical Cure of *Plasmodium vivax* Malaria: Two Systematic Reviews and Individual Patient Data Descriptive Analyses. *Am J Trop Med Hyg*. 2023 Aug 21;109(4):761-769. doi: 10.4269/ajtmh.23-0280.

Author: Dr Chua Hock Hin, Infectious Disease Consultant, Ministry of Health, Malaysia



CHAPTER 10 : Prevention Of Malaria

Important notes

- *Doxycycline is the recommended drug to use for malaria prophylaxis in Malaysia.*
- *Doxycycline dosage is 100mg daily.*
- *Start 1-2 days before departure, continue daily dose at area of concern and only stop after 4 weeks leaving the area.*

Malaria prevention consists of a combination of mosquito avoidance measures and chemoprophylaxis. The mosquito avoidance measures include insect repellent, wearing long sleeves, long pants, sleeping in a mosquito-free setting or using an insecticide-treated bed net. The use of *diethyltoluamide* (DEET) 20-50% in lotions, spray or roll-on formulation is safe and effective when applied to the skin of adults and children. Although *diethyltoluamide* may be used in children > 2 months, it should be used with caution. Because of their increased surface-area-to-body-mass ratio, children may be at increased risk for toxicity due to greater skin absorption. Low-concentration products should be used and applied sparingly. No antimalarial drug is 100% protective and must be combined with the use of personal protective measures

10.1 Precautions

Travellers should note the four principles – the **ABCD** – of malaria protection:

- Be **A**ware of the risk, the incubation period, the possibility of delayed onset, and the main symptoms.
- Avoid being **B**itten by mosquitoes, especially between dusk and dawn.
- Take antimalarial drugs (**C**hemoprophylaxis) when appropriate, to prevent infection from developing into clinical disease.
- Immediately seek **D**iagnosis and treatment if a fever develops 1 week or more after entering an area where there is a malaria risk and up to 3 months (or, rarely, later) after departure from a risk area.

10.2 Risk of malaria

The risk of acquiring malaria differs substantially from region to region and from traveler to traveler, even within a single country as it depends on several factors such as transmission intensity, duration of stay in the endemic area and the efficacy of preventive measures. Often, the risk to the indigenous population is used as a guideline for the risk to the traveler. Travelers with the highest estimated relative risk for infection are those going to West Africa and Oceania. Travelers going to other parts of Africa, South Asia, and South America have a moderate estimated relative risk for infection. Travelers with lower estimated relative risk are those going to Central America and other parts of Asia. (1) All travelers should seek medical attention in the event of fever during or after return from travel to areas with malaria.

At risk groups include:

- 1.Travelers or visitors to endemic areas
- 2.Army personnel
- 3.Loggers/rubber tappers
- 4.Workers in endemic areas e.g. dam construction, plantation

Caution:

Patient with severe splenic dysfunction or pregnant women should avoid traveling to endemic areas.

10.3 Chemoprophylaxis

Prophylaxis is not absolute, and breakthrough infection can occur with any of the drugs recommended. Personal protection against being bitten is very important and is recommended even in malaria-free areas as a preventive measure against other insect vector-borne diseases. There are two types of chemoprophylaxis: suppressive and causal prophylaxis.

10.3.1 Suppressive prophylaxis

Chloroquine, proguanil, mefloquine, and doxycycline are suppressive prophylactics. This means that they are only effective at killing the malaria parasite once it has entered the erythrocytic stage (blood stage) of its life cycle, and therefore have no effect until the liver stage is complete. That is why these prophylactics must continue to be taken for four weeks after leaving the area of risk.

10.3.2 Causal prophylaxis

Causal prophylactics target not only the blood stages of malaria, but the initial liver stage as well. Successful drug activity at this stage prevents the parasite from progressing to infect red blood cells. This means that the user can stop taking the drug seven days after leaving the area of risk. Atovaquone/Proguanil (Malarone) and primaquine are the only causal prophylactics in current use.

10.3.3 Prophylaxis against hypnozoites

P. vivax and *P. ovale* have a dormant stage called the 'hypnozoite'. The hypnozoite remains dormant for months and then "wakes up" to develop into a liver schizont. The dormant hypnozoite explains why attacks can occur long after the end of chemoprophylaxis. This is not due to drug failure, as none of the prophylactic drugs currently recommended acts against the hypnozoite stage of *P. vivax* or *P. ovale*.

10.3.4 Considerations when choosing a drug for malaria prophylaxis

- Check risk of exposure to malaria and the recommended drugs listed in country-specific tables as in Appendix 1 or updated version on-line in the internet.
- When deciding which drug to use, consider length of trip, previous adverse reactions, current medical history, risk of exposure to malaria, extent of drug resistance, efficacy of the recommended drugs, side-effects of the drugs, drug-drug interactions with other medicines that the person might be taking as well, patient-related factors (e.g. age, pregnancy, renal or hepatic impairment, compliance with prophylactic regimen) and availability of drugs.
- All recommended primary prophylaxis regimens involve taking a medicine before, during, and after travel to an area with malaria. Beginning the drug before travel allows the antimalarial agent to be in the blood before the traveler is exposed to malaria parasites.
- Start chemoprophylaxis earlier if there is particular concerns about tolerance to the medications. For example, mefloquine can be started 3–4 weeks in advance to allow potential adverse events (AE) to occur before travel.
- The use of the same or related drugs that have been taken for prophylaxis is not recommended to treat malaria.

Table 10.1: Advantages and Disadvantages of Chemoprophylaxis

DRUG	ADVANTAGES	DISADVANTAGES
Doxycycline* Dose: 100mg daily Start: 1-2 days before departure Stop: 4 weeks after travel Maximum duration: 2 years Pediatric Dosage 1.5mg base/kg once daily (max. 100 mg) <25kg or <8 yr: Do Not Use 25-35kg or 8-10 yr: 50mg 36-50kg or 11-13 yr: 75mg >50kg or >14 yr: 100mg	- It is taken once daily, so convenient and easier to remember - Good for short trip and last-minute travelers because the drug is started 1-2 days before traveling - Tends to be the least expensive - No need additional new medications as some patients might have been on oral doxycycline for acne or hydroxychloroquine for rheumatologic problem - Doxycycline also can prevent some additional infections (e.g., Rickettsiae and leptospirosis) and so it may be preferred by people planning to do lots of hiking, camping, and swimming in fresh water	- Contraindicated in pregnancy and children <8 years old - Photosensitivity – need to avoid considerable sun exposure - Gastrointestinal (GIT) upset - Vaginal candidiasis - Some patients will not want to take medicine daily and 4 weeks after travel.

DRUG	ADVANTAGES	DISADVANTAGES
Atovaquone/Proguanil (Malarone) 100/250mg* Dose: 1 tablet daily Start: 1-2 days before departure Stop: 7 days after travel Maximum duration: 1 year Pediatric Dosage Pediatric tablet of 62.5 mg atovaquone and 25 mg proguanil: 5-8 kg: 1/2 tablet daily >8-10 kg: 3/4 tablet daily >10-20 kg: 1 tablet daily >20-30 kg: 2 tablets daily >30-40 kg: 3 tablets daily >40 kg: 1 adult tablet daily	• Good for last-minute travelers • Good choice for shorter trips • It is taken once daily, so convenient and easier to remember • Very well tolerated medicine – side effects uncommon	• Contraindicated in pregnancy or breastfeeding a child that weighs <5 kg and patient with severe renal impairment (CrCl <30 mL/min) • Expensive
Mefloquine (Lariam)** Dose: 250mg weekly Start: 2-3 weeks before departure Stop: 4 weeks after travel Maximum duration: 1 year Pediatric Dosage <15 kg: 5mg of salt/kg; 15-19 kg: 1/4 tab/week; 20-30 kg: 1/2 tab/week; 31-45 kg: 3/4 tab/week; >45 kg: 1 tab/week	• Good choice for long trips because it is taken only weekly • Can be used in all trimesters of pregnancy if there is no other option	• Cannot be used in areas with mefloquine resistance • Cannot be used in patients with certain psychiatric conditions, patients with epilepsy or cardiac conduction abnormalities • Not a good choice for last-minute travelers and for trips of short duration because drug needs to be started ≥2 weeks before travel. • Some patients will not want to take medications for 4 weeks after travel.

* Preferred choice for Malaysia

** Alternative

References:

1. British National Formulary Edition 83 March-Sept 2022.
2. World Health Organization International travel and health 2012.
3. Public Health England. Guidelines for malaria prevention in travellers from the UK 2021.
4. Malaria - Chapter 4 - 2020 Yellow Book | Travelers' Health | CDC Accessed in 30th July 2022.

Author:
Dr. Chua Hock Hin, Infectious Disease Consultant, Ministry of Health,
Malaysia



CHAPTER 11 : Vector Control Strategies In Malaysia

Malaria vector control strategies have evolved over time, leading to the development of numerous guidelines and protocols. The most recent of these is the Guidelines of Malaria Vector Control ⁶, released in 2022. This document focuses on the essential objectives and planning methods for effective malaria vector control. As Malaysia moves into the Prevention of Malaria Re-establishment phase, a significant development is the use of disease stratification based on foci classification and malariogenic potential. This method helps prioritize resources and efforts for maximum impact.

In the context of malaria vector control, measures should be implemented across Active Foci, Residual Non-Active Foci, and Cleared Foci with high and medium RV index. However, when resources are limited, vector control in Cleared Foci with medium RV index can be given lower priority. This prioritization ensures efficient resource allocation and contributes to the overall success of malaria prevention efforts (**Table 11.1**).

Table 11.1: Intervention Determinations By Foci Classification And Rv Index

Foci Classification	RV Index	Vector Control Interventions
Pseudofoci		No vector control intervention
Active Foci		IRS or ITN/ LLIN for Minimum 6 cycles or 3 years.
Residual Non-active Foci		• IRS or ITN/ LLIN (continuation) • RV assessment after 6 cycles of vector control
Cleared Foci	High	• IRS or ITN/LLIN (Minimum 2 cycles or 1 year). • RV assessment every 2 cycles of vector control*
	Medium	
	Low	No vector control intervention

11.1 Core Intervention

In Malaysia, there are two (2) core interventions for malaria vector control; Insecticide Residual Spray (IRS) and Insecticide Treated Nets/Long-Lasting Insecticidal Nets (ITN/LLIN). The implementation of these methods depends on the type of focus areas, with Active Foci, Residual Non-Active Foci, and Cleared Foci with high or medium RV index being targeted for either IRS or ITN/LLIN. This comprehensive strategy aims to protect the population at risk of malaria. In remote locations, such as interior aboriginal settlements in peninsular Malaysia and the interior regions of Sabah/Sarawak, LLINs are preferred due to communication challenges and unsuitable housing for IRS. The choice of intervention is influenced by operational factors, including cost, insecticide availability, and resources. By addressing these elements, Malaysia can effectively implement the most suitable malaria vector control strategies in different situations.

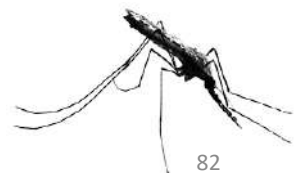


11.2 Supplementary Intervention

Malaysia plans to employ two main supplementary interventions for mosquito control: Larval Source Management (LSM) and Space Spraying. LSM, particularly larviciding, is a cost-effective method used in areas with limited breeding sites near houses. It targets mosquito larvae using biological or chemical compounds. Space spraying is used to eliminate infective adult mosquitoes during outbreaks, involving techniques like thermal fogging and aerosol distribution. It is deployed in Active Foci with reported human malaria cases except in pseudo foci. Timing space spraying during the peak biting time of the vector mosquito is crucial for effectiveness.

References:

1. National Strategic Planning for Prevention of Malaria in Malaysia (NSFPOR) 2021-2026.



CHAPTER 12 :

Antimalaria Drug Resistance Surveillance & Monitoring

DRS_Annex 1

National Surveillance of Drug Response Against Malaria Parasite (Drug Response Surveillance - DRS)

Note: This form is to be completed for all cases of malaria seen and treated with the first line drugs at the all hospitals/clinics.

1	INFORMANT														
	Hospital					State									
	Name of Reporting Officer					Designation of Reporting Officer									
	Medical Officer														
	Coordinator														
2	PATIENT'S DEMOGRAPHIC DATA														
	Patient's Initial			Patient's ID (Reg. No/NRIC)											
	Age (Years)		Ethnic		Gender (✓)			Body weight (kg)							
					Male		Female								
3	DRUG/MEDICATION HISTORY														
	Day of treatment				Actual number of tablet given to and taken by the patient										
					AL		ASMQ		Others (Specify):						
					Adult dose		Paeds dose								
	D0														
	D1														
	D2														
4	PARASITE RESPONSE														
	Type of parasite (✓)			<i>P.falciparum</i>			☐			<i>P.vivax</i>			☐		
				<i>P.malariae</i>			☐			<i>P.ovale</i>			☐		
				<i>P.knowlesi</i>			☐								
	Follow-up slide post treatment					Parasite density (per uL blood)									
	Day		Date Due*		Actual Date**		Asexual			Sexual (gametocyte)					
	D0														
	D3														
	D7														
	D14														
	D21														
	D28														
D42															
5	CASE INVESTIGATION														
	Was this patient treated before for malaria (✓)?				No ☐										
					Yes ☐		What type of medication?(If known)								
					When was medication taken? (eg: 1 week/month prior to current treatment)										
	Probable Source of Infection in This Case (✓)				Local ☐										
Imported ☐					From where?										
6	VERIFICATION AND VALIDATION (Epid.Officer/MOH)														
	Name					Signature									

AL=Artemether-Lumefantrine (Riamet) ASMQ=Artesunate-Mefloquine

D0 = Day of starting the treatment

* Enter the date due for the subsequent post-treatment follow-up (D3, D7, D14, D21, D28, D42)

**Enter the actual date of slide collection during follow-up

**STANDARD OPERATING PROCEDURE
NATIONAL SURVEILLANCE OF DRUG RESPONSE AGAINST MALARIA PARASITE,
MINISTRY OF HEALTH MALAYSIA**

1. Organisation and Standard Operating Procedure (refer DRS_Annex 3)

- 1.1 Medical Officer of Health (MOH) at the District Health Office is responsible to coordinate and monitor surveillance activity.
- 1.2 Appoint a dedicated staff to act as coordinator (e.g: PPKP).
- 1.3 Identify all Medical Officers who manage malaria cases (from medical & paediatric wards) at all hospitals in the district.
- 1.4 Brief the medical officers, Sisters and senior Laboratory Technologist on the surveillance mechanism while patient admitted in the ward and provide the coordinator's contact phone number. The coordinator will continue the surveillance process after patient being discharged from the ward.
- 1.5 Coordinator to provide and distribute the Surveillance Form (DRS_Annex 1) to MOs/Sisters of the relevant wards
- 1.6 MOH to instruct the MOs to carry out the surveillance activities.
- 1.7 MOs complete the Surveillance Form, inform coordinator via phone and return the form to the coordinator once patient is discharge. The coordinator will continue the follow-up procedure and complete the form.
- 1.8 Coordinator compiles all Surveillance Form and key in data into e-Vekpro completely.
- 1.9 Coordinator checks the total number of *P.falciparum*, *P.vivax*, *P.malariae*, *P.ovale* or *P.knowlesi* malaria cases admitted for the month and prepares Reporting Format (DRS_Annex 4) for the MOH to verify and validate.
- 1.10 Send the Reporting Format (Drs_Annex 4) to State Health Office.
- 1.11 State KPP will compile reports from all districts, prepares Reporting Format (Drs_Annex 5) and submit to Vector Borne Disease Section, on a monthly basis to :-
National Anti-Malarial Surveillance,
C/o Malaria Unit,
Vector Borne Diseases Section,
Disease Control Division, Block E10, Parcel E,
Ministry of Health Malaysia, Precint 1,
62590, Putrajaya.
- 1.12 Vector Borne Disease Section will compile reports from all states and prepare monthly report.

2. Patient enrolment

- 2.1 Inclusion criteria :
 - 2.1.1 Admit patient
 - 2.1.2 Confirmed laboratory diagnosis of the presence of mono-infection asexual stages of *P.falciparum*, *P.vivax*, *P.malariae*, *P.ovale* or *P.knowlesi*.
 - 2.1.3 Ability to swallow oral medication.
 - 2.1.4 Non-severe and uncomplicated malaria cases treated with first-line treatment regime (refer Guidelines for Management of Malaria 2013).
- 2.2 Attach the Surveillance Form (**DRS_Annex 1**) to the patient Case Note and enter the information accordingly.
- 2.3 From the history, acquire the information on previous history of malaria treatment and probable source of infection. Enter the information in Section 5 of the Surveillance Form accordingly.

3. Treatment regime and medication history

- 3.1 Calculate the artemether-lumefantrine (AL-Riamet) and artesunate-mefloquine (ASMQ-FDC) dosage based on the body weight according to the following regime:-

Treatment regime : 3 days treatment schedule (D0, D1 and D2).

(refer Guidelines for Management of Malaria 2013).

Drug	artemether-lumefantrine (AL-Riamet)	artesunate-mefloquine (ASMQ-FDC)
Dose per day	2 doses (BD)	1 dose (OD)
Total dose	2 X 3 days = total 6 doses	1 X 3 days = total 3 doses

Dosage calculation	5 - < 15 kg : 1 tablet per dose 15 - < 25 kg : 2 tablets per dose 25 - < 35 kg : 3 tablets per dose ≥ 35 kg : 4 tablets per dose	< 10 kg : 1 tablet AS 25mg OD X 3 days, MQ 125mg single dose 10 -20 kg : 2 tablets ASMQ 25/55mg per dose 20 -40 kg : 1 tablet ASMQ 100/220mg per dose ≥ 40kg : 2 tablets ASMQ 100/220mg per dose
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3.2 Ascertain the strength of the drug component for each tablet and determine the number of tablet required for each drug.

3.3 Instruct the nurse to supervise the taking of the medication. If the patient vomits within ½ hour of consumption, repeat the dose and indicates that in the appropriate drug column of Section 3 with the initial "rpt".

3.4 Record the actual number of tablet given to and taken by the patient in Section 3. Record also other concomitants treatment (e.g. paracetamol, antibiotics, etc).

(NOTE: D0 is the first day of treatment)

4. Parasite response and monitoring

4.1 Original slide that confirm the diagnosis (D0) must be kept for future reference.

4.2 Clinically monitor the case according to the standard management as per guidelines protocol.

4.3 Repeat BFMP one day after the last dose of AL/ASMQ (D3). Enter the result in Section 4 of the Surveillance Form. (You may repeat BFMP on D1 and D2 if clinically justified e.g. no evidence of clinical improvement or clinically getting worse).

4.4 If the result on D3 is negative, repeat BFMP on D5, D6 and D7. If continue to be negative, discharge the patient. Enter the result of D7 in the Surveillance Form. MOs inform the coordinator and return the form to the coordinator.

4.5 Coordinator arrange for the patient to be followed-up on D14 at patient's house. Inform the patient of the appointed date for the BFMP taking.

4.6 If BFMP negative, enter the result in the Surveillance Form and inform the patient for D21, D28 and D42 visit for similar follow-up. Stress the importance of complete monitoring to ensure compliance.

4.7 If BFMP continues to be negative until D42 or if patient failed to return after 42 days of initiation of treatment, conclude the surveillance.

4.8 If :-

4.8.1 BFMP is positive during any of the follow-up (D3, D5-D7, D14, D21, D28, D42), or

4.8.2 BFMP is repeated and the asexual density is not less than 25% and 50% on D1 and D2 respectively compared to D0 density, or

4.8.3 Patient shows clinical deterioration, concludes the surveillance. If surveillance is concluded because of clinical deterioration, indicate such reason (e.g. "Treatment failure on D2) in Section 4 (just write anywhere).

4.8.4 If conclusion of treatment failure is made on D14, D21, D28 and D42; send both of D0 and the current positive slides to IMR for genotyping. Preferably, send spotted blood sample on filter paper.

5. Submission of surveillance format

5.1 Once the surveillance is concluded, the coordinator check all the information entered into the Surveillance Form and key in data into e-Vekpro completely. Prepare the report with verification and validation by the MOH.

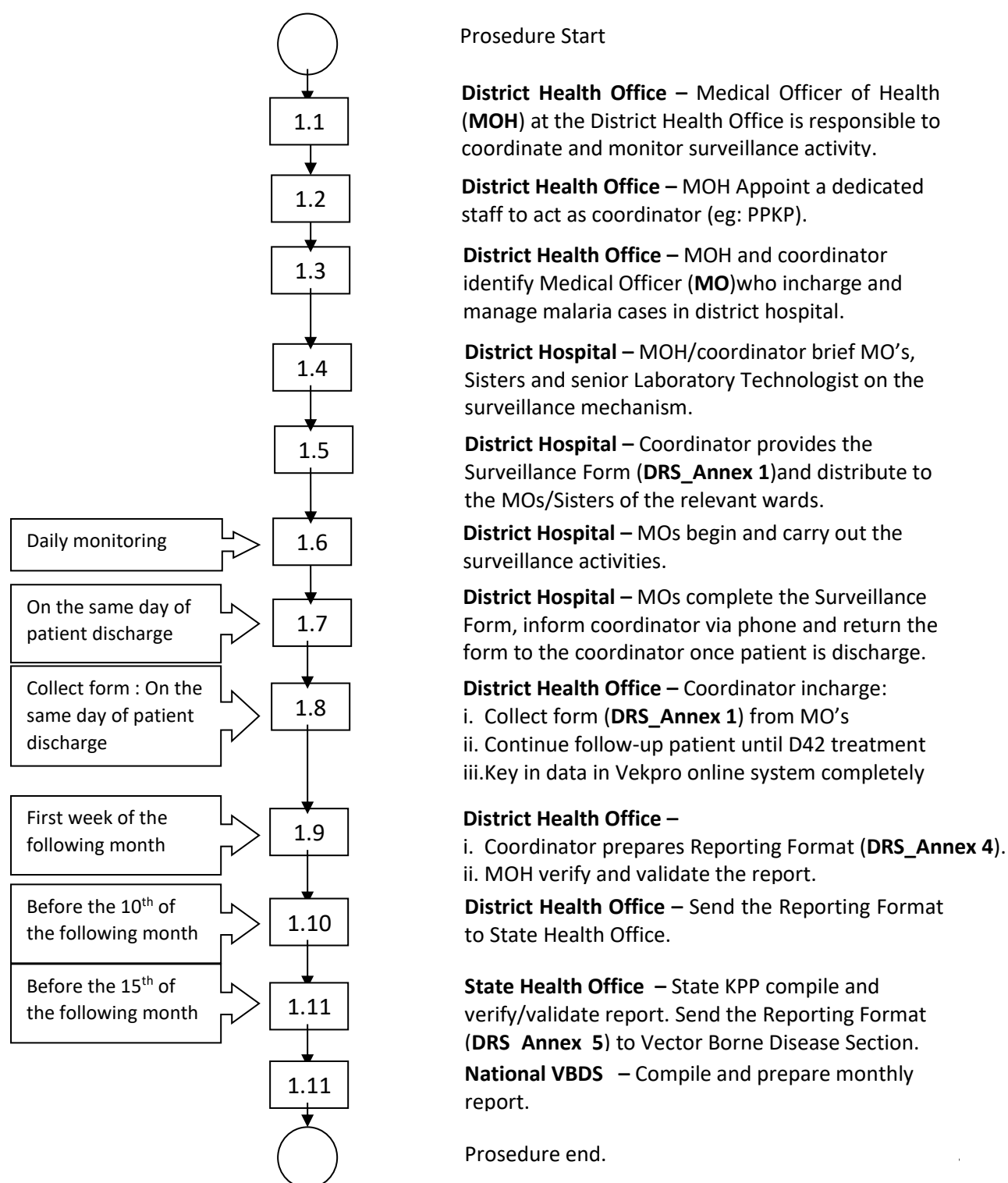
5.2 Send the report Form to the national coordinator, close the case for surveillance.

Author: Dr. Mohd Hafizi Abdul Hamid, MD, MPH. Malaria Unit,
Vector Borne Disease Sector, Disease Control Division, Ministry
of Health, Malaysia



National Surveillance of Drug Response Against Malaria Parasite

Standard Operating Procedure Flow Chart :



**National Surveillance of Drug Response Against
Malaria Parasite (*All species*)
REPORTING FORMAT (District)**

Month Year

District

Type of Parasite: _____

No.	Hospital	No. of patient admitted for the month	No. of patient enrolled for anti-malaria surveillance for the month	No. of surveillance concluded in the month	No. of surveillance form submitted for the month	No. of resistance cases
Total						

Prepared by:

(Coordinator)

Verified and validated by:

(District health officer)

**National Surveillance of Drug Response Against
Malaria Parasite (*All species*)
REPORTING FORMAT (State)**

Month Year

District

Type of Parasite: _____

No.	Hospital	No. of patient admitted for the month	No. of patient enrolled for anti-malaria surveillance for the month	No. of surveillance concluded in the month	No. of surveillance form submitted for the month	No. of resistance cases
Total						

Prepared by:

Verified and validated by:

(State KPP-Vector)

CHAPTER 13 :

Quality Assurance Program For Malaria Microscopy

Important notes

- ✚ All case of suspected malaria should have a parasitological test to confirm diagnosis and supported with quality assurance program.
- ✚ Microscopic examination of both thick and thin film remains the gold standard for confirmation of malaria in Malaysia.

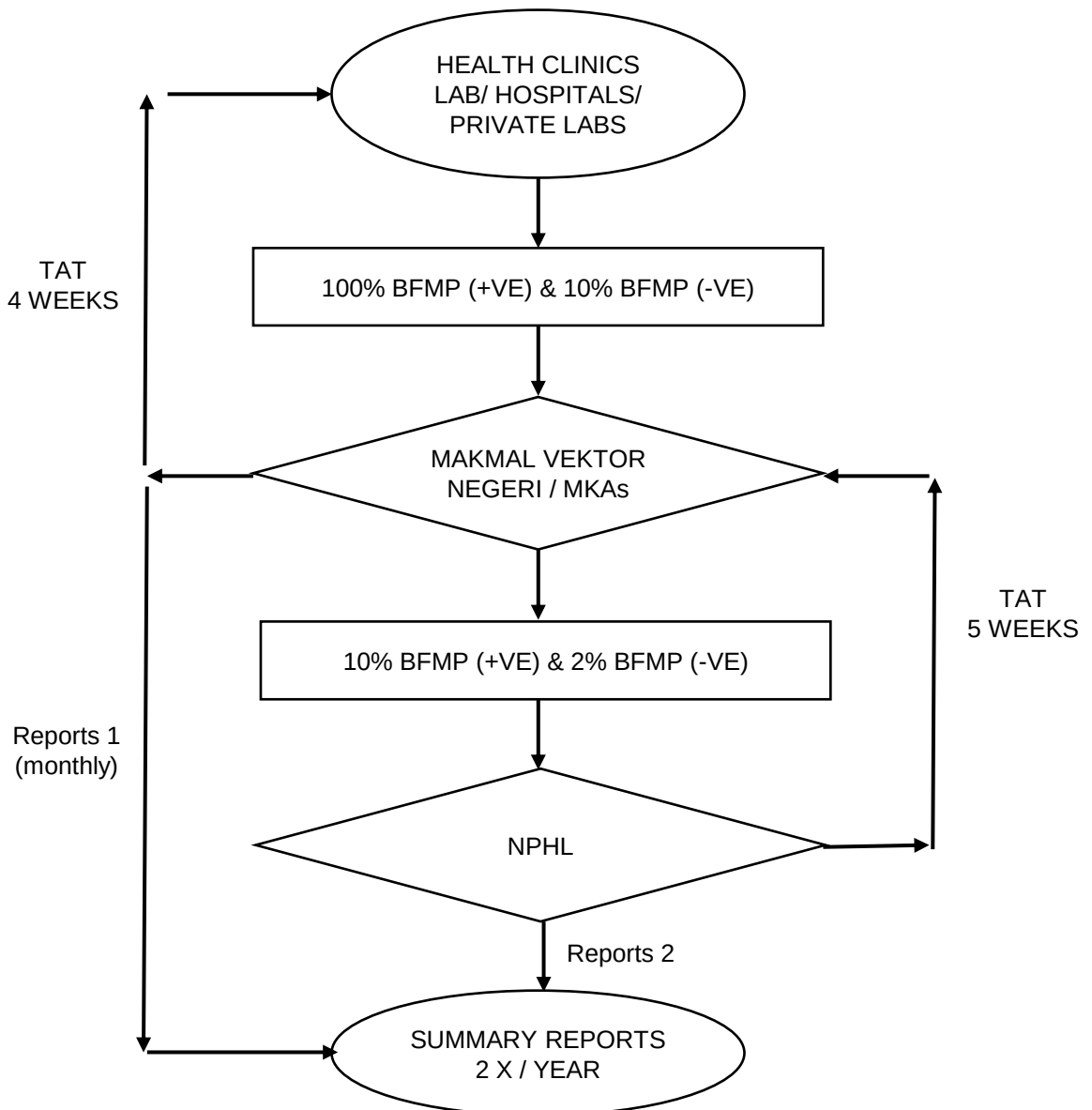
13.1 Quality Checking

The microscopy quality assurance and quality control are carried out as outlined in the WHO Malaria Microscopy Quality Assurance Manual, Version 1 (2009) and Version 2 (2016).

Table 13.1: Quality checking of blood film malaria parasite

Responsible bodies	Actions
Field sampling/Health Care Clinic /Hospitals	Sending of 100% of microscopy-positive slides and 10% of negative slides to State Vector Laboratory.
State Vector Laboratory/Kota Kinabalu Public Health Laboratory	Rechecking of 100% of positive slides and 10% of negative slides for quality control. Results are reported back to the respective laboratory optimally within one month. Subsequently all the slides have to be sent to National Public Health Laboratories for quality checking.
Parasitology & Mycology Section, National Public Health Laboratory (NPHL)	Rechecking 100% of positive slides and 2% of negative slides, which were sent from State Vector Laboratory/other Public Health Laboratories for quality control. Results are reported back to the respective laboratory optimally within 5 weeks upon receipt of slides. Slides with discrepancy or “errors” identified by NPHL will proceed with PCR assay. Results shall be reported immediately using the “errors/discrepancy” notification form.

**FLOW CHART
QUALITY CHECKING (QC) FOR BFMP
[10% BFMP (+VE) & 2% BFMP (-VE)]**



13.2 Training programme and supervision of malaria diagnostics

All malaria microscopists at all health clinics and hospitals must undergo a formal malaria microscopy training prior performing to the microscopy works and must attend the re-fresher training of malaria microscopy at respective State Vector Laboratories or Public Health Laboratories at least once in three years. It is also mandatory that all the reference malaria microscopists at State Vector and Public Health Laboratories undergo a continuous refresher training once in three to four years at National Public Health Laboratory. The 5-days refresher training is considered essential for maintaining competency and commitment of the malaria microscopists.

In addition to the continuous training and refresher course, malaria microscopists are recommended to participate and attend the National Public Health Laboratory Competency Assessment of Malaria Microscopist (NPHL ECAMM) and the WHO-WPR External Competency Assessment of Malaria Microscopist (WHO-WPR ECAMM) and to be certified as a malaria microscopist and become A National Core Group of Expert Microscopist. The NPHL ECAMM and WHO-WPR ECAMM course are organized and coordinate by the National Public Health Laboratory, Ministry of Health Malaysia.

References:

1. WHO Expert Committee on Malaria. Twentieth report. Geneva, World Health Organization, 2000 (WHO Technical Report Series, No. 892).
2. WHO Expert Committee on Malaria. Twentieth report. Geneva, World Health Organization, 2000 in WHO Technical Report Series, No. 892.
3. World Health Organization. Guidelines for the treatment of malaria Second edition 2010.



CHAPTER 14 :

ENHANCED MALARIA CASE SURVEILLANCE TO PREVENT RE-ESTABLISHMENT OF MALARIA IN MALAYSIA

14.1 INTRODUCTION

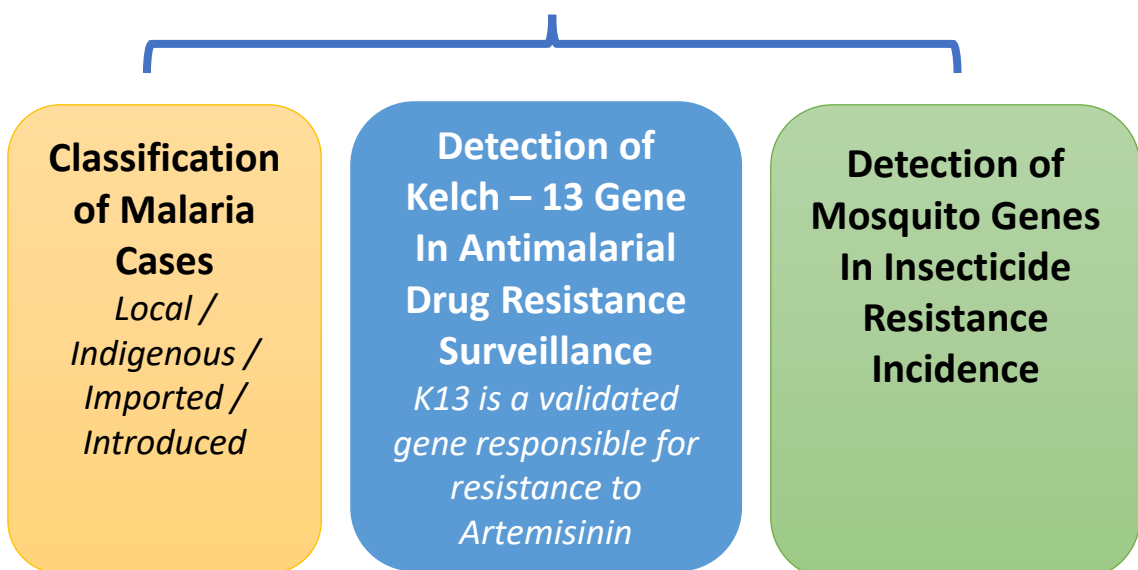
Molecular epidemiology of malaria is the study of malaria genotypes and their gene expressions that is utilized to study disease occurrence and transmission, evolution and adaptation of zoonotic malaria into human subjects and reactions to antimalarial agents.

Human and mosquito vector genetics vary but at the same time, offers a symbiotic relationship when explaining the transmission dynamics in malaria. Without the mosquitos, no new transmission can occur. Without human subjects, then there are no malaria parasites to introduce to the mosquitos.

Malaria genotypes can be studied using molecular genotyping methods that shows the relationship between malaria species and the infected hosts. Using malaria genotyping methods, in Malaysia, we are able to:

- Correlate whether an index (primary) case of malaria is related to other contacts during a malaria outbreak scenario, in combination with epidemiological link,
- Identify whether the parasite population infecting several localities are of the same or different genotypes,
- Build a molecular phylogenetic tree to compare whether the parasite species infecting the local community is closely related to other geographical areas or actually of indigenous species,
- Monitor and/or detect at an earlier onset of allelic differences in fragments of genes that are associated or known to demonstrate antimalarial drug resistance

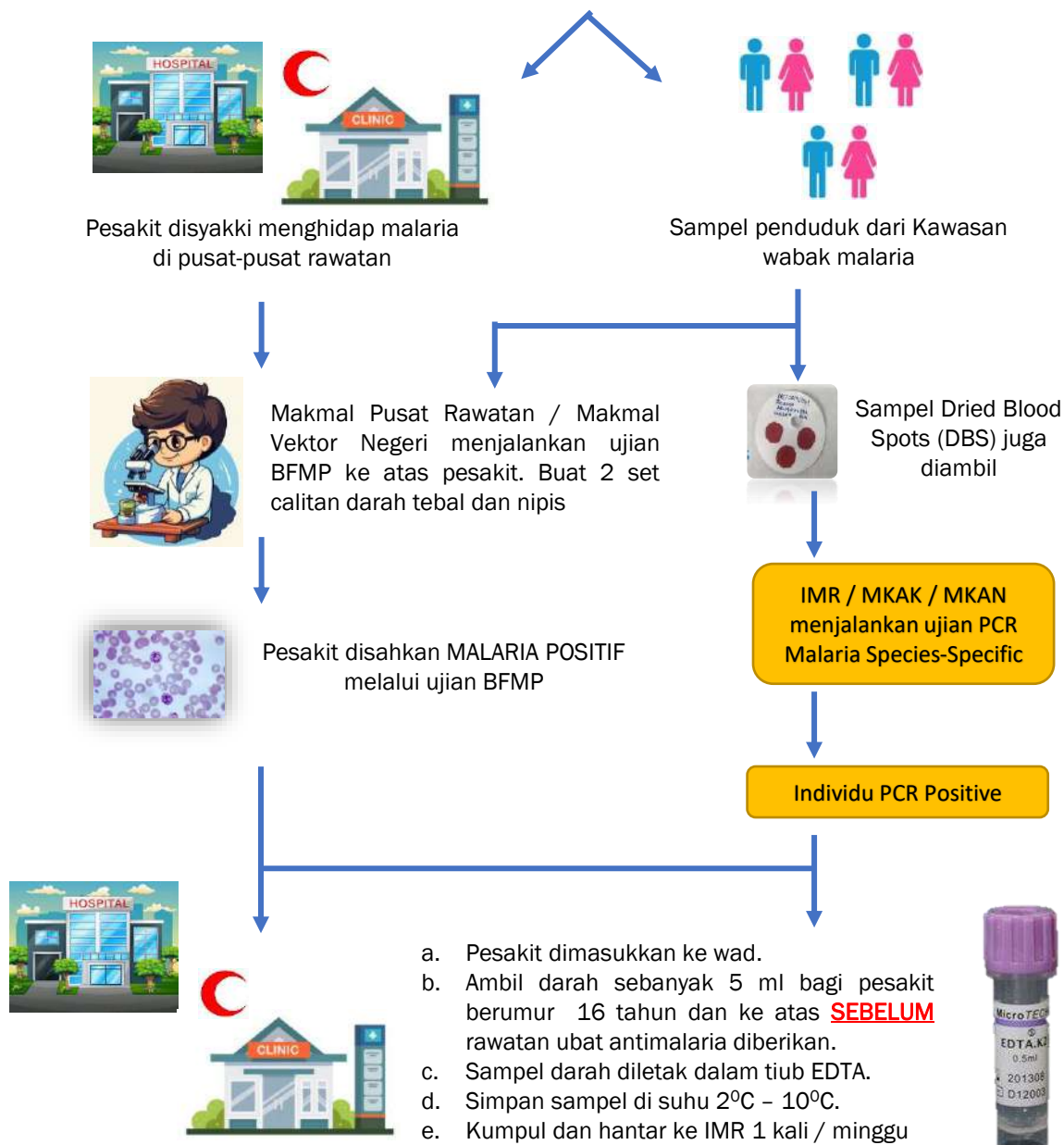
In Malaysia, molecular genotyping methods are used in:



DEFINISI PUSAT RAWATAN

Hospital / Klinik Kesihatan / Wad/ Jabatan Kecemasan / Jabatan Pesakit Luar
(Kerajaan, Universiti, Swasta)

SUMBER SAMPEL MALARIA UNTUK UJIAN MALARIA GENOTYPING (PENKELASAN MALARIA IMPORT DAN SIASATAN WABAK MALARIA)



Hantar ke IMR:

- Sampel darah EDTA di suhu 3°C – 10°C
- 1 set slaid mikroskopi positif
- Borang Permohonan Malaria Genotyping (Pengelasan Malaria Impot / Siasatan Wabak Malaria)



UNIT PENGURUSAN SPESIMEN (Blok C3 Aras 1)

Inst. Penyelidikan Perubatan @ Kompleks NIH
No. 1, Jln Setia Murni U13/52, Bandar Setia Alam,
40170 SHAH ALAM, SELANGOR
u.p. UNIT PARASITOLOGI
Tel: 03-3362-7660\7661\7663\ 7667\ 7668

DEFINISI PUSAT RAWATAN

Hospital / Klinik Kesihatan / Wad/ Jabatan Kecemasan / Jabatan Pesakit Luar
(Kerajaan, Universiti, Swasta)

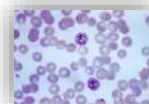
SUMBER SAMPEL MALARIA UNTUK UJIAN MALARIA GENOTYPING UNTUK:

- SURVEILAN KERINTANGAN UBAT ANTI MALARIA
- RECRUDESCENCE/ RE-INFECTION/ RELAPSE

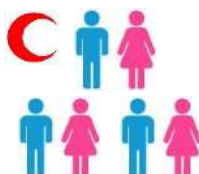
01



Pesakit disahkan **MALARIA POSITIF**
melalui ujian BFMP



02



Pesakit dimasukkan ke wad untuk
terima rawatan ubat antimalaria

03

MASA AMBIL SAMPEL	SURVEILAN KERINTANGAN UBAT ANTIMALARIA	RECRUDESCENCE / RE-INFECTION / RELAPSE
D-0 (sebelum mula rawatan)	Darah EDTA. Simpan di suhu 2 ⁰ C – 10 ⁰ C	Darah EDTA. Simpan di suhu 2 ⁰ C – 10 ⁰ C
D-3 (Hari ke-3 rawatan)	Darah EDTA. Simpan di suhu 2 ⁰ C – 10 ⁰ C	TIADA KEPERLUAN
D-7	DRIED BLOOD SPOT. Simpan di suhu 2 ⁰ C – 10 ⁰ C	TIADA KEPERLUAN
D-14	DRIED BLOOD SPOT. Simpan di suhu 2 ⁰ C – 10 ⁰ C	TIADA KEPERLUAN
D-21	DRIED BLOOD SPOT. Simpan di suhu 2 ⁰ C – 10 ⁰ C	Darah EDTA. Simpan di suhu 2 ⁰ C – 10 ⁰ C, ATAU di D-28
D-28	DRIED BLOOD SPOT. Simpan di suhu 2 ⁰ C – 10 ⁰ C	Darah EDTA. Simpan di suhu 2 ⁰ C – 10 ⁰ C, ATAU di D-42
D-42	DRIED BLOOD SPOT (bagi kes P. vivax sahaja). Simpan di suhu 2 ⁰ C – 10 ⁰ C	Darah EDTA. Simpan di suhu 2 ⁰ C – 10 ⁰ C, ATAU > D-42
> D-42 → TAHUN X	TIADA KEPERLUAN	Darah EDTA. Simpan di suhu 2 ⁰ C – 10 ⁰ C

04

Hantar ke IMR:

- Sampel darah EDTA di suhu 2⁰C – 10⁰C
- 1 set slaid mikroskopi positif
- Borang Permohonan Malaria Genotyping (Surveilan DRS / Recrudescence / Relapse)



UNIT PENGURUSAN SPESIMEN
(Blok C3 Aras 1)

Inst. Penyelidikan Perubatan @ Kompleks NIH
No. 1, Jln Setia Murni U13/52, Bandar Setia Alam,
40170 SHAH ALAM, SELANGOR

u.p. UNIT PARASITOLOGI

Tel: 03-3362-7660\7661\7663\ 7667\ 7668

Example of Laboratory Analysis of Malaria Genotyping During Malaria Outbreak Investigation

Step 1: Confirmation of Malaria Species By Nested PCR Assay

						BFMP		PCR	
59	WBK 2022/0566			01/07/2022	DARAH EDTA	PKD LIPI	PV 208/16 parasite per ul of blood	Plasmodium vivax	Proceed to Genotyping
60	WBK 2022/0567			01/07/2022	DARAH EDTA	PKD LIPI	PV 16/0 parasite per ul of blood	Plasmodium vivax	Proceed to Genotyping
61	WBK 2022/0568			01/07/2022	DARAH EDTA	PKD LIPI	PV 96/0 parasite per ul of blood	Plasmodium vivax	Proceed to Genotyping

Step 2: Malaria Genotyping Assays, Fragment Alleles Analysis for P. falciparum

Patient ID	Target Gene	Target Gene			Target Gene		Conclusion (Rujukan kepada Sampel- 1)
	CSP/ MSP1	MSP 1			MSP 2		
	FL	Fragment Allele			Fragment Allele		
	-	K1	MAD20	RO33	FC27	3D71C	
*IMR/PST/1		●					Sampel 1 Kes indeks/kontek/kluster
*IMR/PST/2		●					Sampel 2: Kes indeks/kontek/kluster mungkin terjangkit dari populasi parasit sama dan dari lokaliti sama
*IMR/PST/3		●		●			Sampel 3: Kes indeks/kontek/kluster mungkin terjangkit dari populasi parasit sama dari lokaliti sama atau lokaliti lain
*IMR/PST/4					●	●	Sampel 4: Kes indeks/kontek/kluster terjangkit dari populasi parasite yang berbeza. Punca jangkitan mungkin dari lokaliti selain disyakki
*IMR/PST/5			●		●		Sampel 5: Kes indeks/kontek/kluster terjangkit dari populasi parasit yang berbeza dari lokaliti sama atau lokaliti lain

EXAMPLE OF MALARIA GENOTYPING ANALYSIS IN MALARIA DRUG RESISTANCE SURVEILLANCE

BIL.	NAMA PESAKIT/KES	IMR LAB ID	CROSS REF-ID	IC / PASPORT NO	HOSP/ JKN ID	BFMP JKN/MKA	BFMP IMR	PCR IMR	DRS- HARI SAMPEL DIAMBIL	KEPUTUSAN SURVEILAN DRS	NOTA
1		2022/ DRS 01	2022/G132		NA	P.falciparum	P.falciparum	P.falciparum	Slaid (Day-0)		Tiada mutasi pada K13 yang dikesan menunjukkan tiada Artemisinin Drug Resistance.
2		2022/ DRS 02	2022/G132						Day 1		
3		2022/ DRS 03	2022/G132						Day 3		
4		2022/ DRS 04	2022/G132						Day 7		

Author:

Dr. Shamilah Hisam, MD, MScPHDC, DAPE. Parasitology Unit, Infectious Diseases Research Center, Institute for Medical Research, NIH Setia Alam, Selangor, Malaysia

Dr. Norazian, MScPHDC, DAPE. Parasitology Unit, Infectious Diseases Research Center, Institute for Medical Research, NIH Setia Alam, Selangor, Malaysia



CHAPTER 15 : Training for Malaria Management

Malaysia has established a comprehensive training program for health and medical personnel, including specialists, medical officers, nurses, environmental health officers, entomologists, and laboratory technologists.

The program, led by the Ministry of Health's (MOH) Malaria Elimination Program, is implemented in collaboration with the Institute for Medical Research, the Malaysia Space Agency (MYSA), universities, and other partners.

MOH also partners with international organizations such as the World Health Organization (WHO), the Asia Pacific Malaria Elimination Network (APMEN), the Asian Collaborative Training Network for Malaria (ACTMalaria), the Malaria Consortium, and other WHO Collaborating Centers to expand training opportunities for Malaysian malaria personnel. Regular training, refresher courses, and participation in national and international programs ensure the continued expertise of the country's malaria health workforce.

15.1 : Malaria Training Program

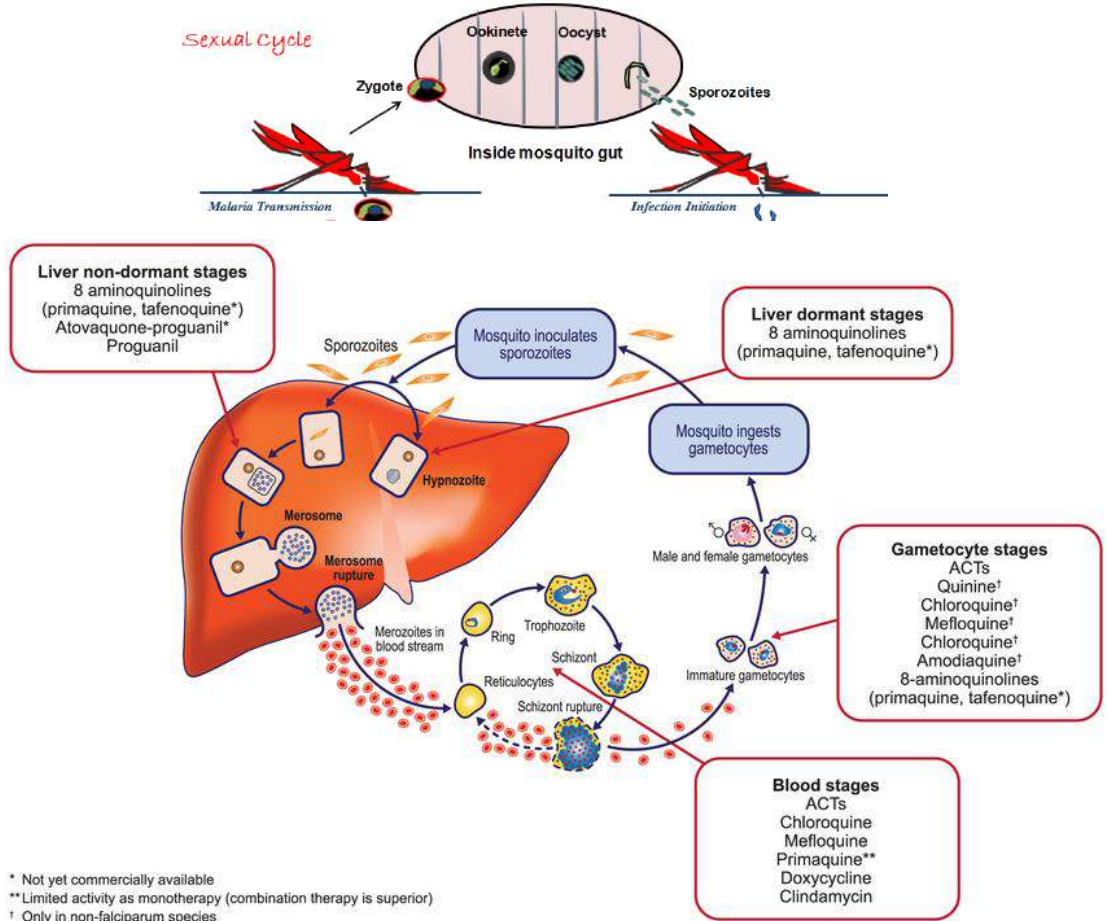
Name of Malaria Training	Organizers	Medical personnel in hospitals / clinics	Environmental Health Officers	Entomologist	Medical Lab Technologist
Integrated Vector Management Course (IVM)	CDC MOH Malaysia	√	√	√	
Malaria Management Field Operation Course (MMFO)	WHO ACT Malaria DOH Thailand	√	√	√	√
Malaria Elimination Training Course	WHO APMEN CDC China	√	√	√	√
Malaria Prevention & Control Course	WHO APMEN	√	√	√	√
Advanced Malaria Microscopy Training	WHO RITM, Philippines	√	√	√	√
Applied Parasitology and Entomology (DAP & E) Course	Institute for Medical Research (IMR), MOH Malaysia	√	√	√	√
Clinical Malaria Management Course	Hospitals / Clinics	√	√	√	√

Name of Malaria Training	Organizers	Medical personnel in hospitals / clinics	Environmental Health Officers	Entomologist	Medical Lab Technologist
National Public Health Laboratory Competency Assessment of Malaria Microscopist (NPHL ECAMM)	NPHL State PHL				√
WHO-WPR External Competency Assessment of Malaria Microscopist (WHO-WPR ECAMM)	NPHL WHO				√
Malaria Microscopy Refresher Training	State Vector Lab Hospital's and clinic's Lab				√
Vector Control Management Training	VBDS MOH Malaysia	√	√	√	
Malaria Vector Surveillance for Elimination (MVSE) Course	WHO			√	
Malaria Data Management System Training	VBDS MOH Malaysia MYSA	√	√	√	
Field Entomology Training	Entomology & Pest Sector State Health Office			√	



CHAPTER 16 : Research (Potential Area)

16.1 Anti-Malaria Drugs: Current Choice and Future Drug Candidates



Above groups of known antimalarial drugs that are targeted to kill / inhibit specific stage of malaria parasite development

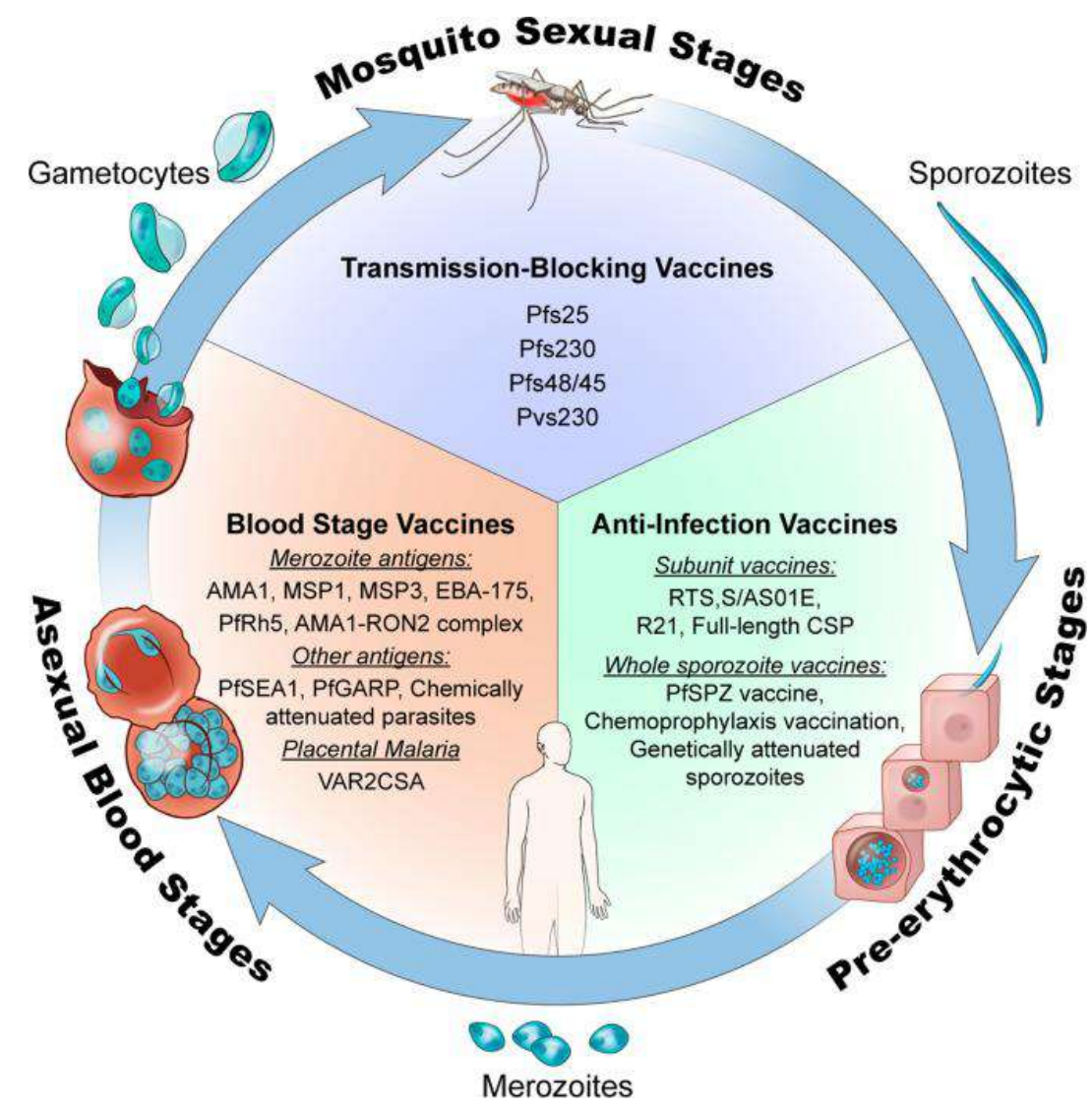
Current Situation and Research Potential In Use of Anti-Malarial Drugs:

- Drugs to block sporozoites from mosquito to enter liver
- Drugs to render mosquito non-fertile
- No alternative to Primaquine as hypnozoitocidal
- Tafenoquine (anti-hypnozoite) specific for PF, under trial
- Rising resistance to Artemisinin derivatives
- Known resistance to non-artemisinin agents
- Artemisinin not effective against mature gametocytes of PF

Future Direction of Transmission Blocking Drugs / Insecticide that have the ability to BLOCK:

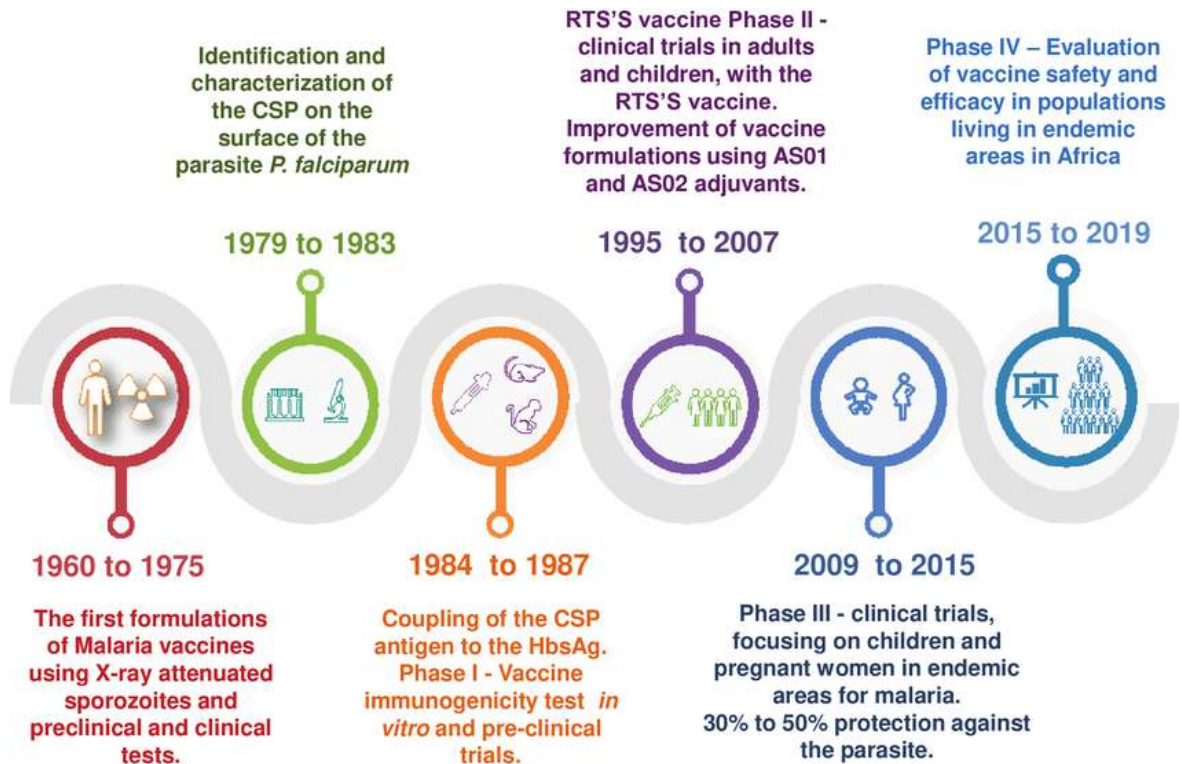
- sporozoites to enter human liver
- gametocytes to become zygote
- Ookinete to enter midgut
- Formation of oocyst

16.2 Malaria Vaccines



Form	Life Cycle Stage	Vaccine Strategy
✦ Sporozoite	✦ Transmitted by mosquitoes ✦ Infects liver hepatocytes ✦ Develops into thousands of merozoite forms	✦ Vaccines targeting sporozoites or infected hepatocytes aim to prevent infection
✦ Merozoite	✦ Released into blood stream ✦ Replicates in red blood cells ✦ Exponential rise in parasitemia associated with clinical illness	✦ Vaccines targeting merozoites or infected red blood cells aim to reduce disease severity
✦ Gametocyte	✦ Some merozoites instead develop into gametocytes ✦ Can be taken up by a feeding mosquito and transmitted	✦ Vaccines targeting gametocytes aim to block transmission in a population

Progression of Development of Malaria Vaccines



Process of Development of Malaria Vaccines

